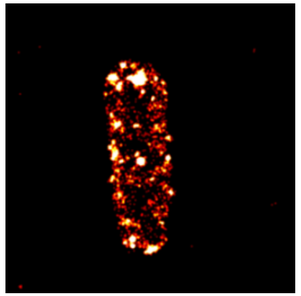
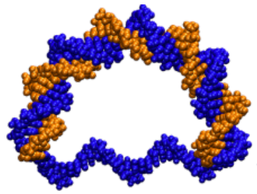
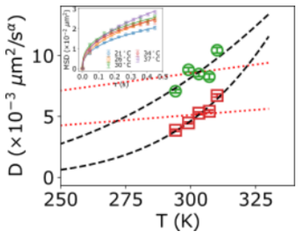
Protein Diffusion in Live Bacteria and Bacterial Motility in Porous Structures

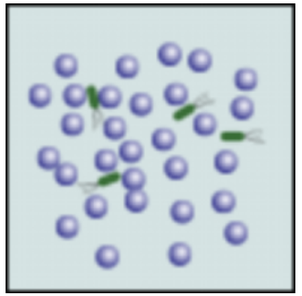
Dr. Yong Wang

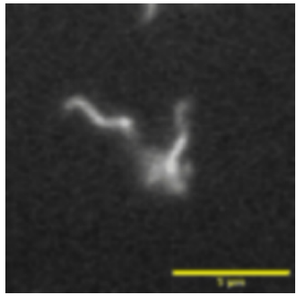
Department of Physics
University of Arkansas











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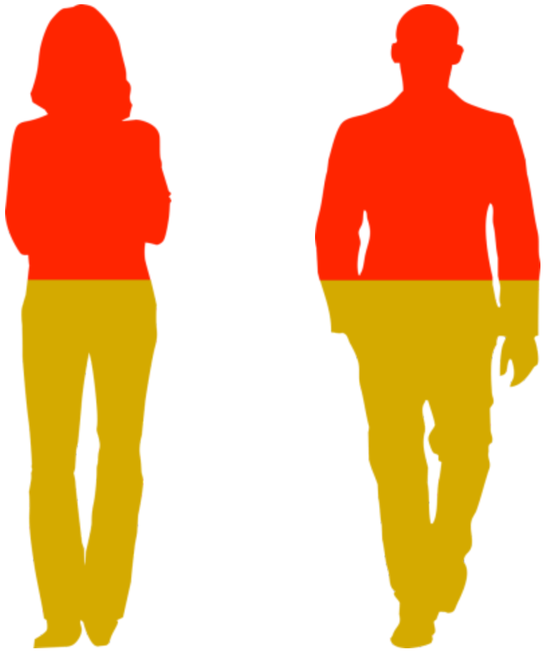
Why Bacteria?

43% human cells



57% bacterial cells
(many are beneficial)

Sender et al. PLoS Bio (2016)

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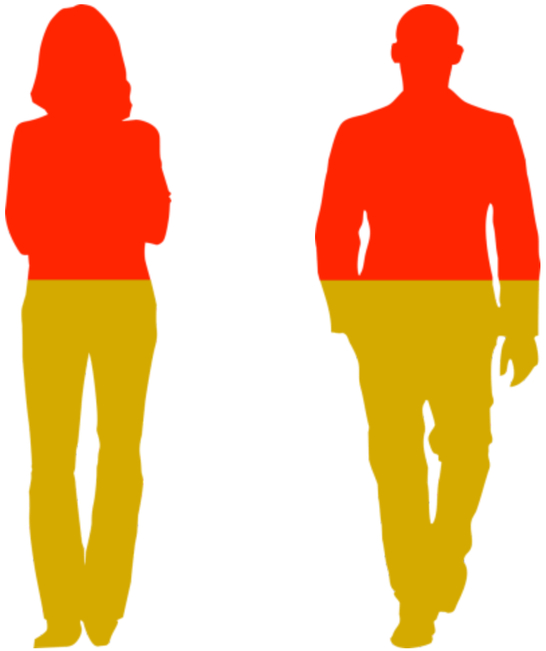

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Harmful bacteria cause infections, diseases, and death!

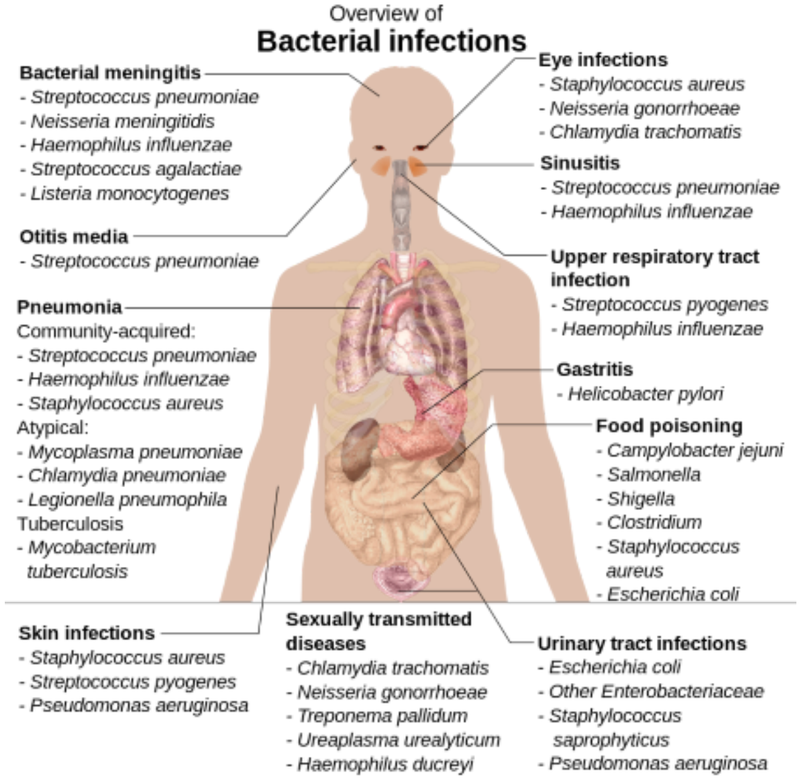
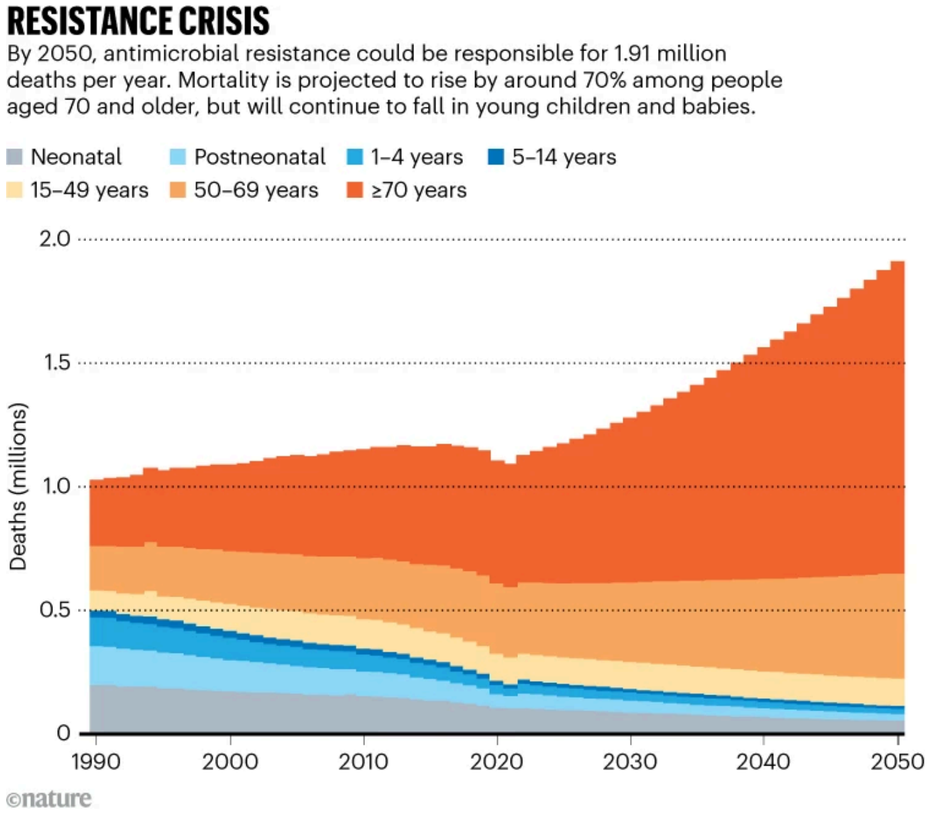


Image from study.com



Naghavi et al. The Lancet (2024)

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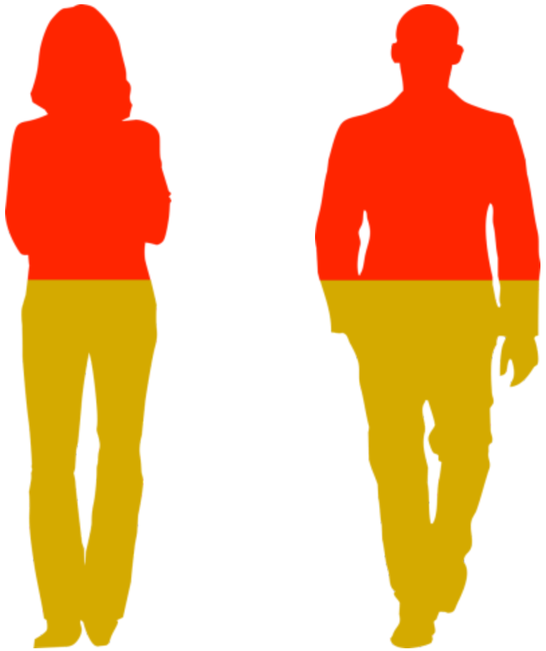

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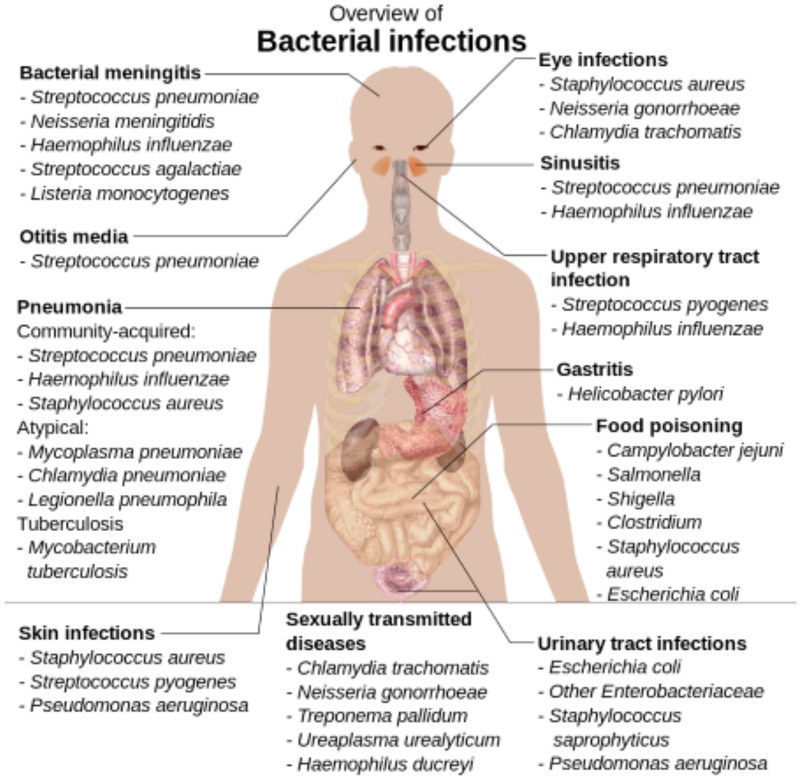

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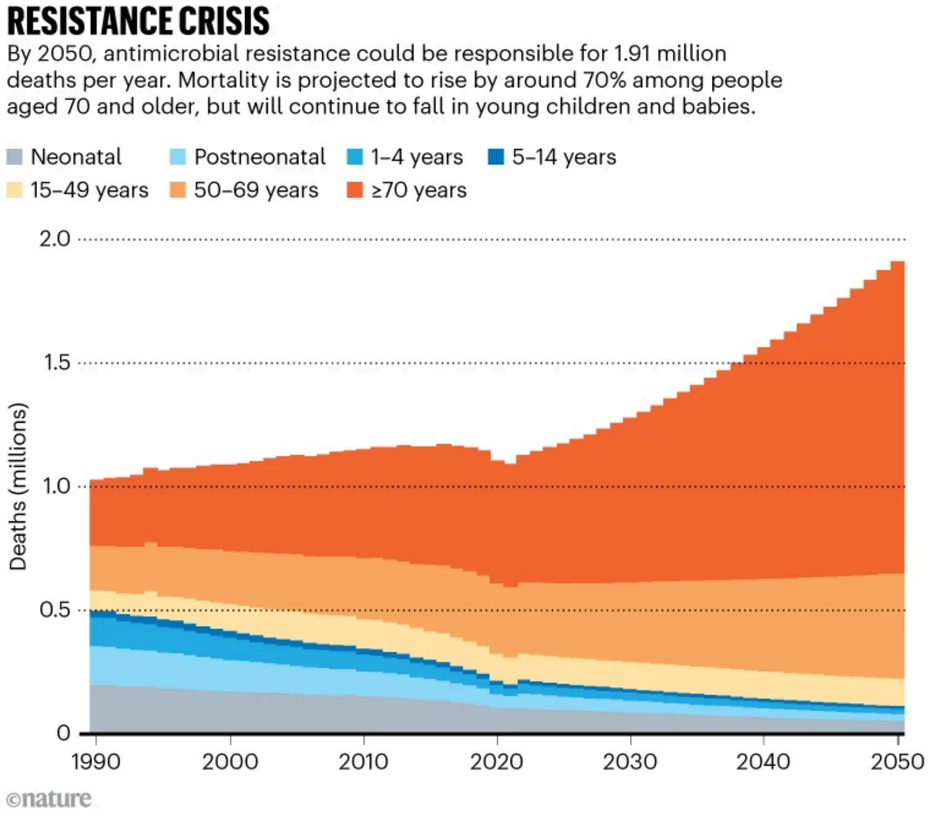

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If we can understand the properties and functions of bacteria,
we can help beneficial bacteria and fight against harmful ones!

Diffusion of Single Proteins in Bacteria

Protein Diffusion is Important for Bacteria

Bacteria Lack Active Transport Mechanism

Video from YouTube

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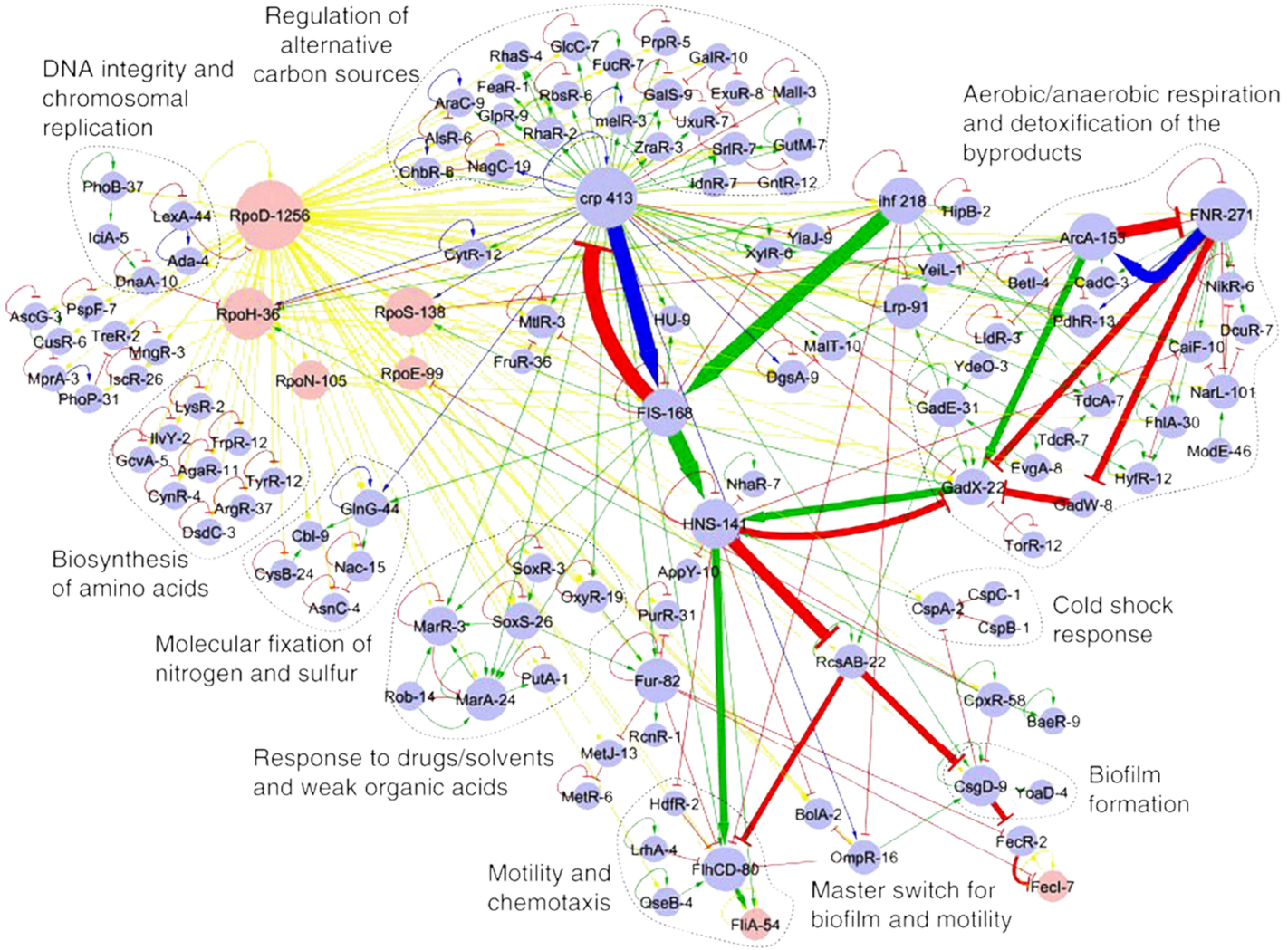
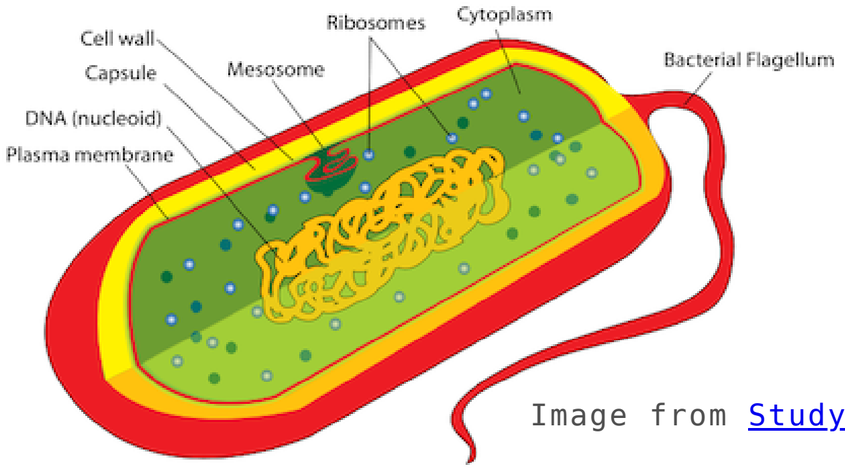
Regulatory Networks in Bacteria Rely On Diffusion



Martínez-Antonio et al., JMB (2008)

Protein Diffusion in Bacteria is Interesting in Physics

70% of bacterial content is water



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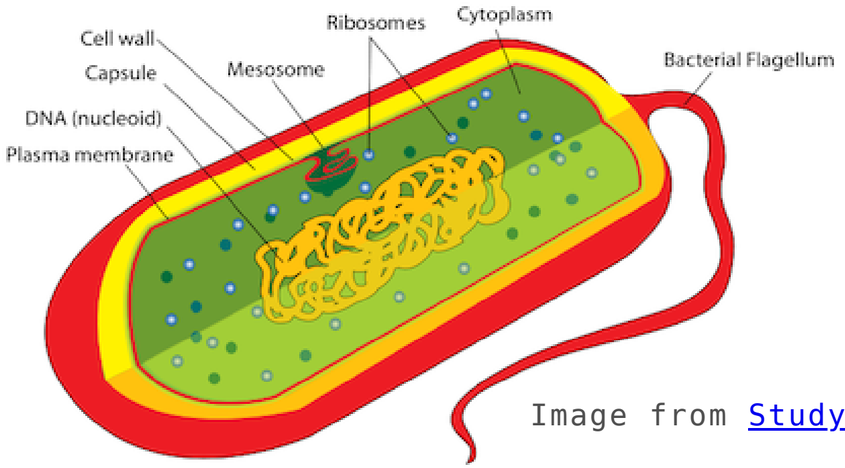

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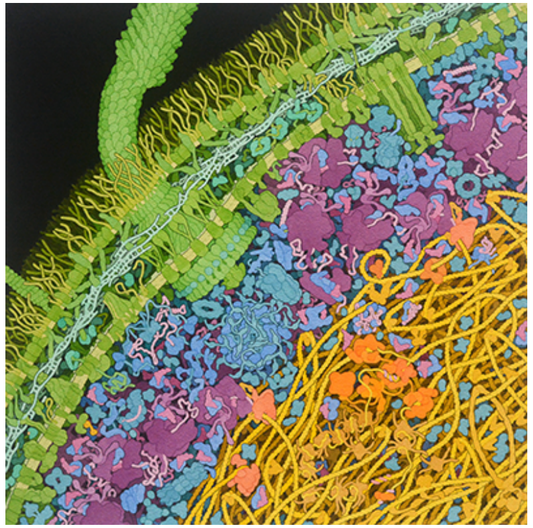
Full of molecules and polymers



Goodsell, Nat Struct Mol Biol (2021)

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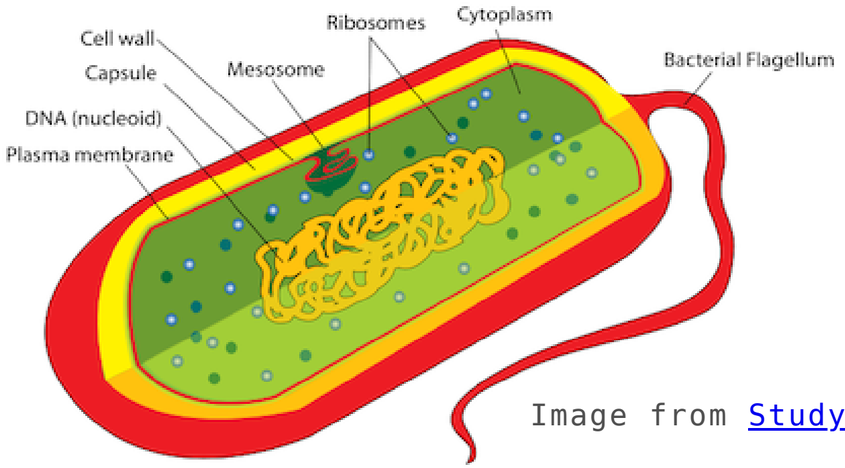

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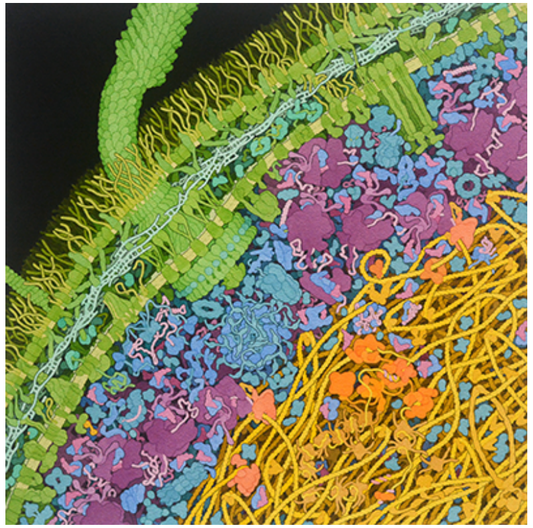

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Constant matter/energy exchanges

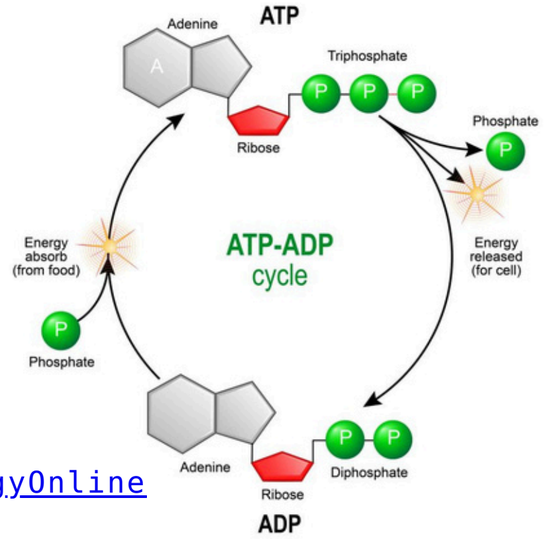

Image from BiologyOnline

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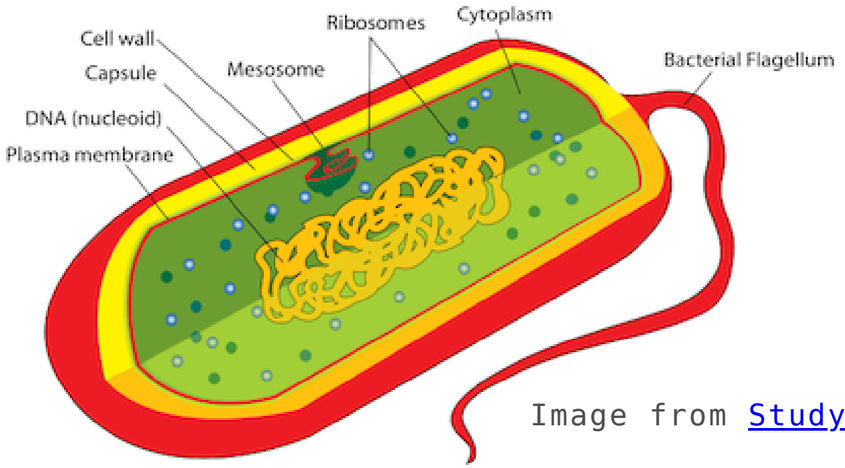

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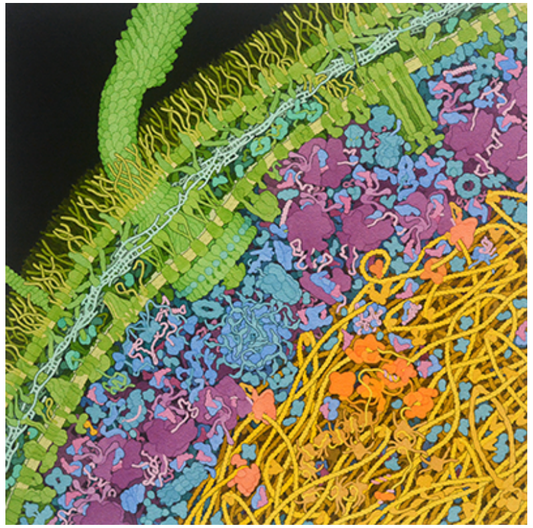

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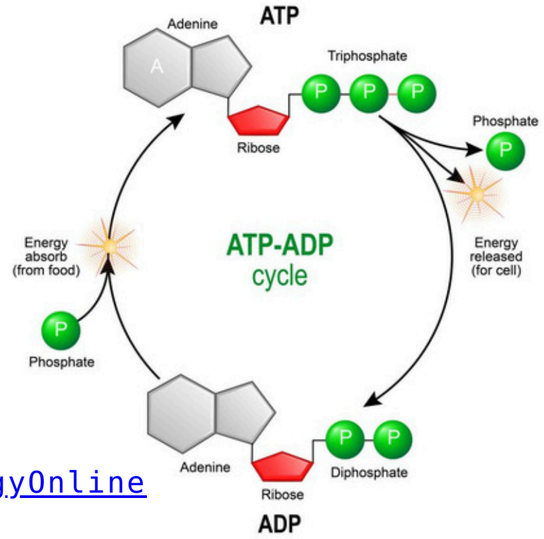


Image from BiologyOnline

"Always" out of equilibrium

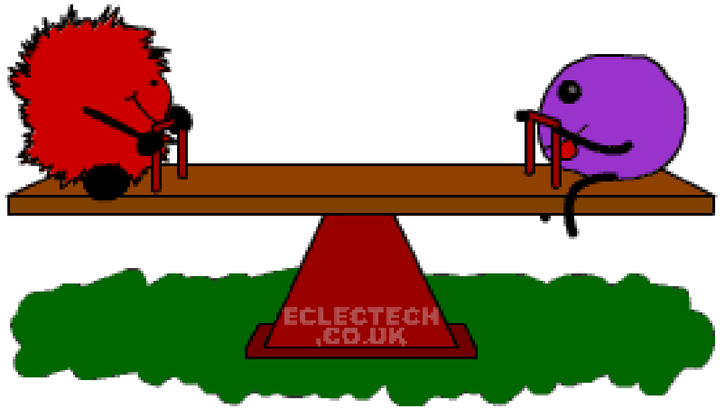
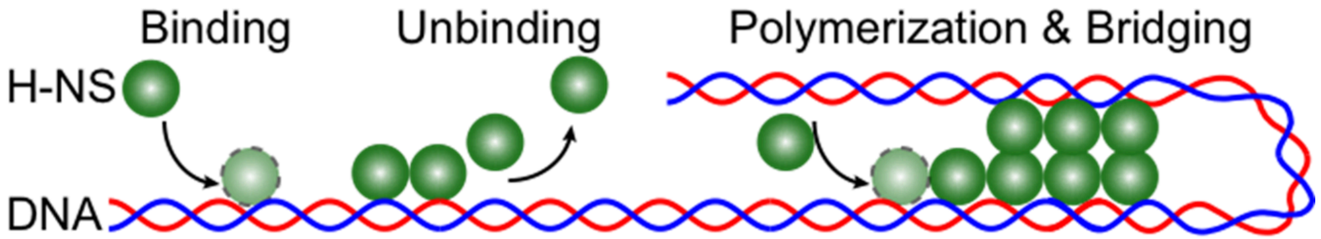


Image from Eclectech

Histone-like Nucleoid Structuring (HNS) Protein



- DNA-binding protein (nonspecific)
- Several thousand copies per bacterium
- Regulate 5% of all genes in *E. coli*
- Modulate stress responses
- Involved in DNA compaction and chemotaxis

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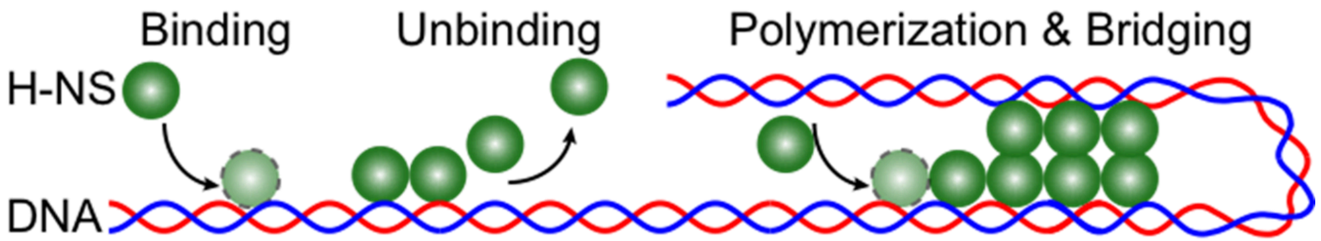

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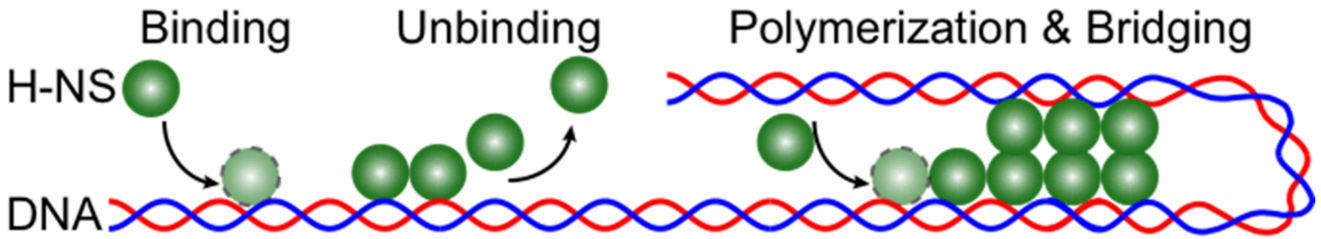

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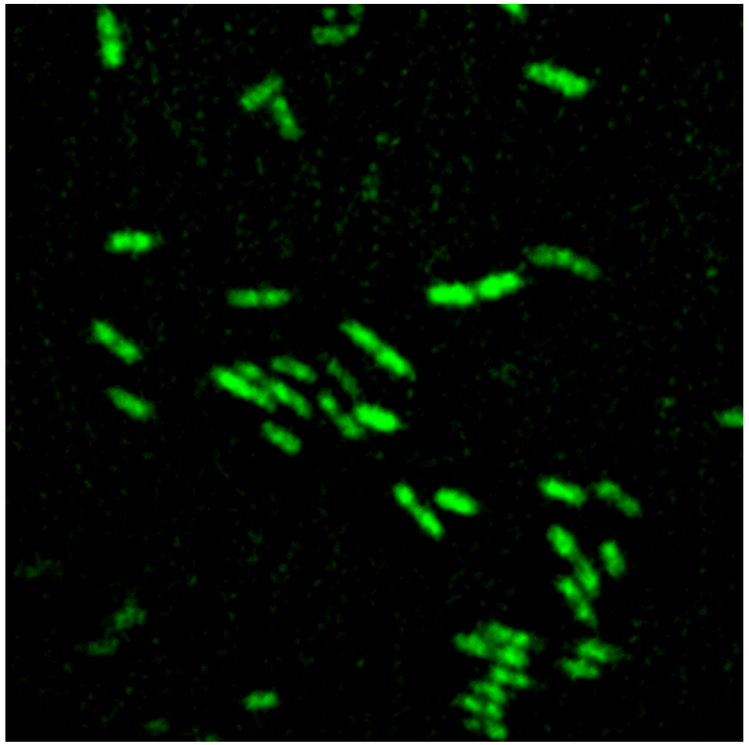
- **Small size vs resolution**
- **Crowdedness (high copy number)**



Dorman, Nat. Rev. Microbiology (2004); **Sadoon & Wang, PRE (2018); Sadoon et al., AEM (2020); Sadoon et al., PRL (2022)**

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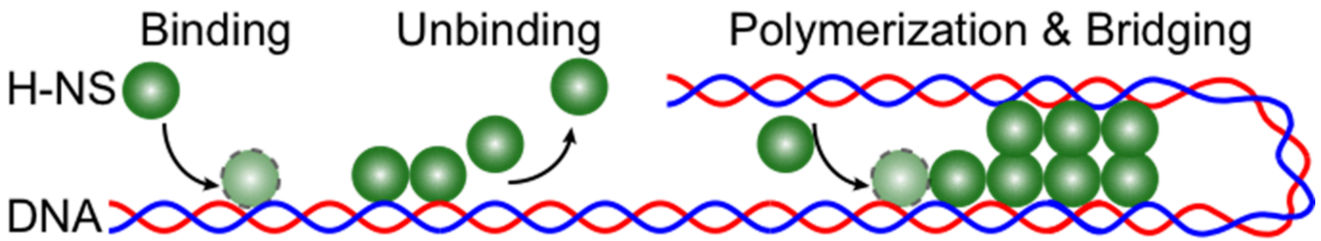

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A better tool is needed!

- **Small size vs resolution**
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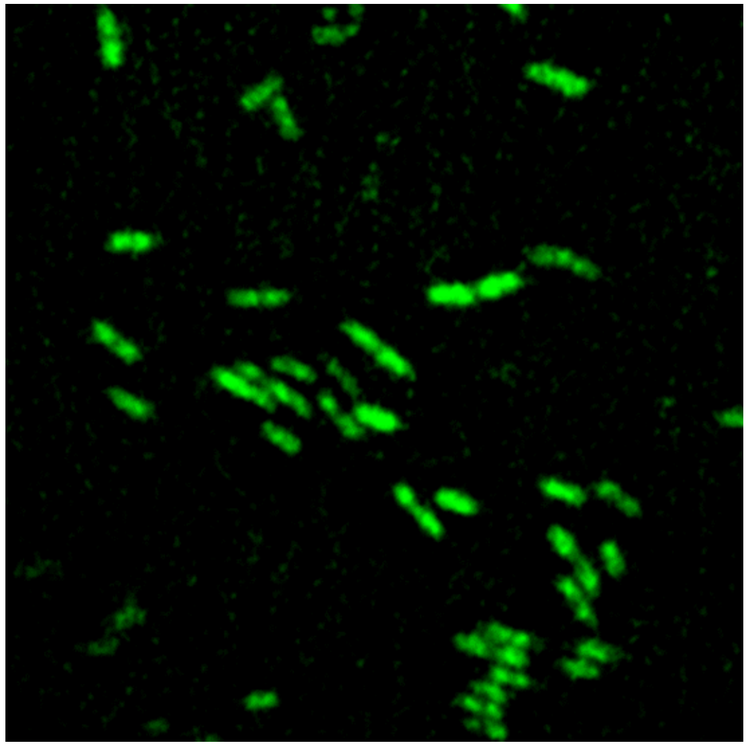

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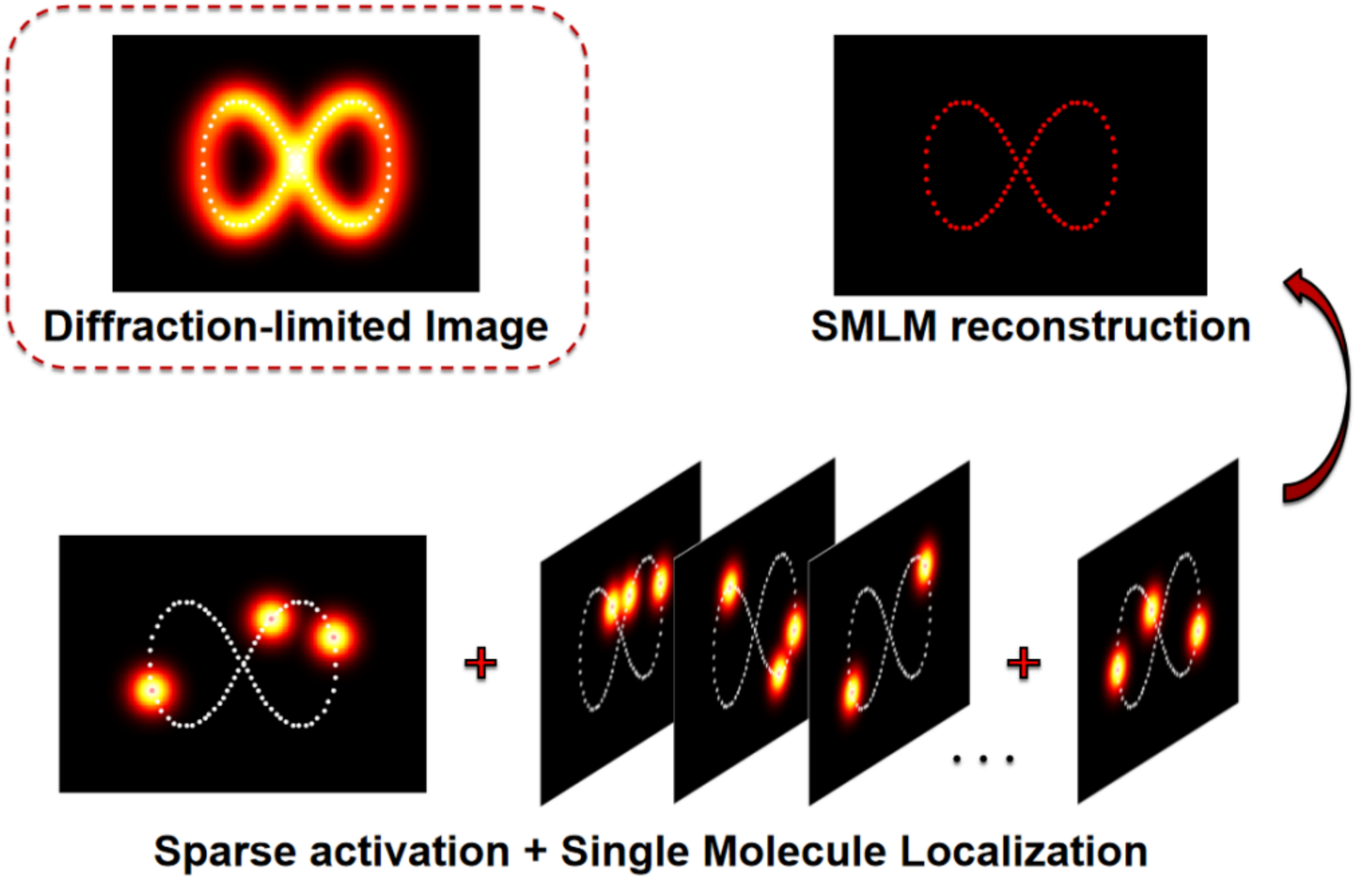
Super-Resolution Fluorescence Microscopy (SRFM)

Single Molecule Localization Microscopy (SMLM)
(Nobel Prize in 2014)



Lelek et al., Nat. Rev (2021); Wu, ECMI (2025)

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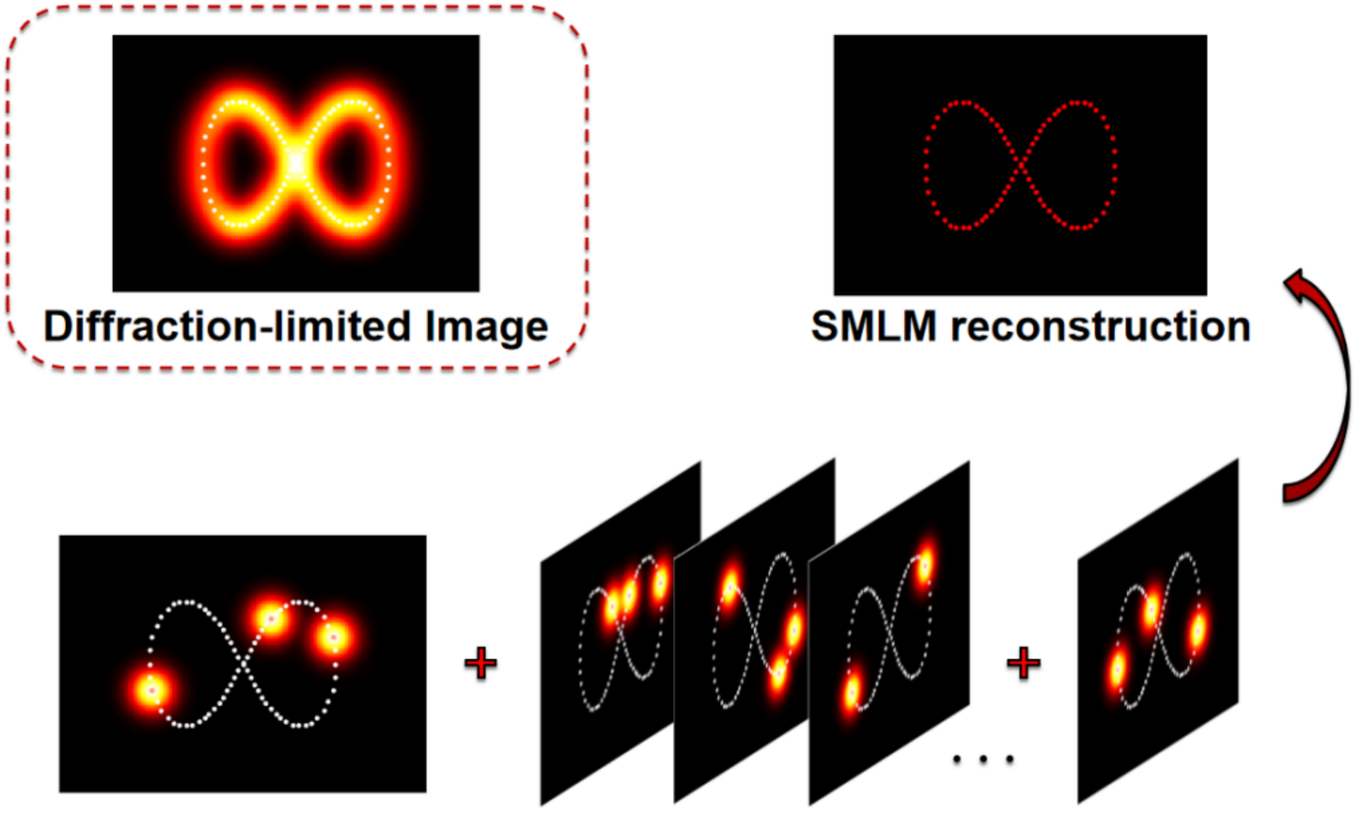

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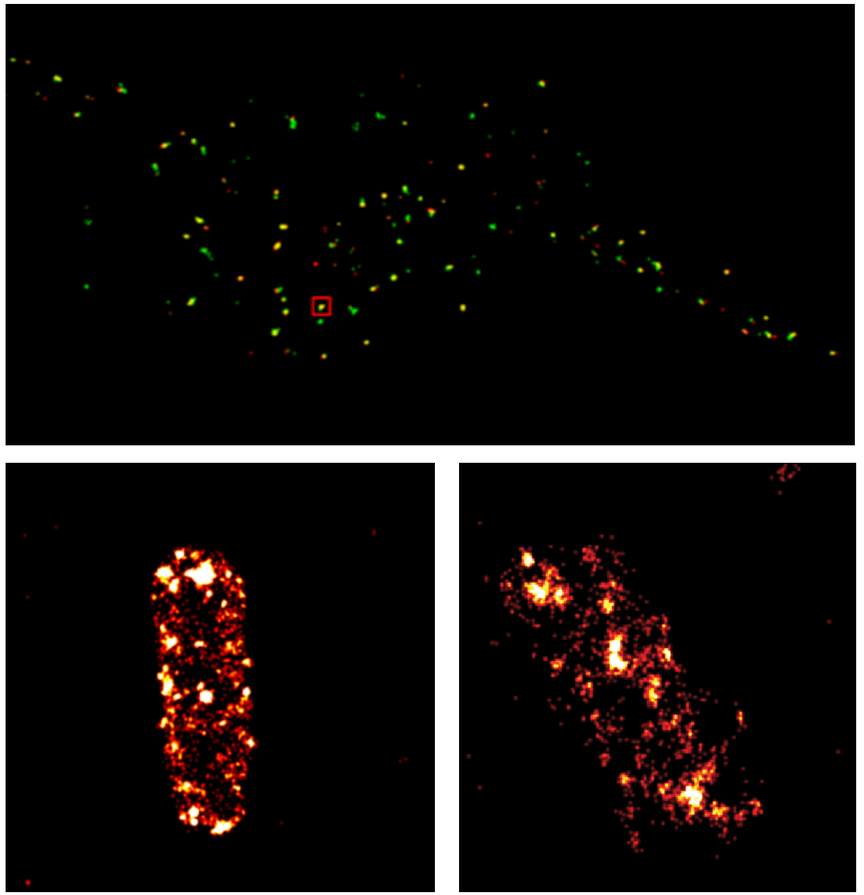

Wang et al., Nano Lett (2013)
Wang et al., Biophys J (2016)
Alqahtany et al., Nanotechnology (2019)

SMLM + Single-Particle Tracking (SPT)

Sadoon & Wang, PRE (2018)
Sadoon et al., AEM (2020)
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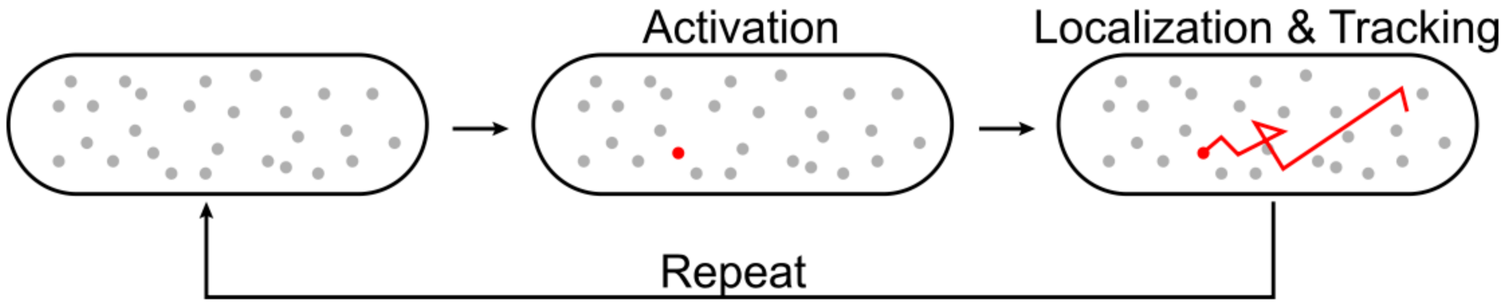

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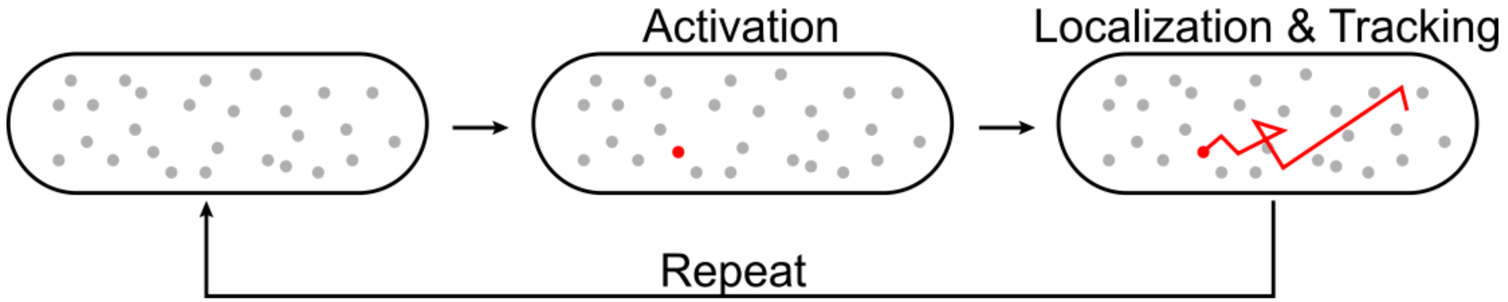

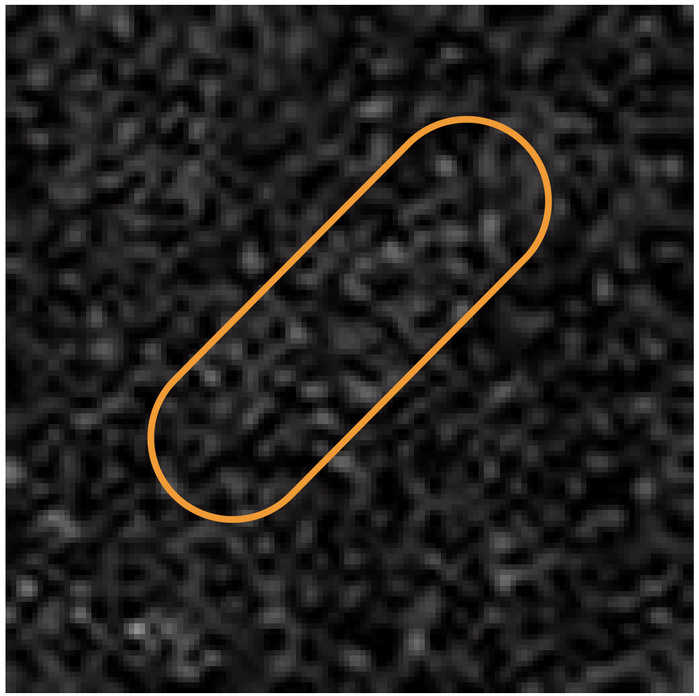

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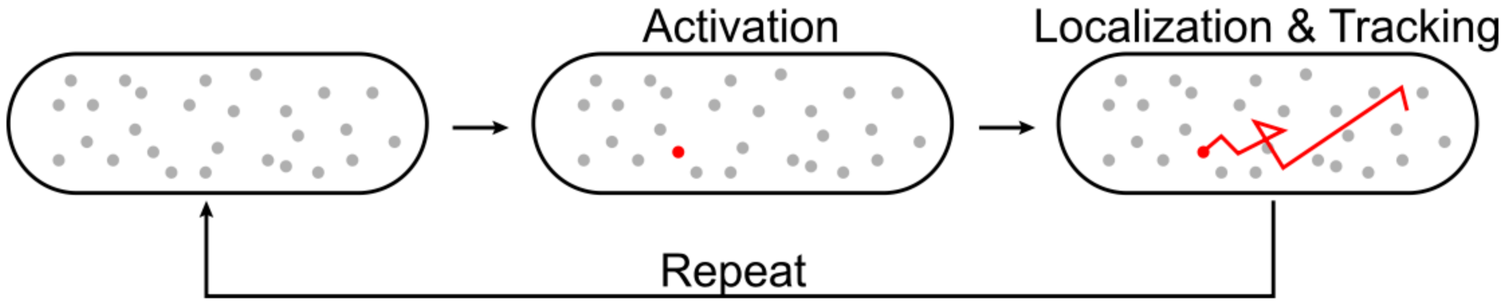

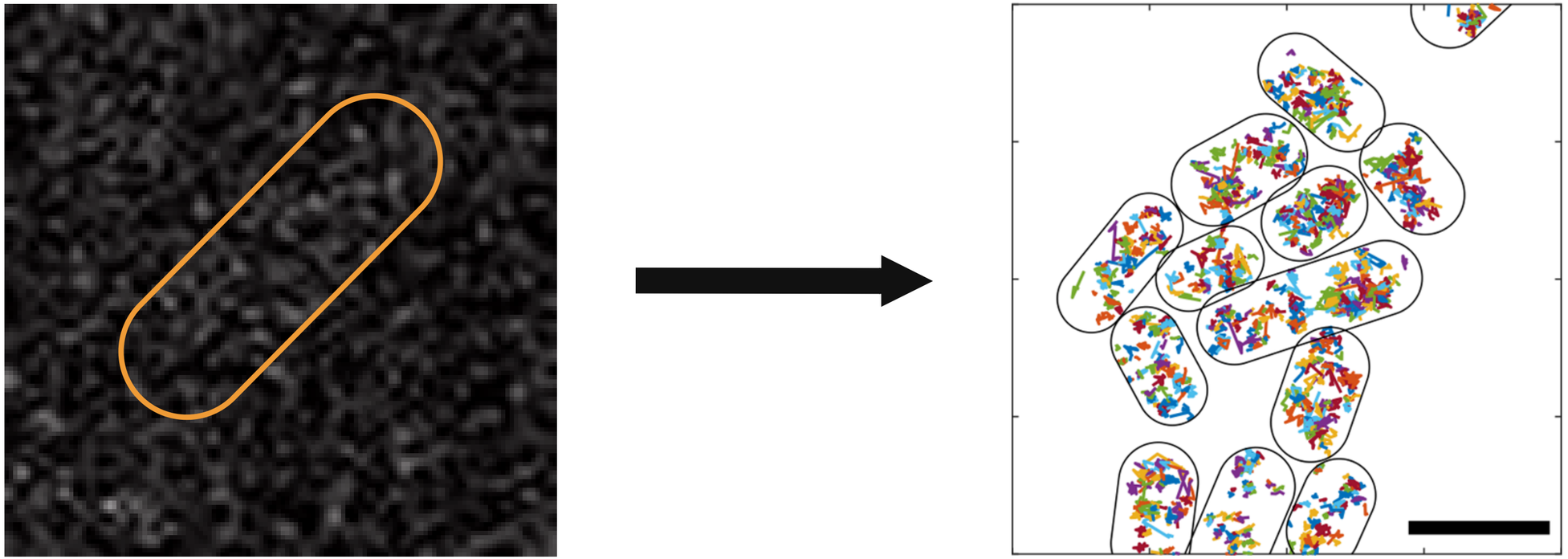

Sub-Diffusion of HNS

Mean-square-displacement (MSD):

$$\mathrm{MSD}(\tau) = \langle (\vec{r}(t+\tau) - \vec{r}(t))^2 \rangle = 4D\tau^{\alpha}$$

- $\alpha = 1$: normal diffusion
- $\alpha < 1$: sub-diffusion
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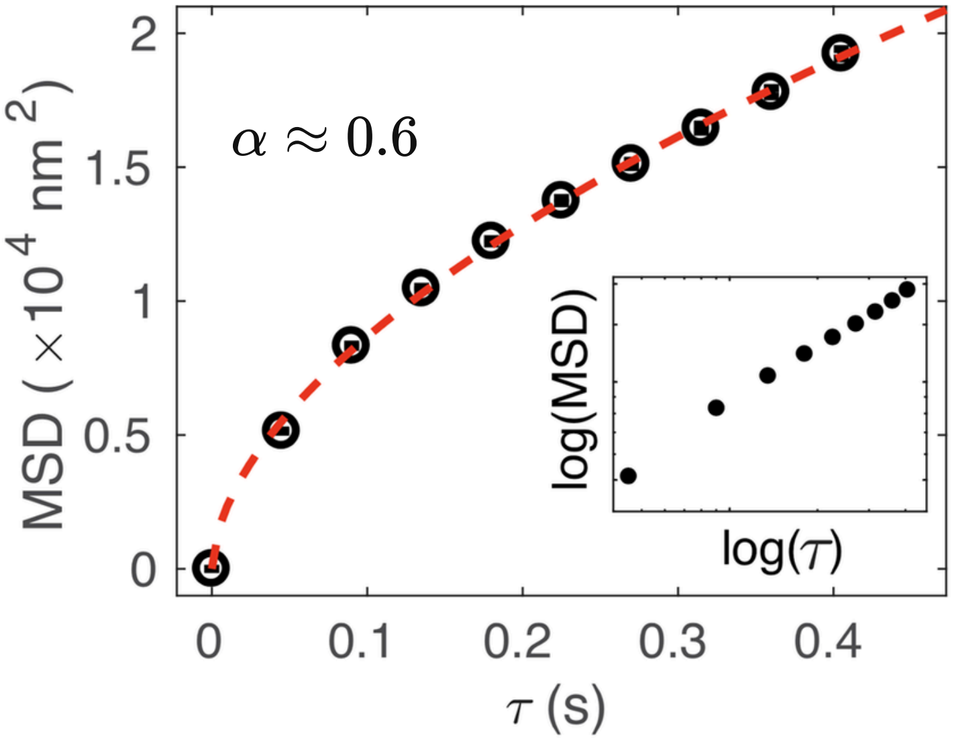

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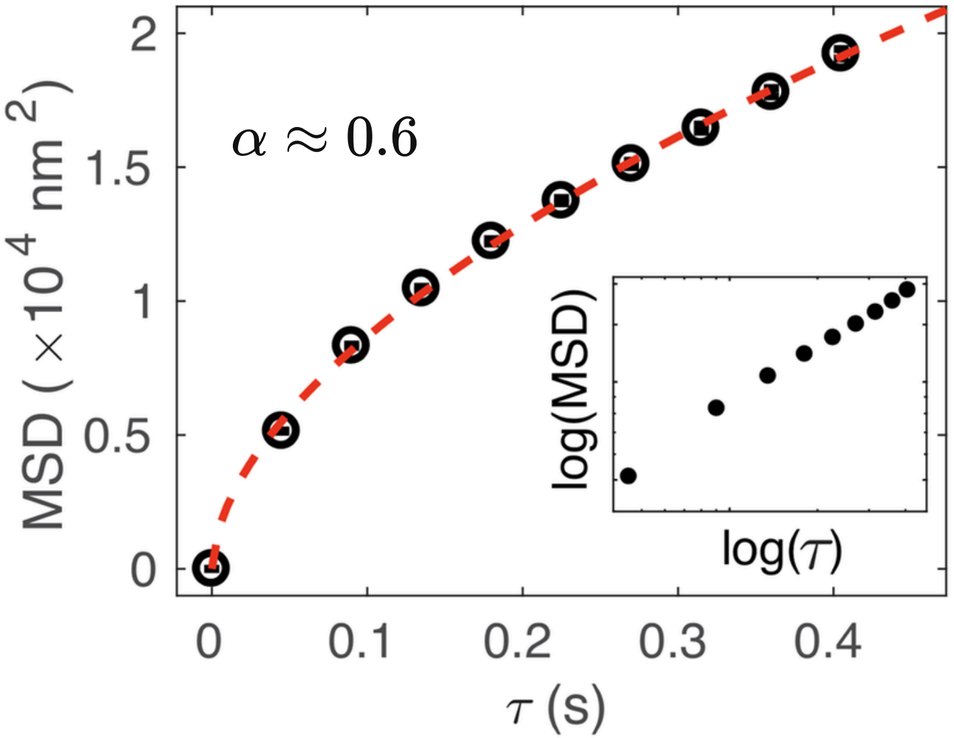

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Motion of HNS proteins is confined inside bacteria!

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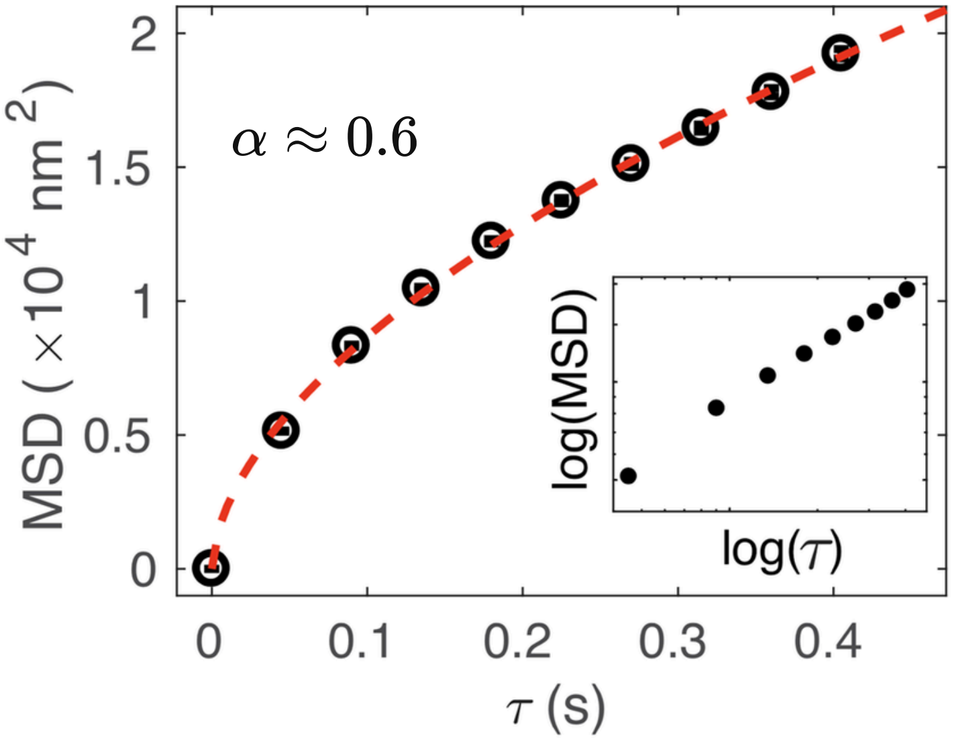

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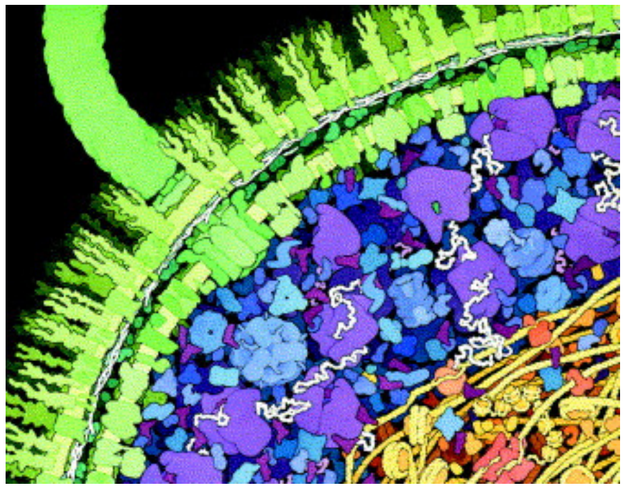
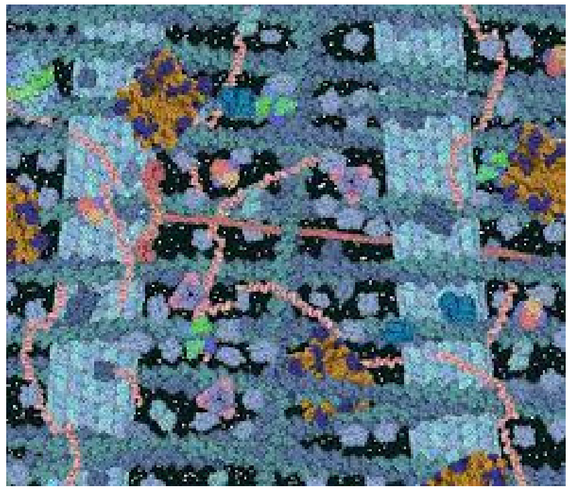
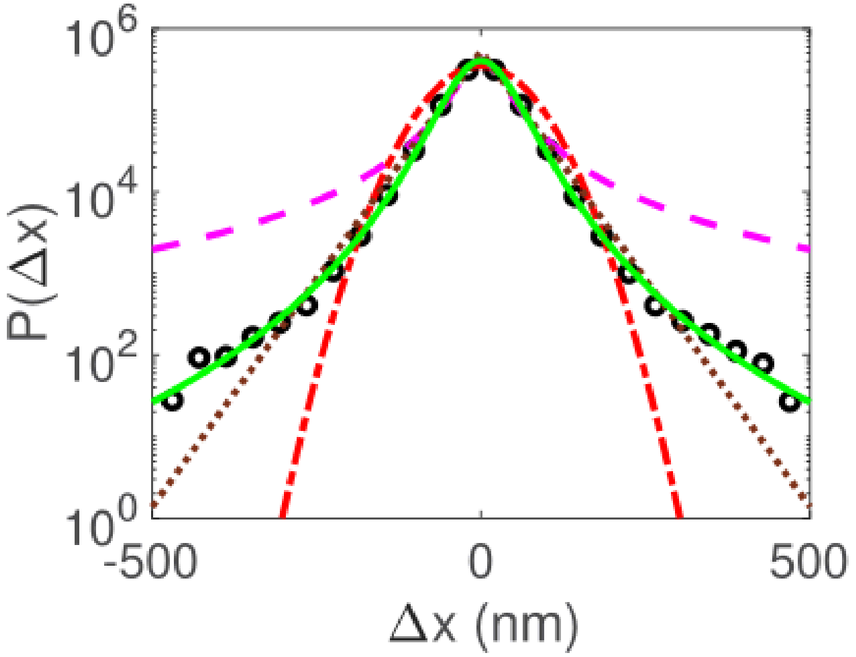
The Inside of Bacteria is Crowded!





Ellis, Trends in Biochemical Sciences (2001)

Step Size Distributions

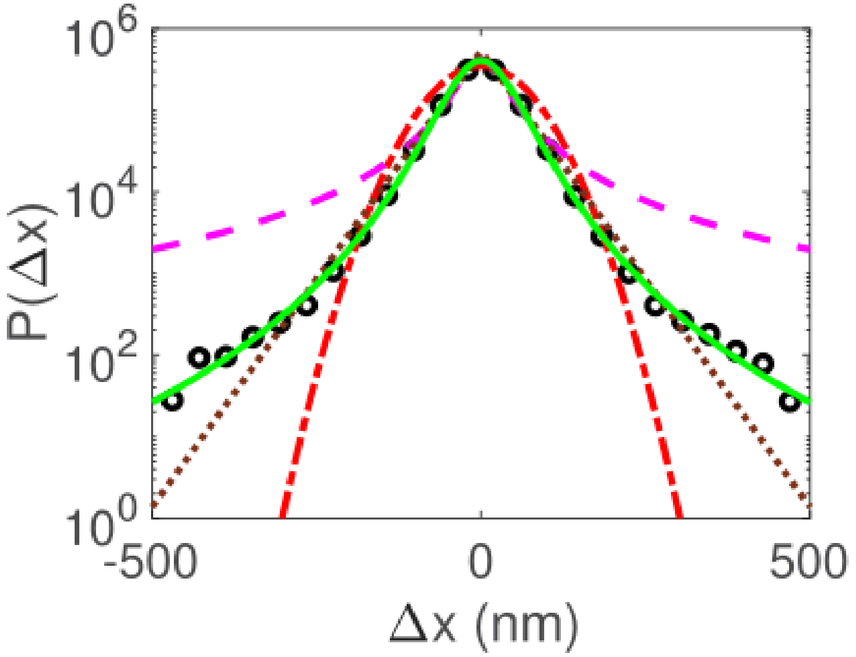


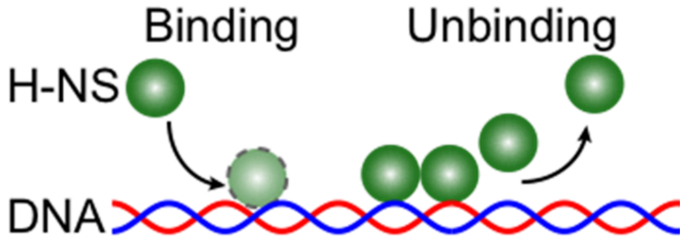
$$P(\Delta x) \propto \left(1 + \frac{\Delta x^2}{w^2}\right)^{-m}$$

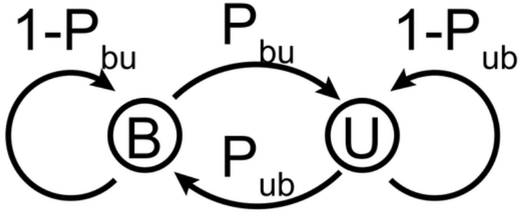
Pearson Type VII Distribution
instead of Guassian, Laplace or Cauchy

Sadoon & Wang, PRE (2018)

Step Size Distributions





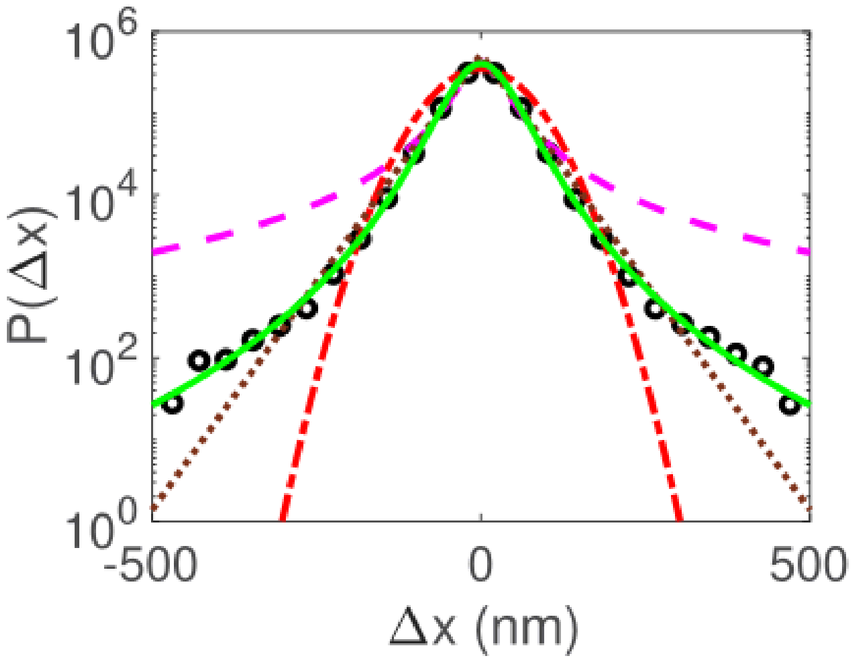


$$P(\Delta x) \propto \left(1 + \frac{\Delta x^2}{w^2}\right)^{-m}$$

Pearson Type VII Distribution
instead of Guassian, Laplace or Cauchy

Sadoon & Wang, PRE (2018)

Step Size Distributions



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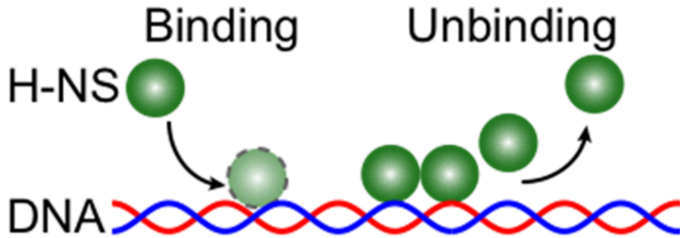
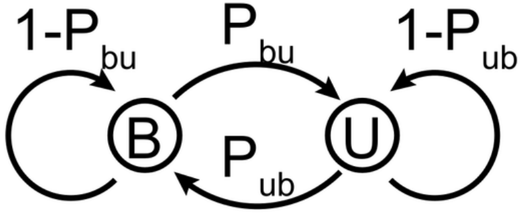
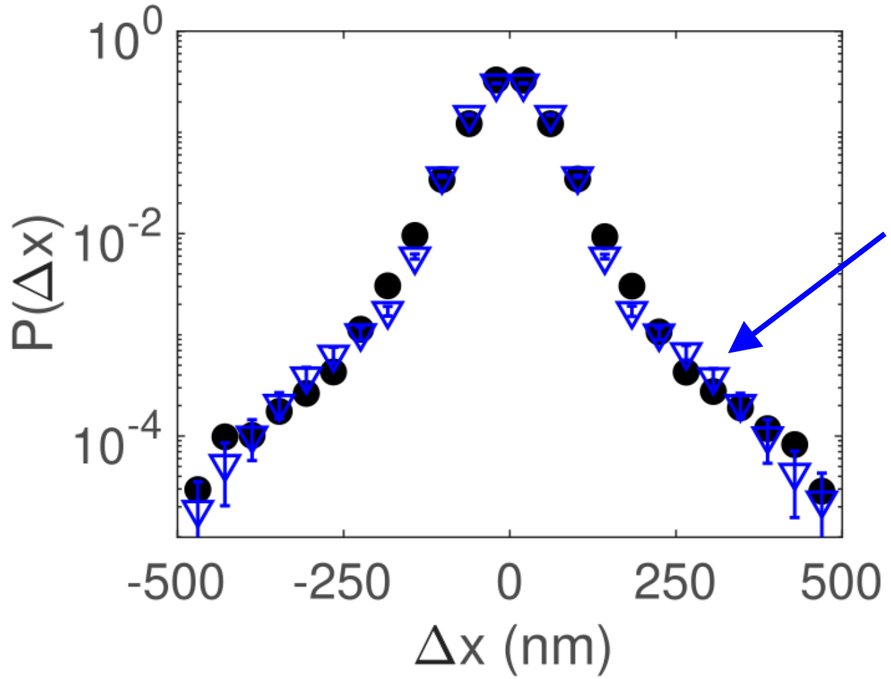


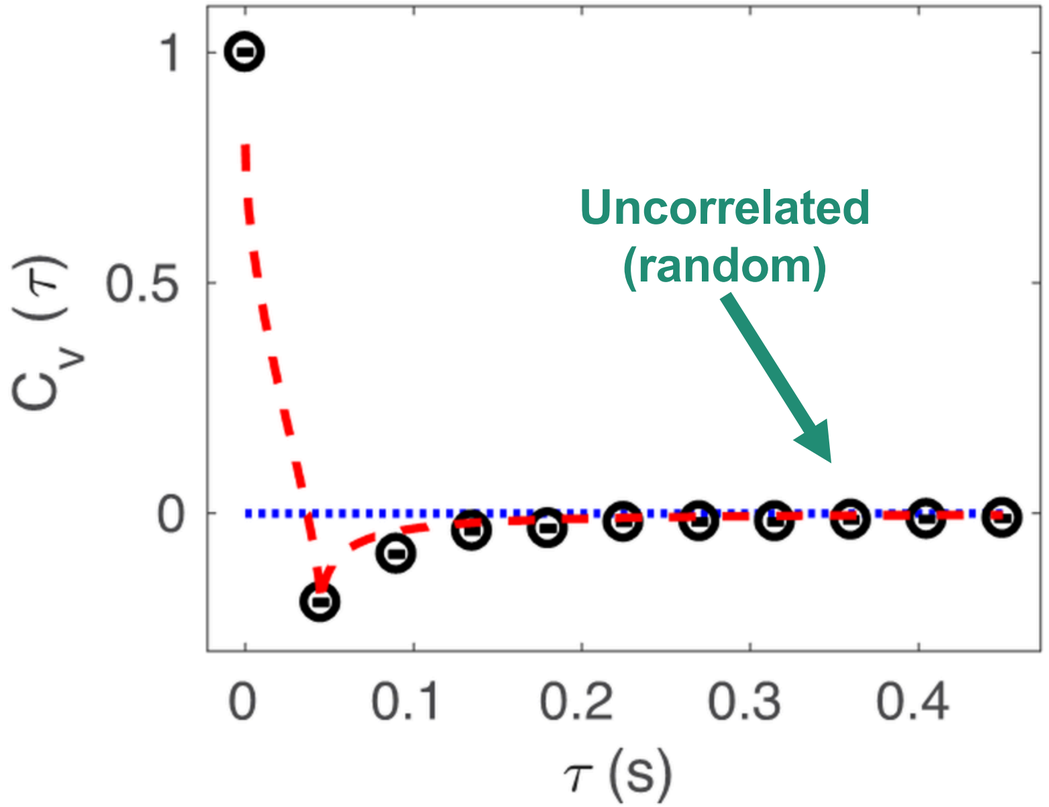




Sadoon & Wang, PRE (2018)

Viscoelasticity of Bacterial Cytoplasm

Autocorrelation of HNS speed: $C_v(\tau) = \langle v(t+\tau)v(t)\rangle$



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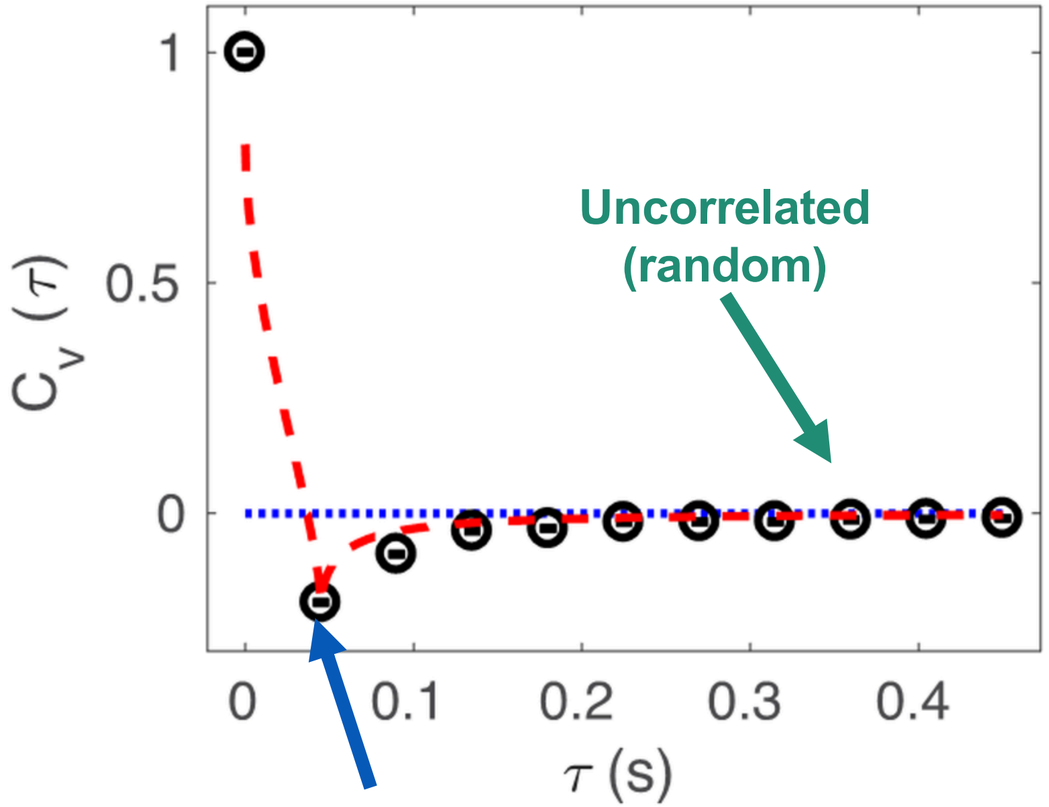

Negative correlation ⇒ Push-back ⇒ Elasticity

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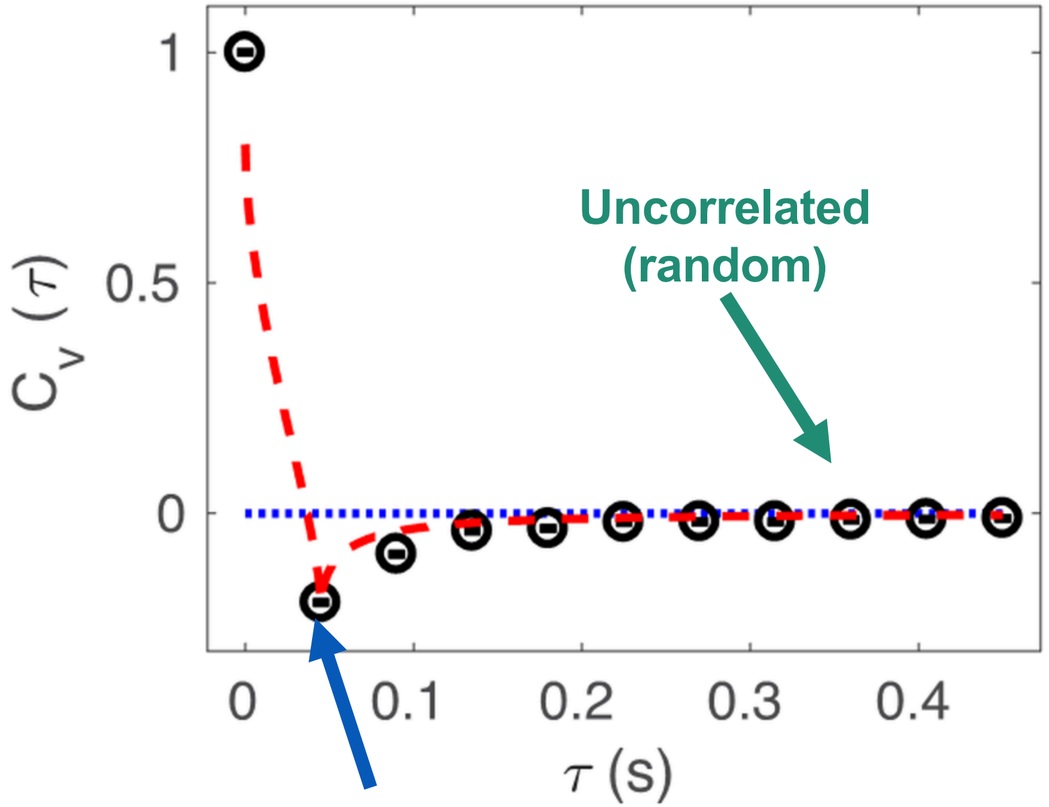

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- Real part: storage modulus ⇔ elasticity
- Imaginary part: loss modulus ⇔ viscosity

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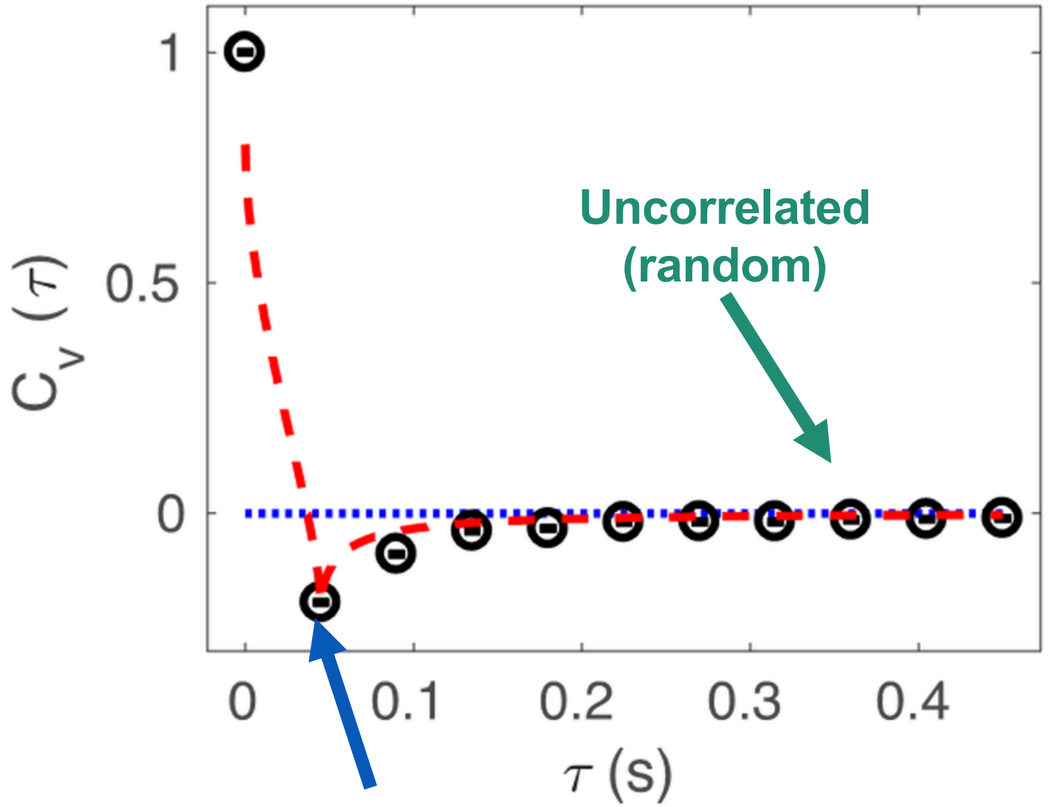

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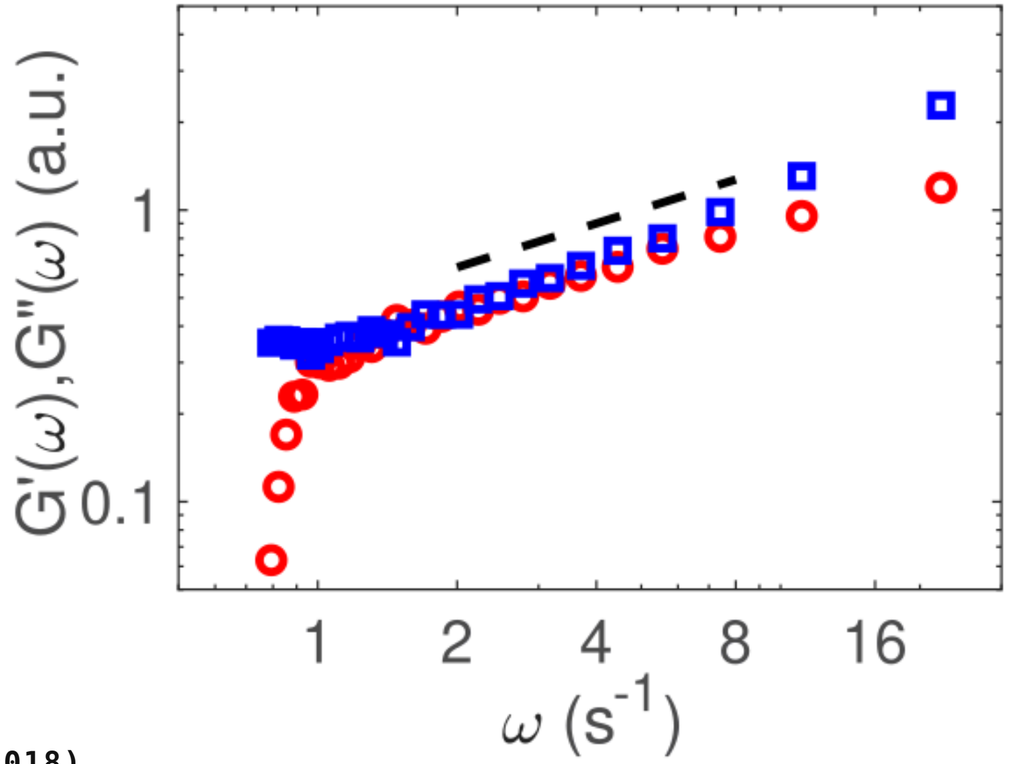

Sadoon & Wang, PRE (2018)

Effect of Silver on Diffusion of HNS Proteins

Image from Silver Coins

Image from VCHRI

Sadoon et al, AEM (2020)

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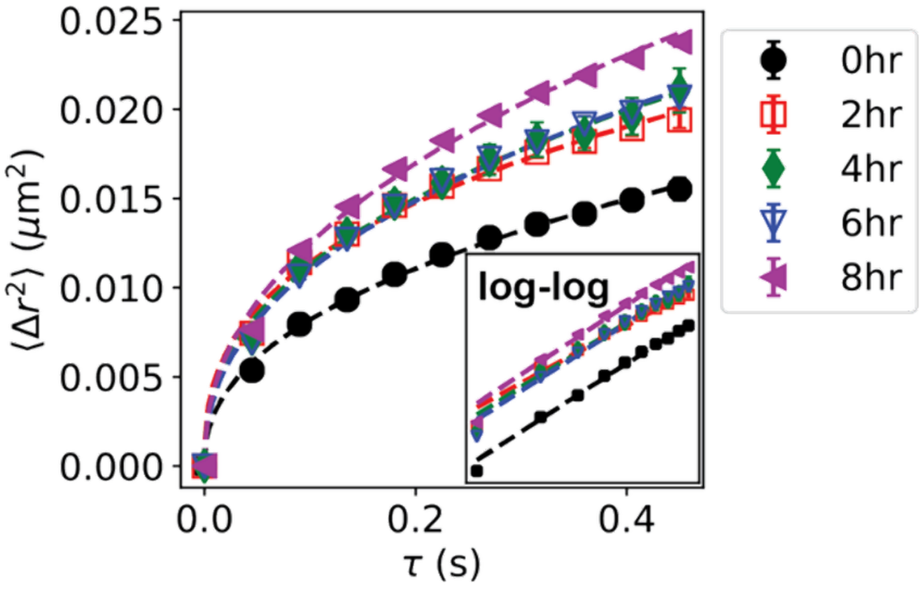

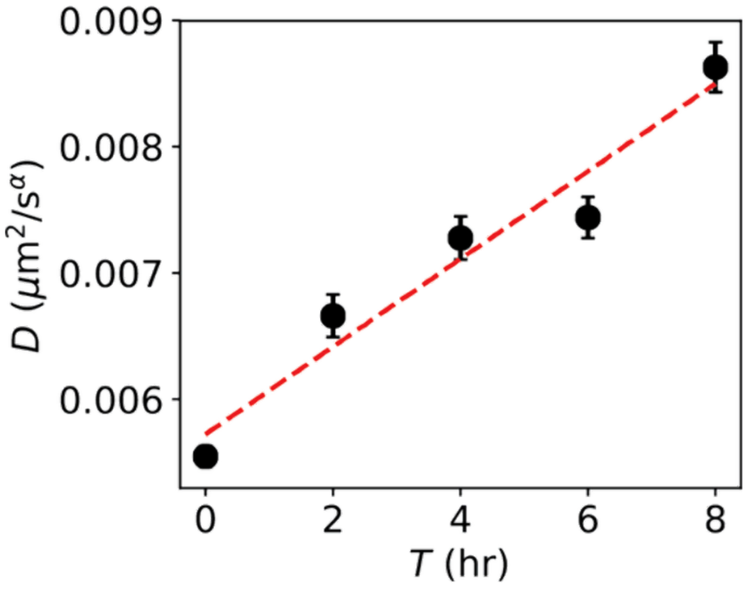

Silver causes faster diffusion of HNS

Sadoon et al, AEM (2020)

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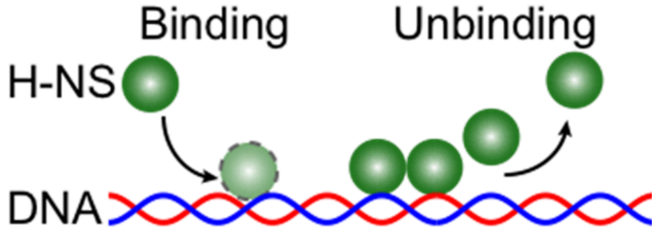

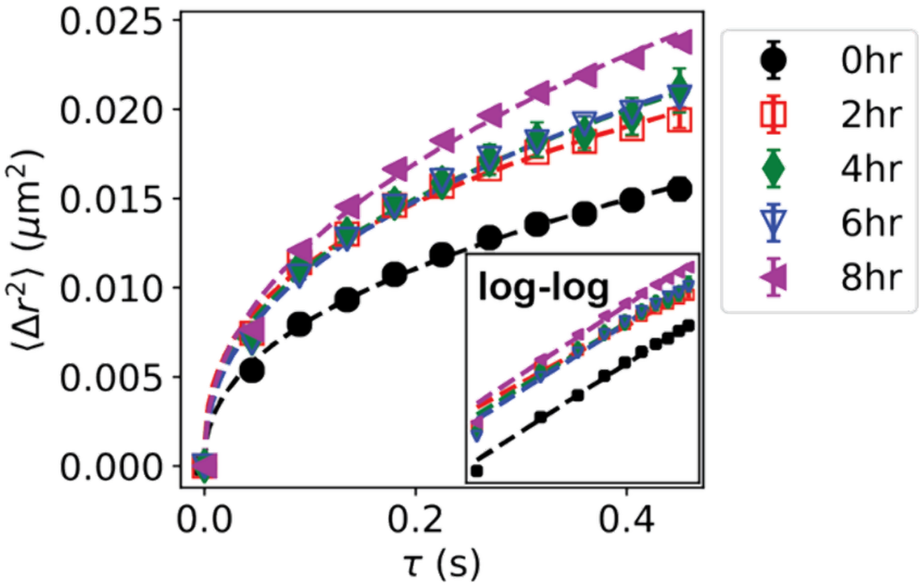

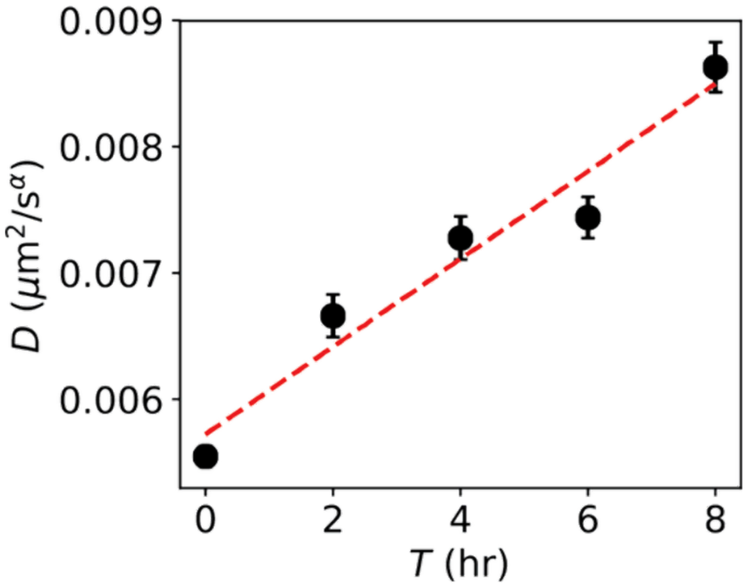

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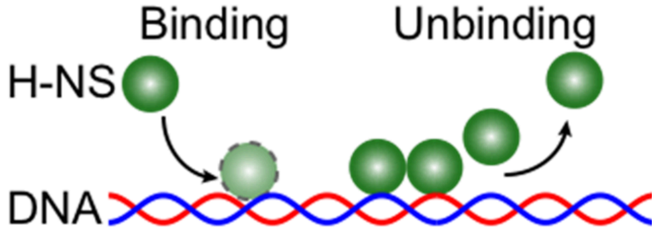
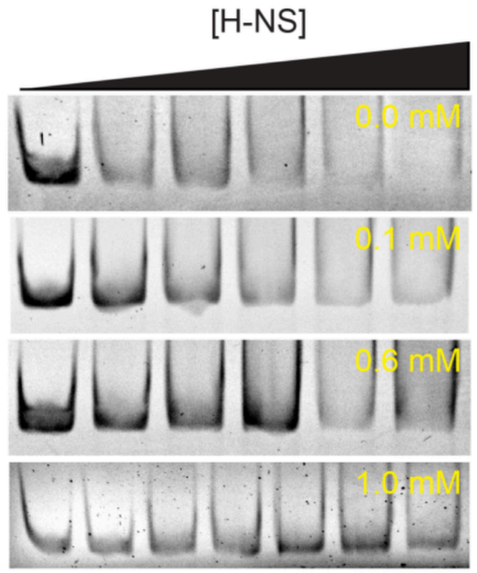
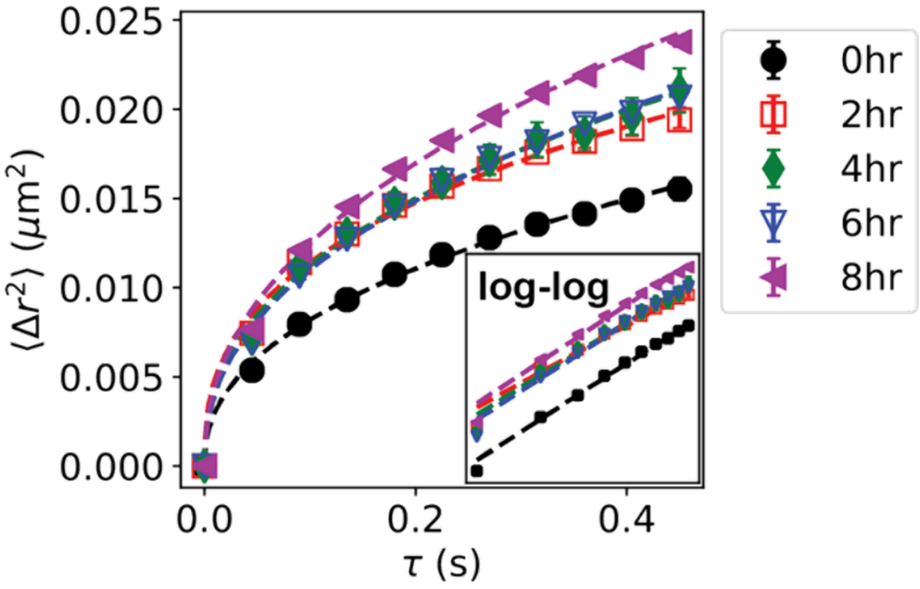
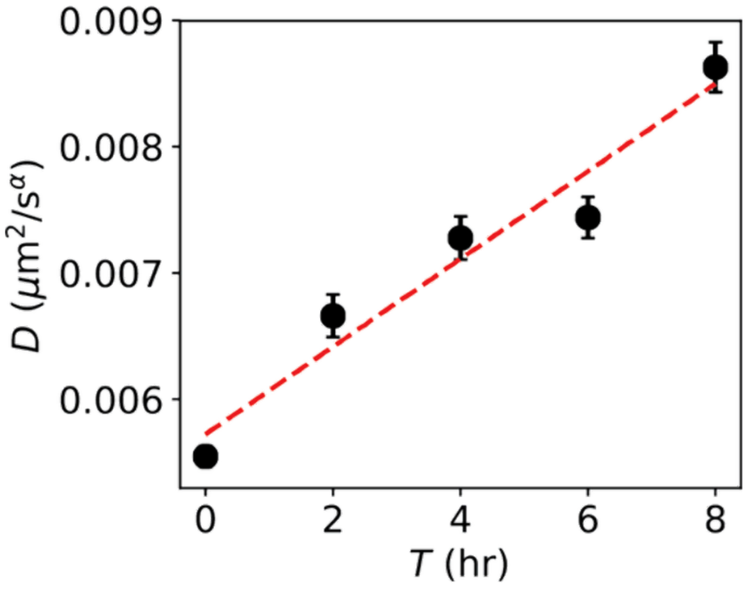
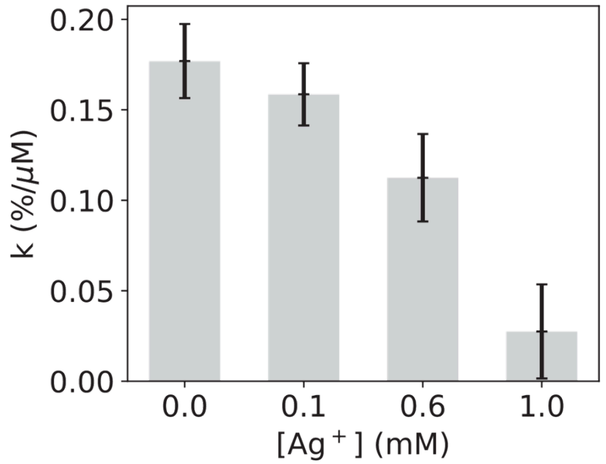










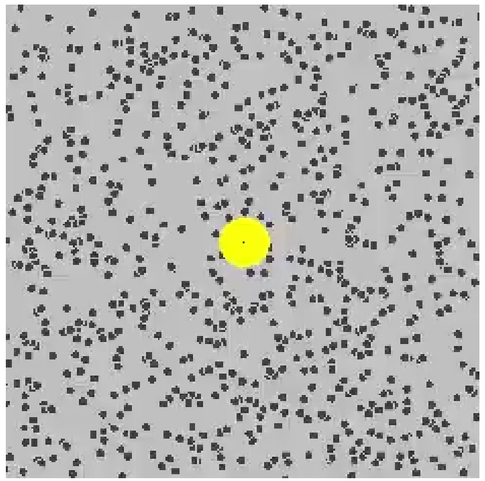
Silver causes faster diffusion of HNS

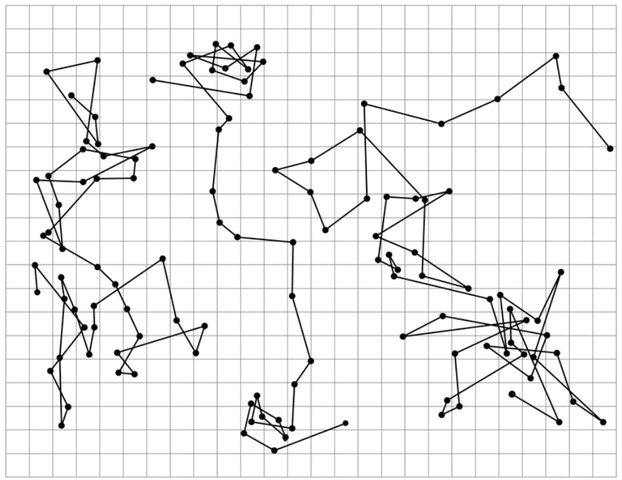
Silver weakens binding between HNS and DNA!

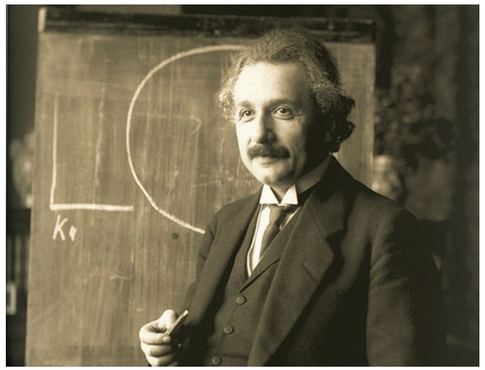
Sadoon et al, AEM (2020)

Temperature Dependence of Diffusion of HNS

Diffusion and Brownian Motion in Simple Fluid







Images from [Wikipedia]

The Einstein Relation

$$D = \mu k_B T = \frac{k_B T}{6\pi\eta r}$$

$$D \propto T \;\; !!$$

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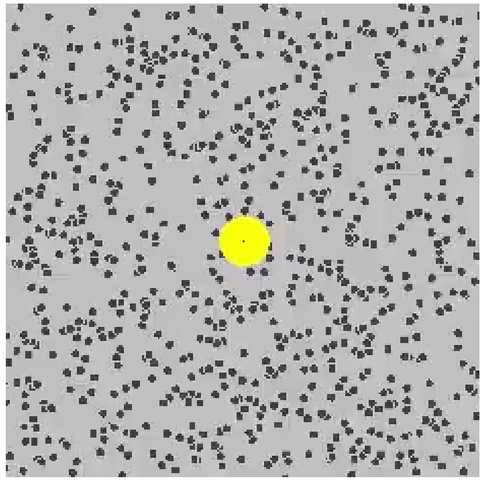

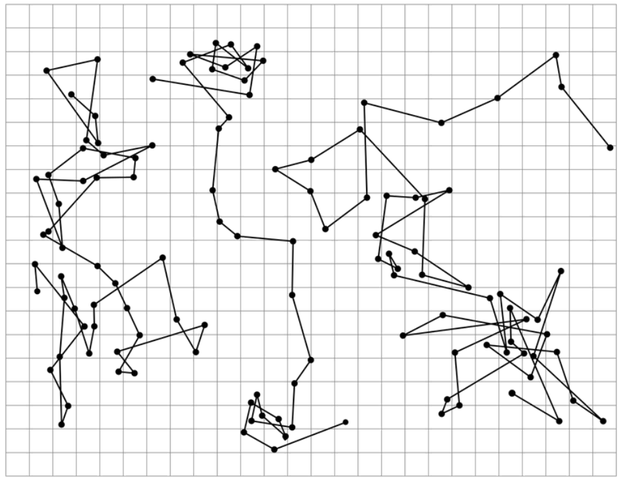

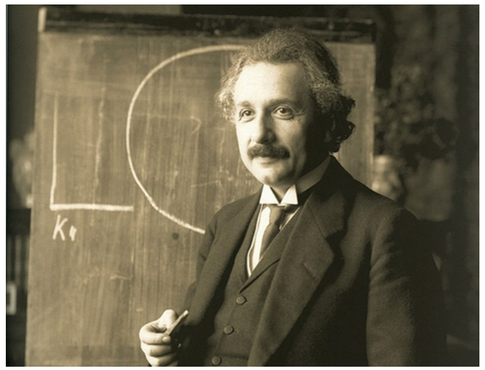

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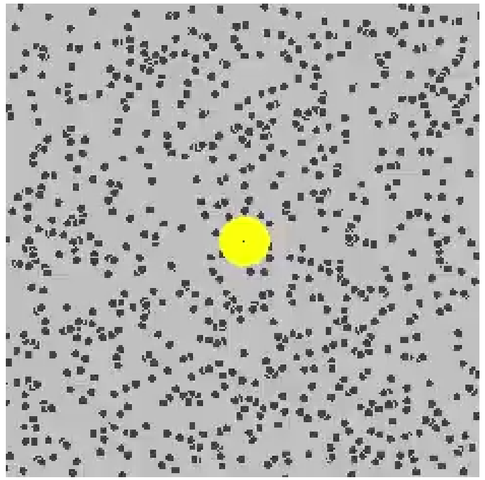

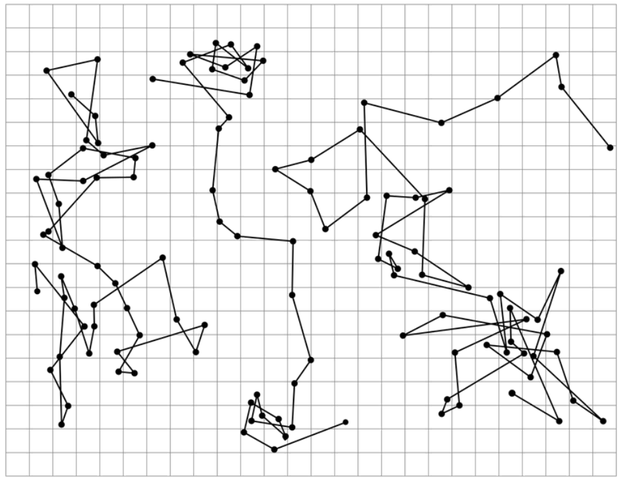

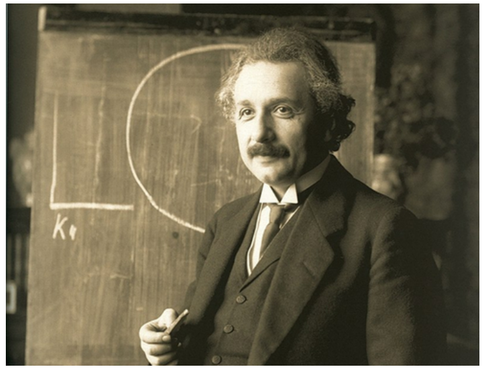

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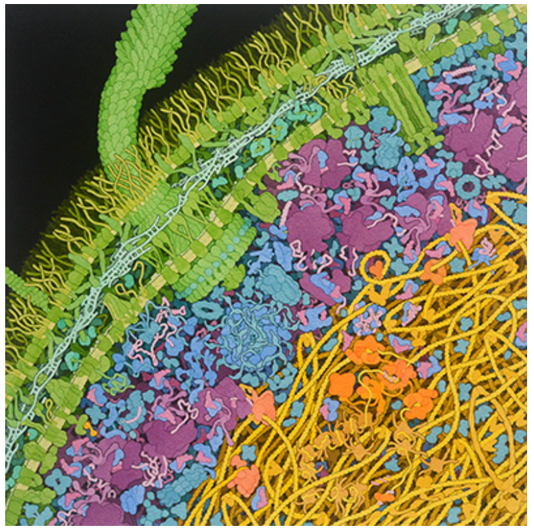


Goodsell, Nat Struct Mol Biol (2021)

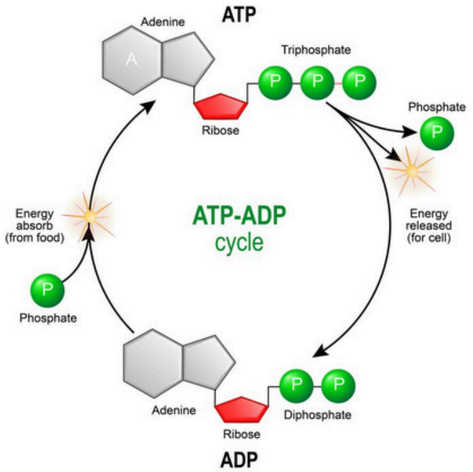


Image from BiologyOnline

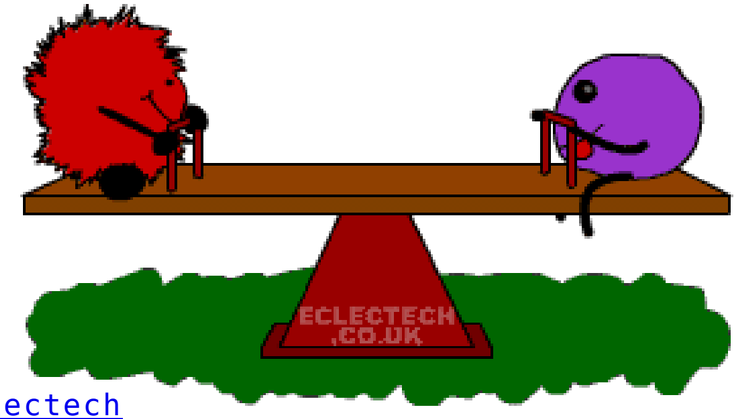
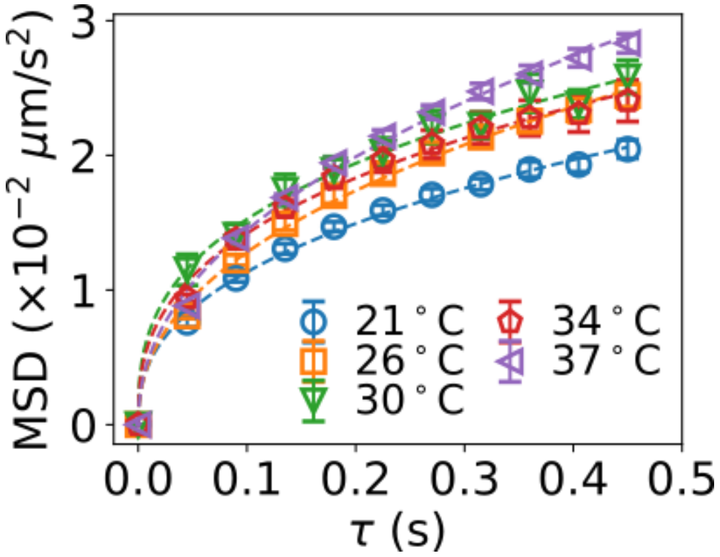
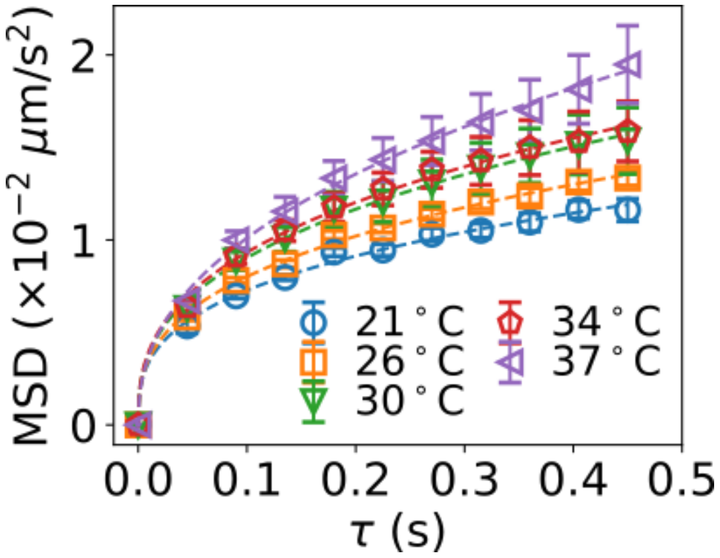
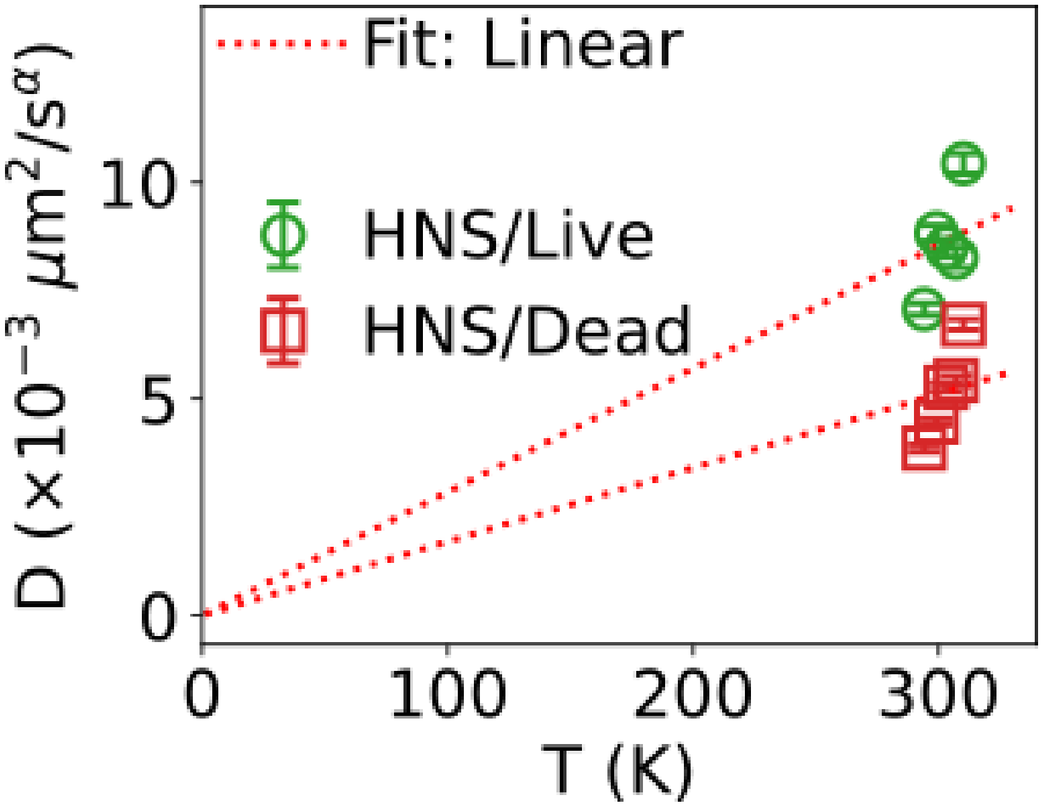


Image from Eclectech

Diffusion of HNS at Different Temperatures







Sadoon et al., PRL (2022)

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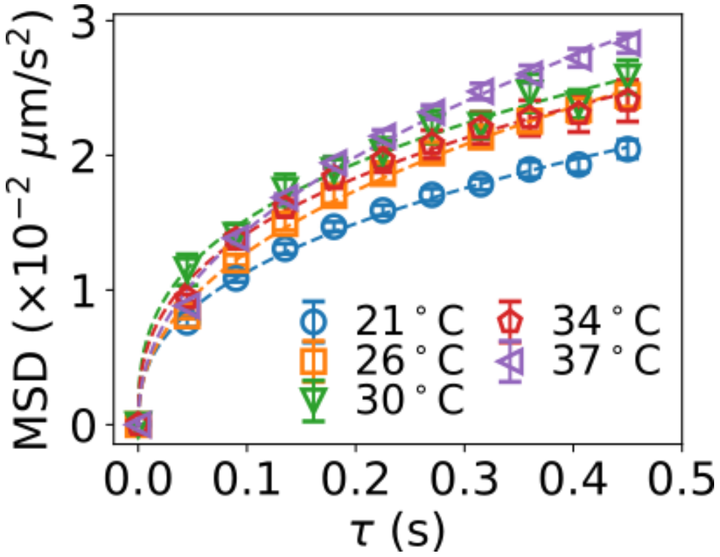

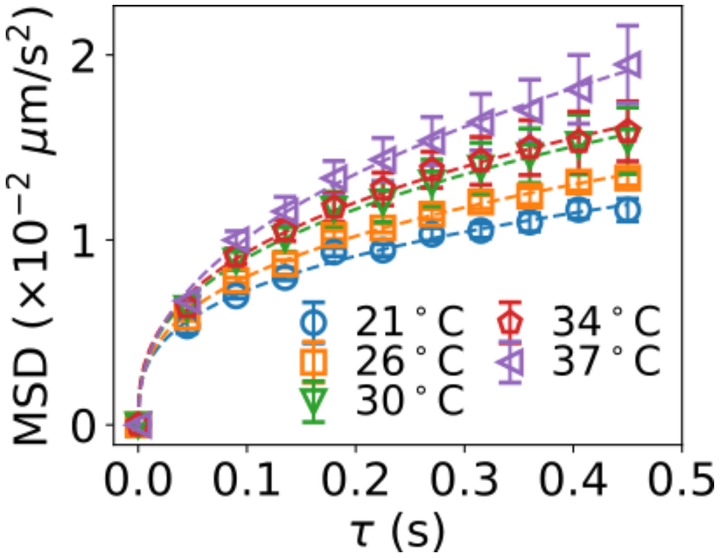

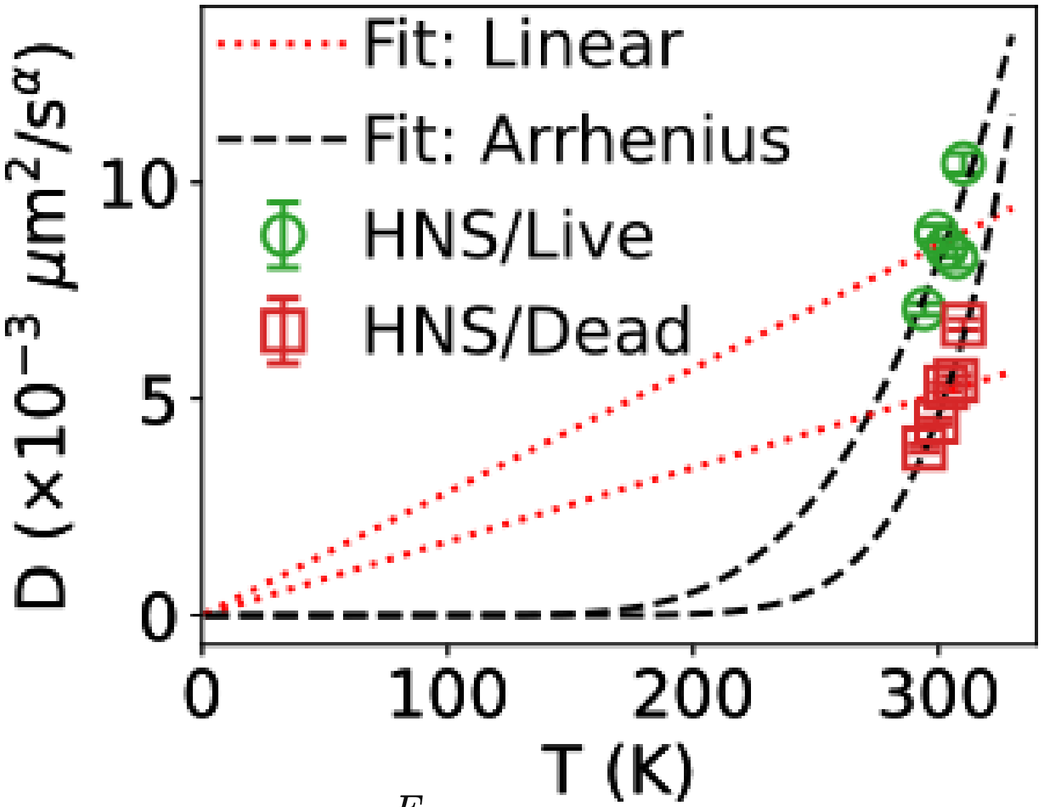

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Sadoon et al., PRL (2022)

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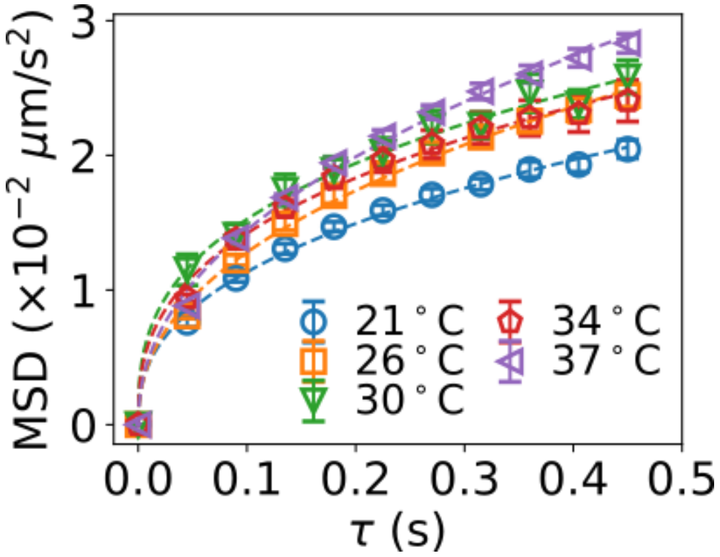

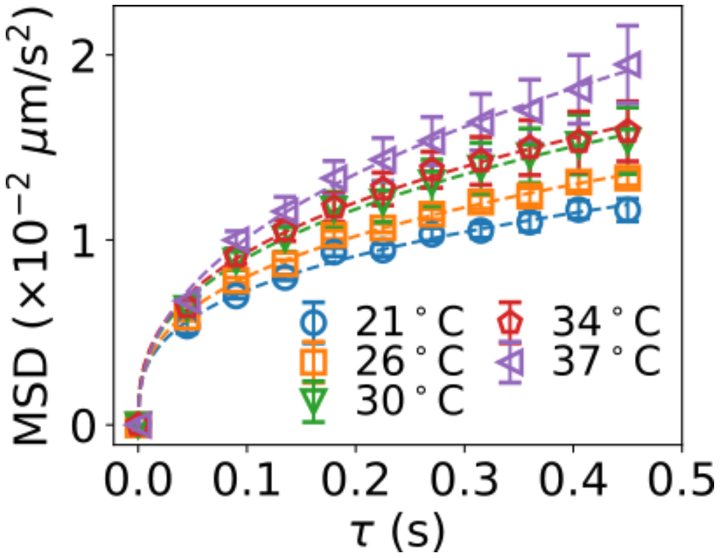

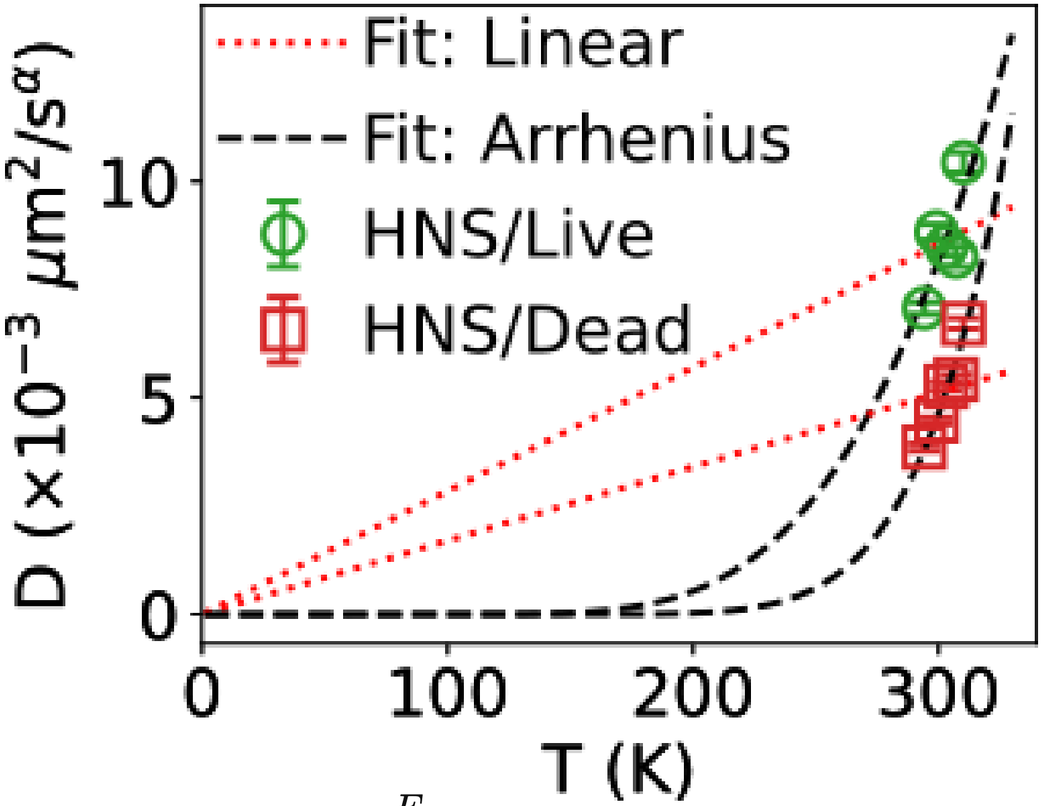

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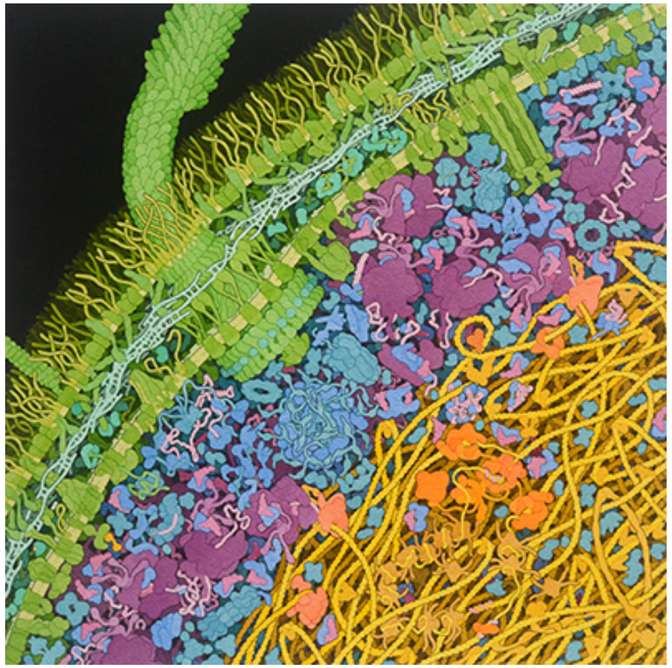
Why Arrhenius equation?

What is E_a?

Sadoon et al., PRL (2022)

Theory: Additional Viscosity

Interactions & Crowdedness



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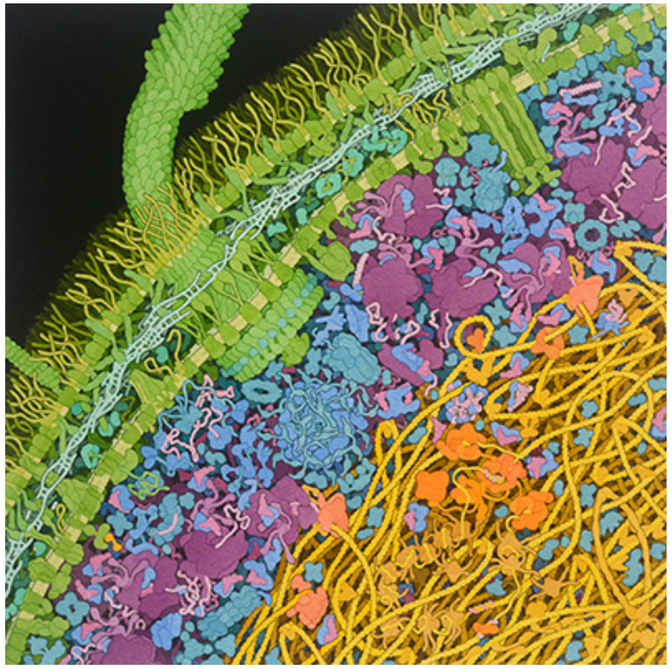

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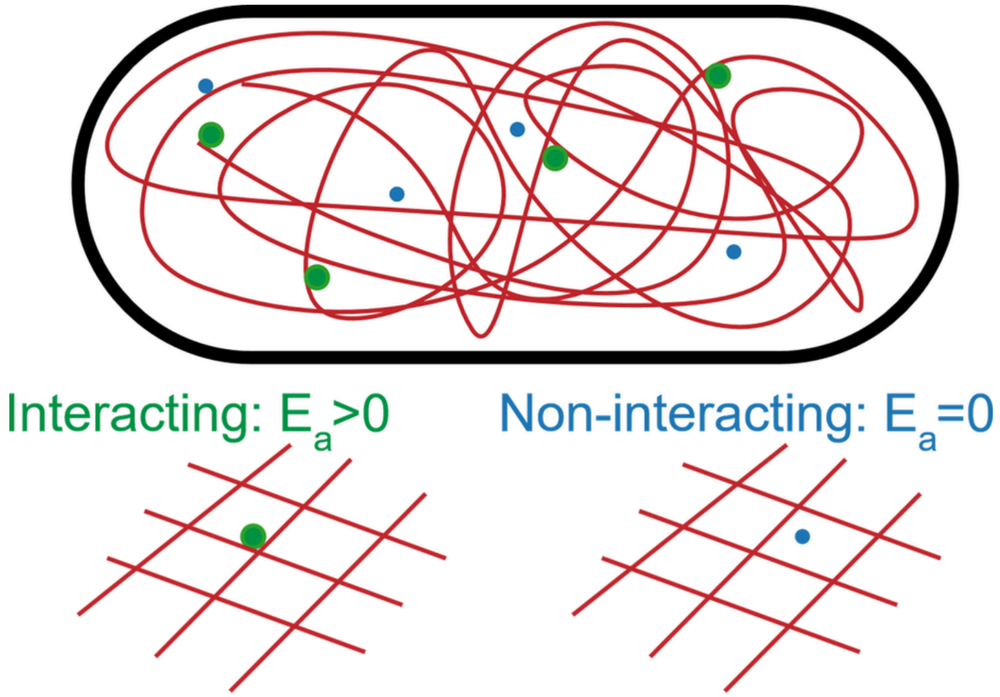

1) Interaction energy needs to be overcome

2) Target site needs to be empty

Sadoon et al., PRL (2022)

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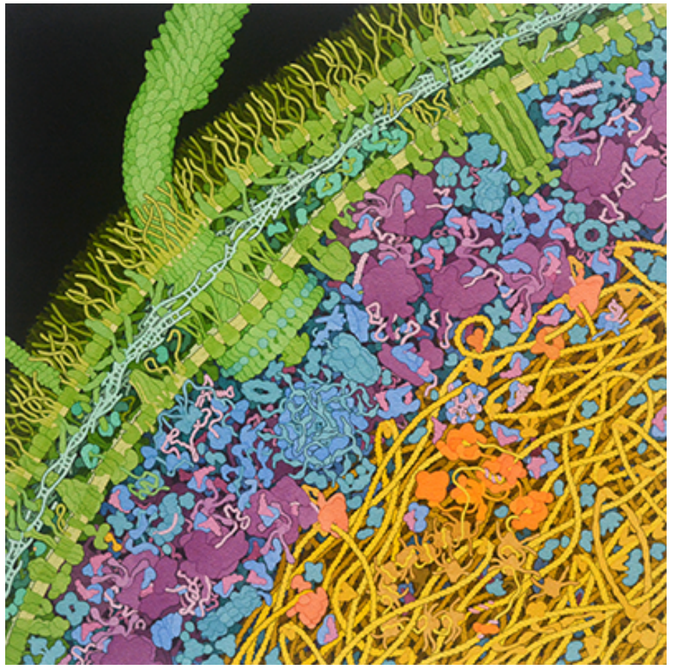

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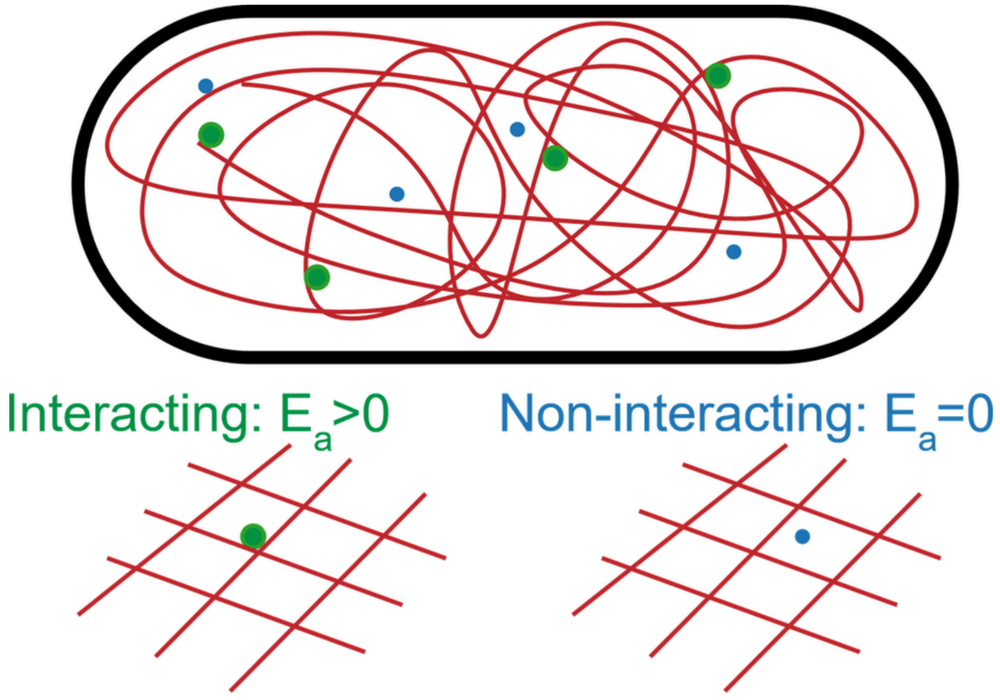

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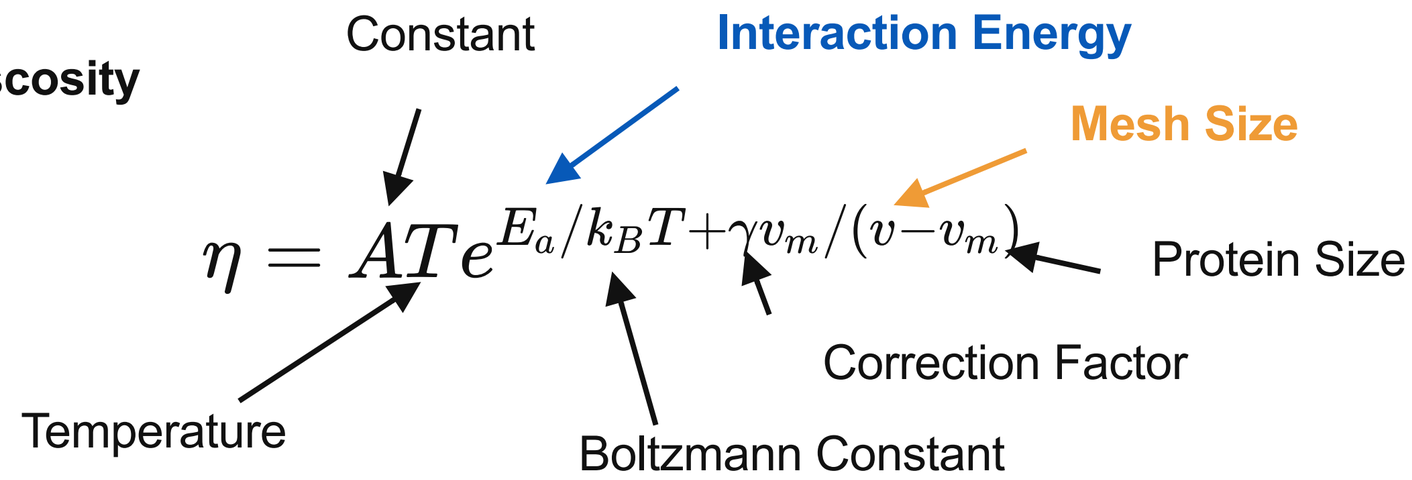
Sadoon et al., PRL (2022)

3) Mesh size is T dependent: $v = -\xi(T - T_0)$

Zhao et al., Polym J (2012)

Theory: Additional Viscosity

Viscosity



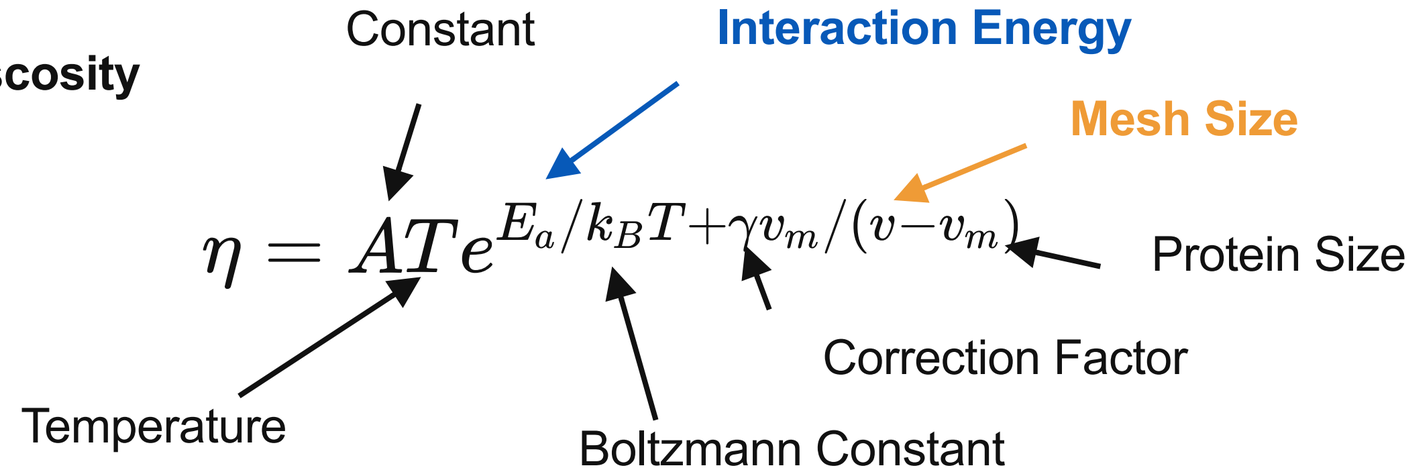
Sadoon et al., PRL (2022)

Diffusion Coefficient

$$D = \frac{k_BT}{6\pi\eta r} = D_0e^{-E_a/k_BT+T_m/(T-T_c)}$$

Theory: Additional Viscosity

Viscosity



Sadoon et al., PRL (2022)

Diffusion Coefficient

$$D = \frac{k_BT}{6\pi\eta r} = D_0 e^{-E_a/k_BT+T_m/(T-T_c)}$$

Theory: Additional Viscosity

Viscosity

$$\eta = ATe^{E_a/k_BT+\gamma v_m/(v-v_m)}$$

Constant

Interaction Energy

Mesh Size

Protein Size

Correction Factor

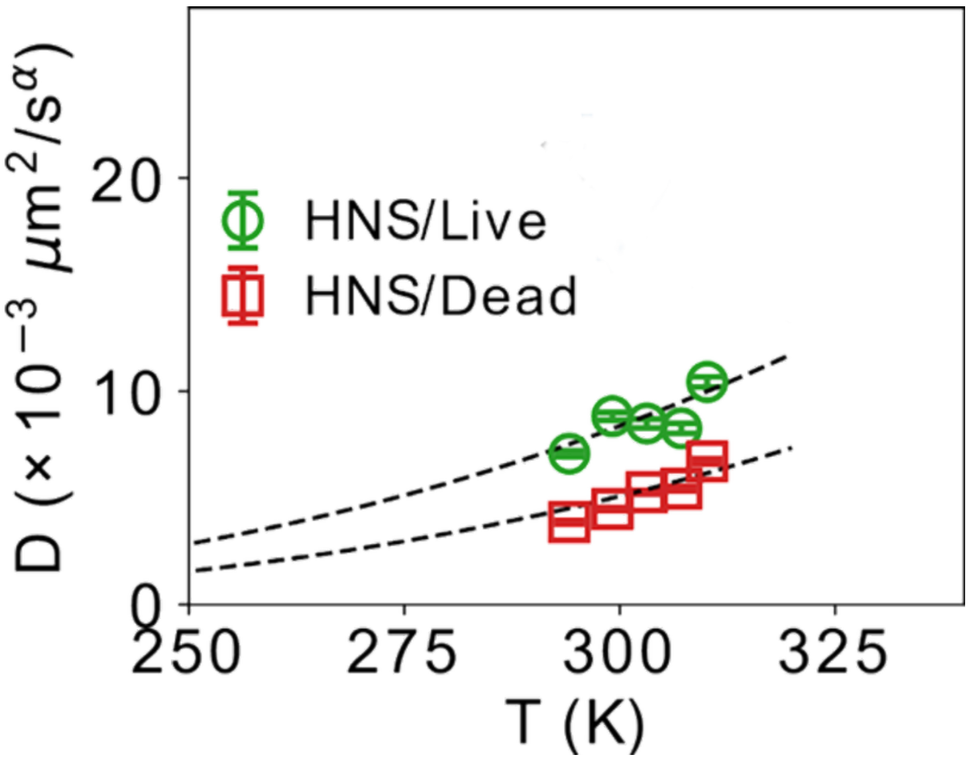
Temperature

Boltzmann Constant

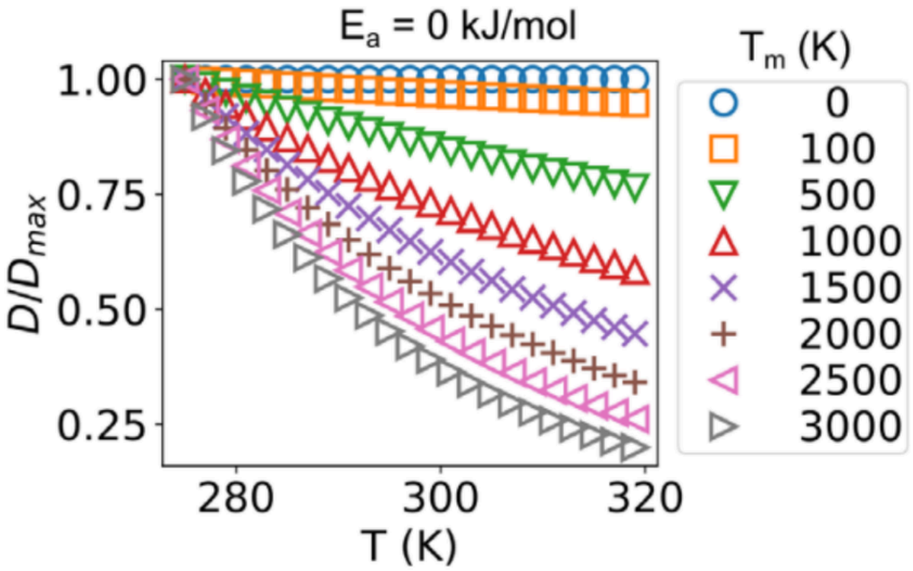
Sadoon et al., PRL (2022)

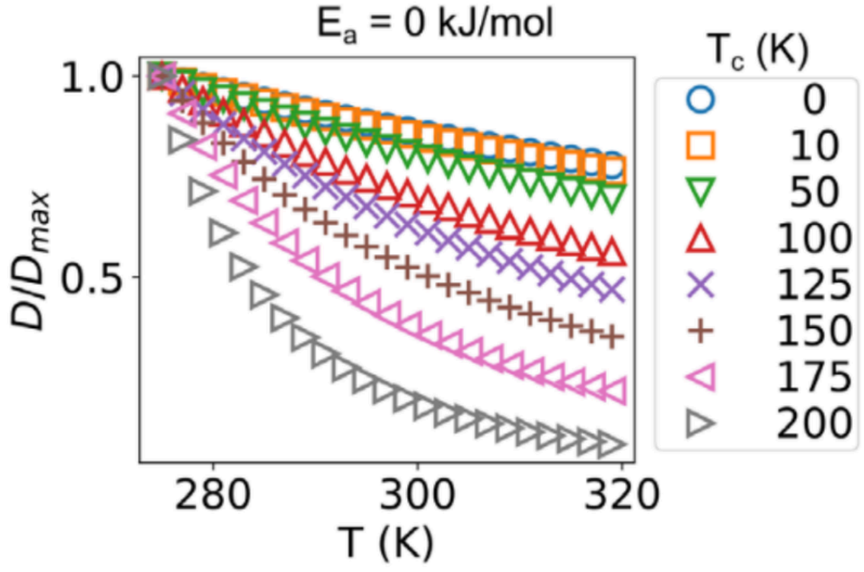
Diffusion Coefficient

$$D = \frac{k_BT}{6\pi\eta r} = D_0 e^{-E_a/k_BT+T_m/(T-T_c)}$$



What if $E_a = 0$?





Sadoon et al., PRL (2022)

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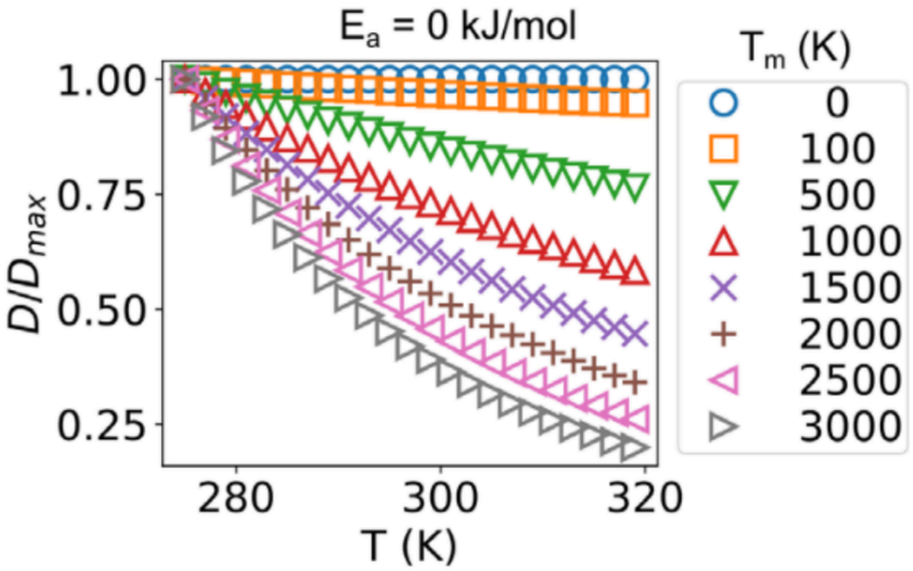
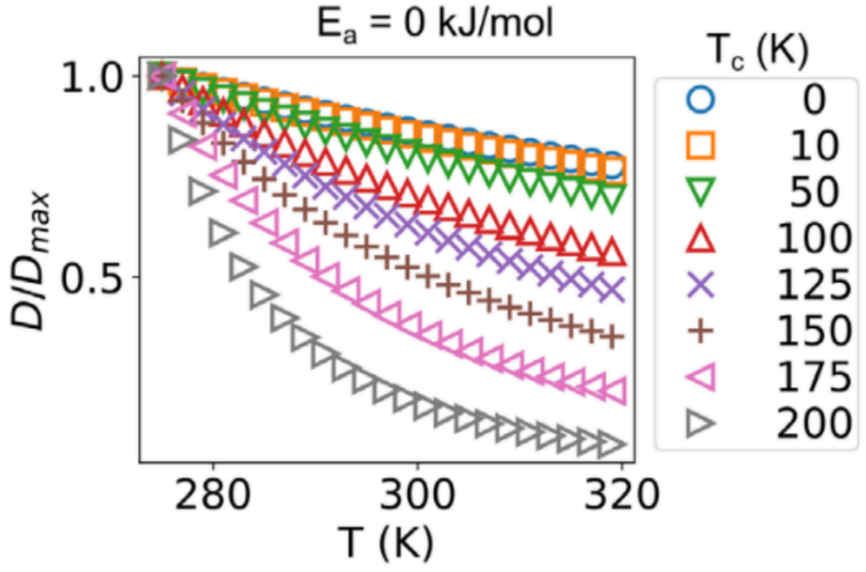
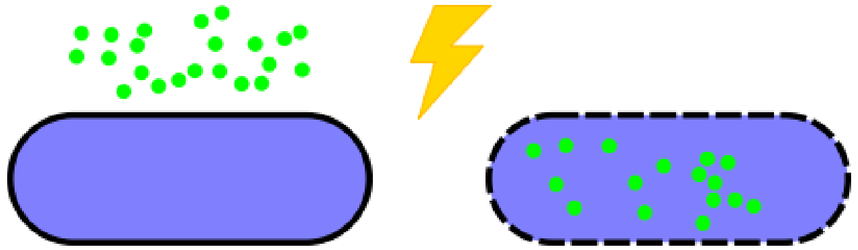




Deliver free dyes into bacteria by electroporation



Sadoon et al., PRL (2022)

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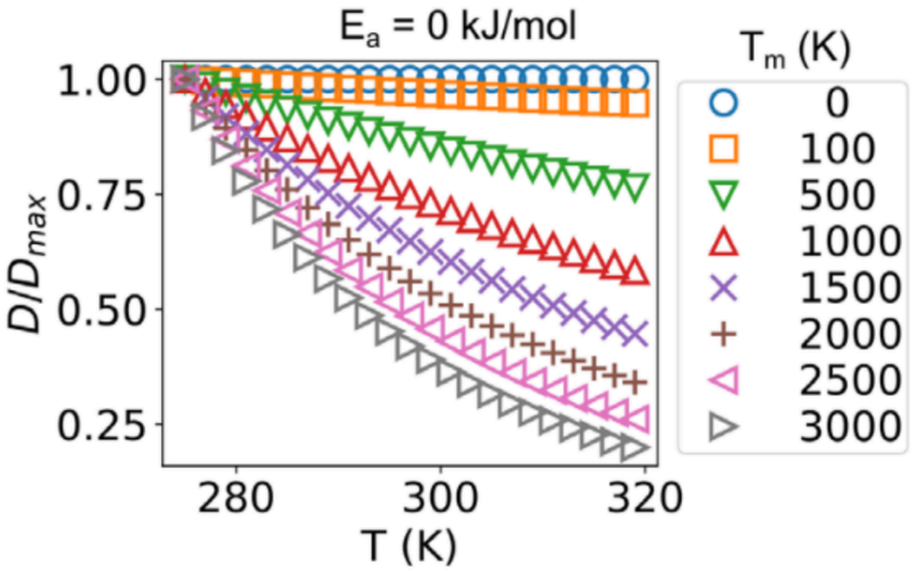
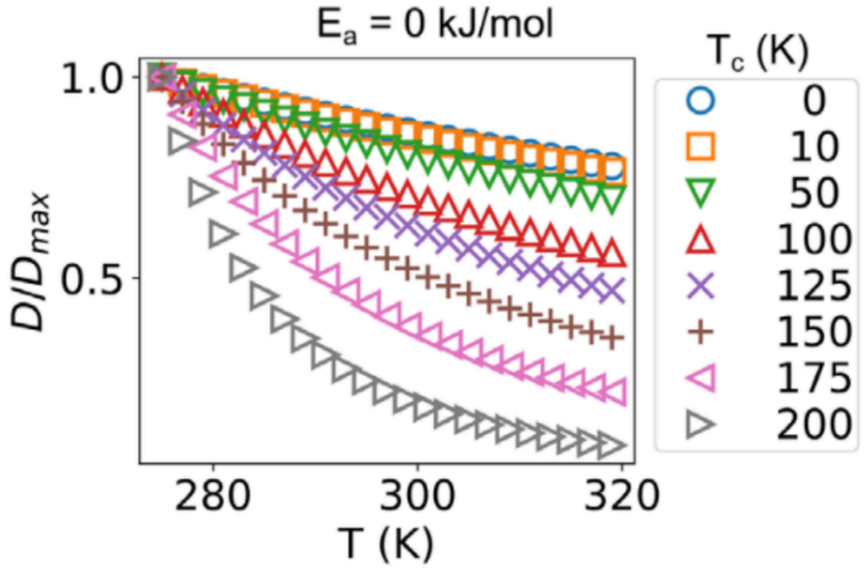
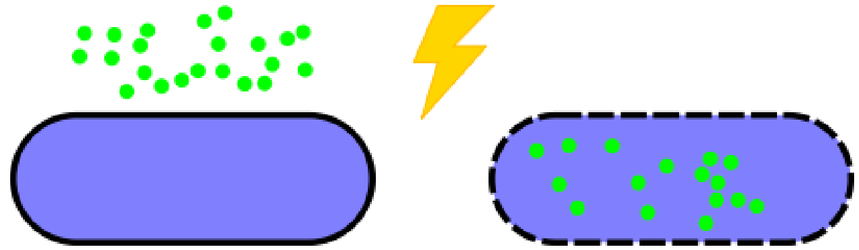
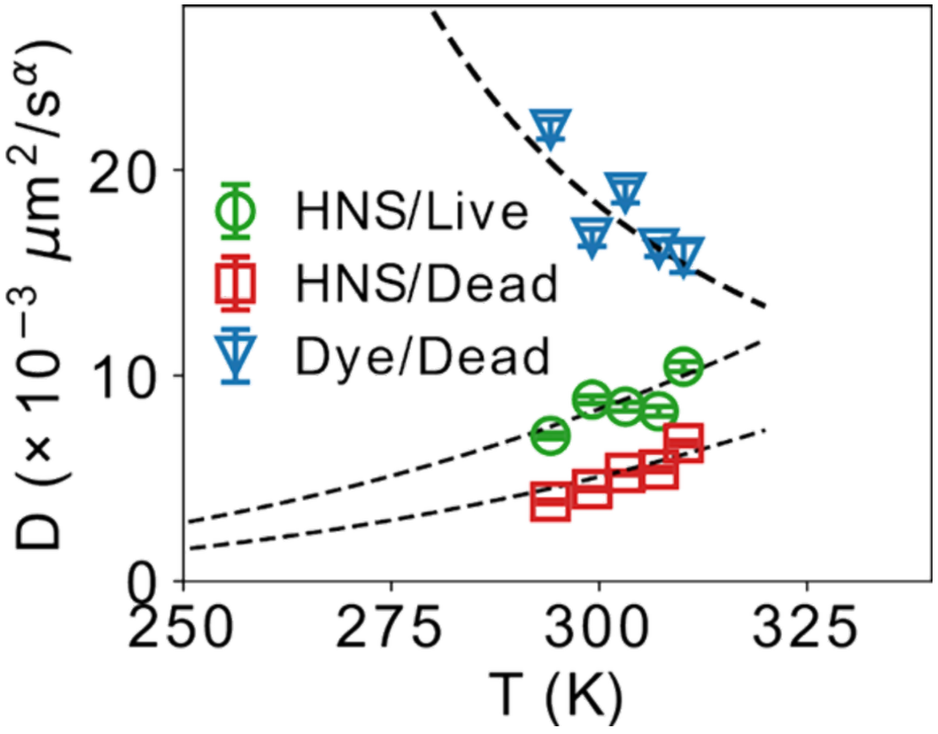




Deliver free dyes into bacteria by electroporation





Sadoon et al., PRL (2022)

Motility of Single Bacteria with Stresses or in Complex Environments

How *E. coli* Bacteria Move in Aqueous Liquid

Image from Shutterstock

How *E. coli* Bacteria Move in Aqueous Liquid

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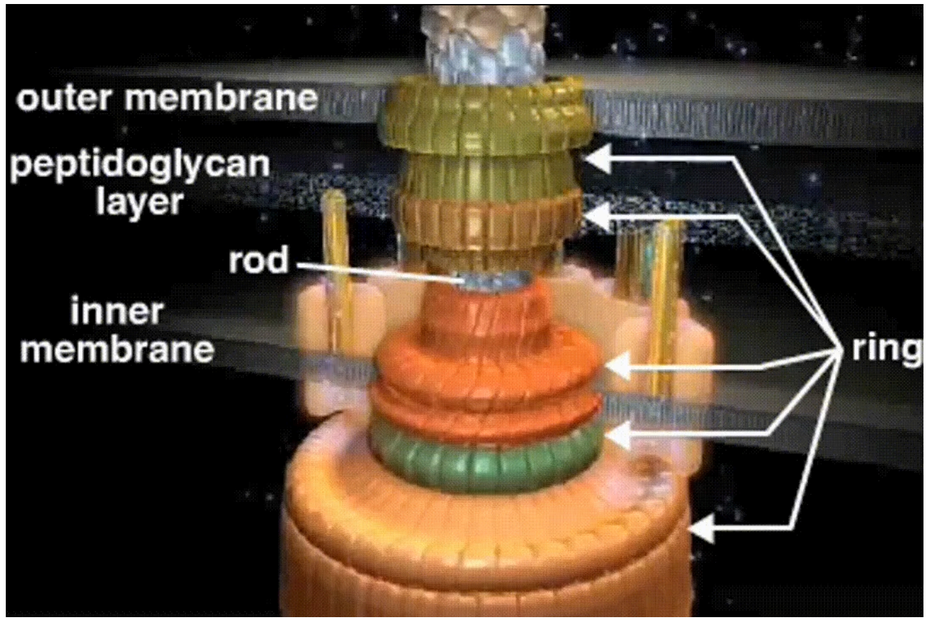
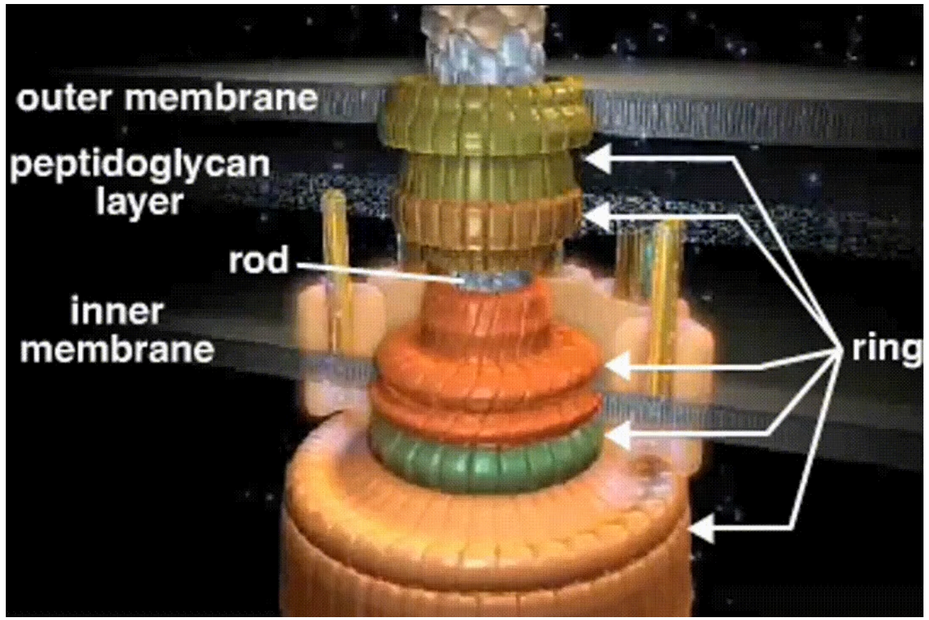


Video from Youtube

How *E. coli* Bacteria Move in Aqueous Liquid

Image from Shutterstock



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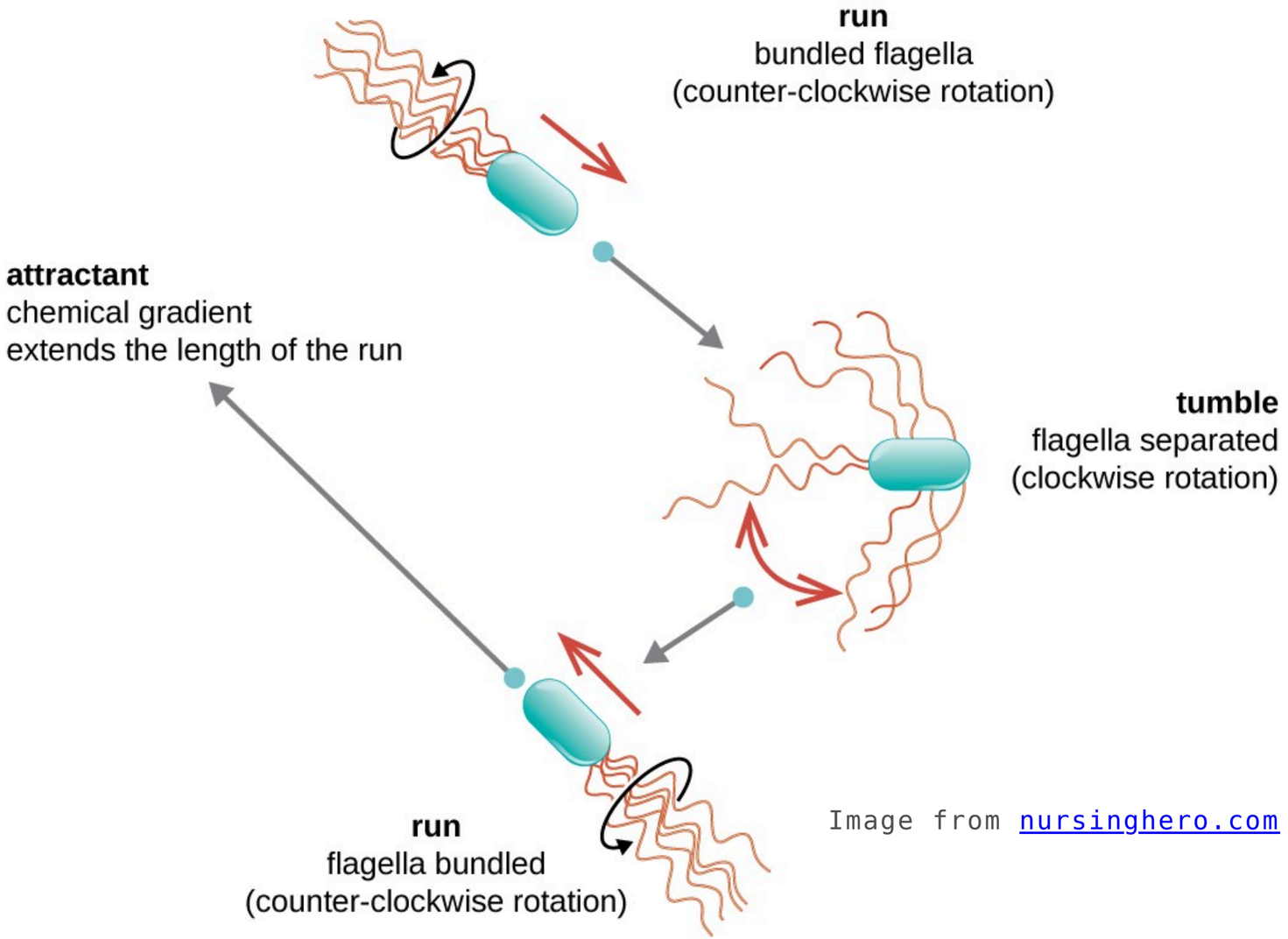

Image from nursinghero.com

Effect of Silver on Motility of *E. coli* Bacteria

Image from [Silver Coins]

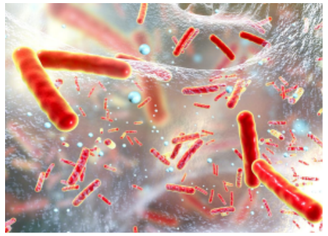

Image from VCHRI

Russel et al., IJMS (2023)

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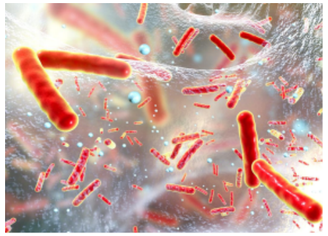

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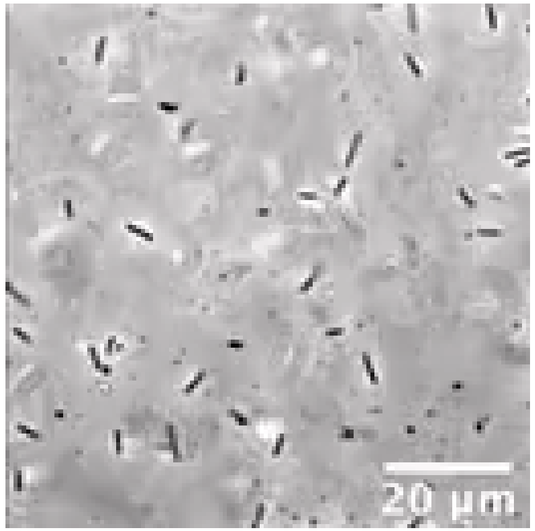

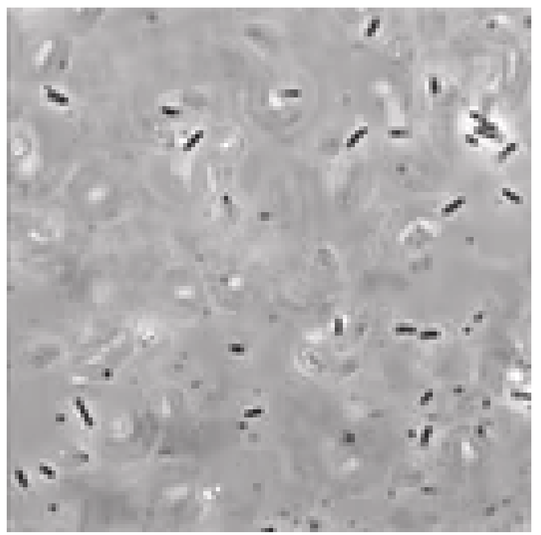

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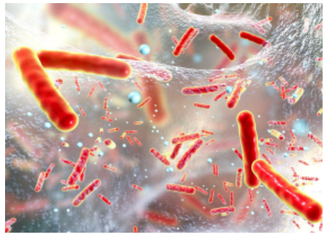

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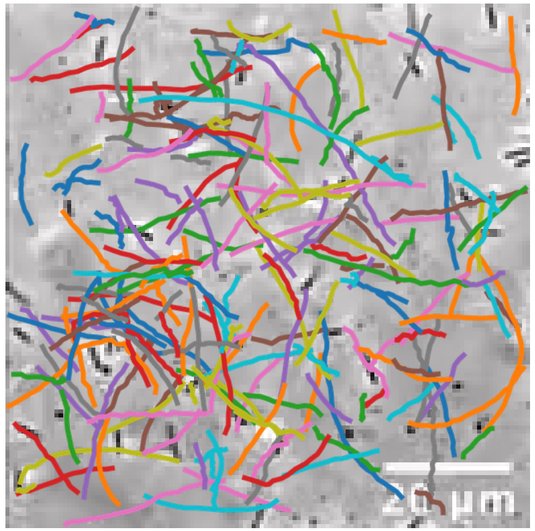

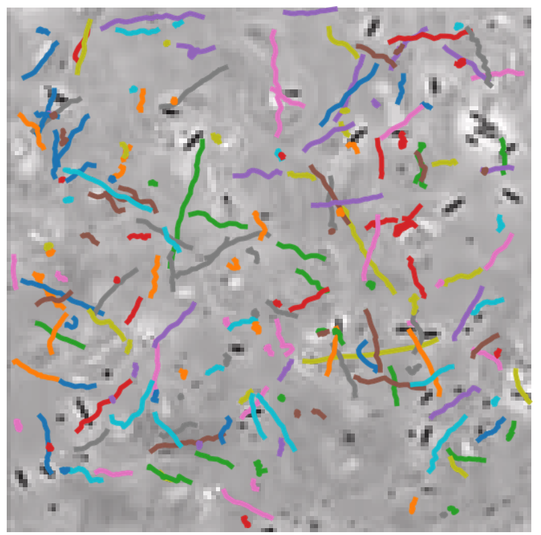

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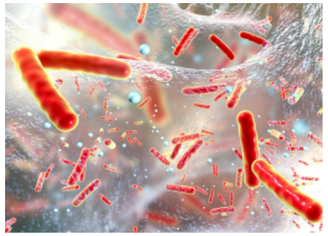

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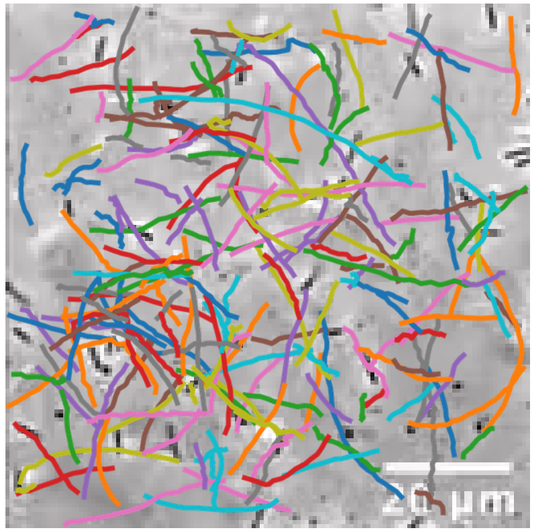

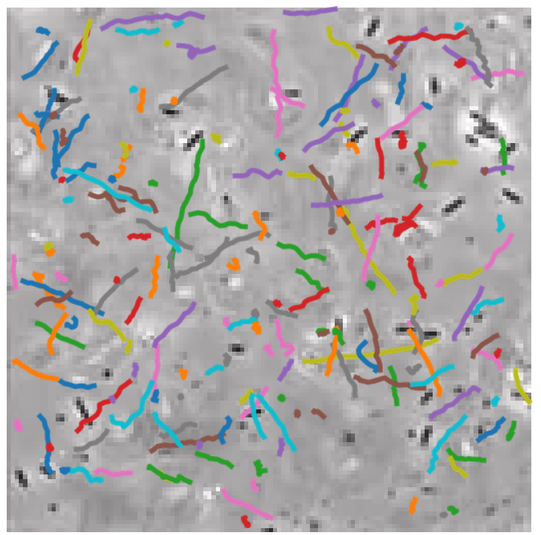

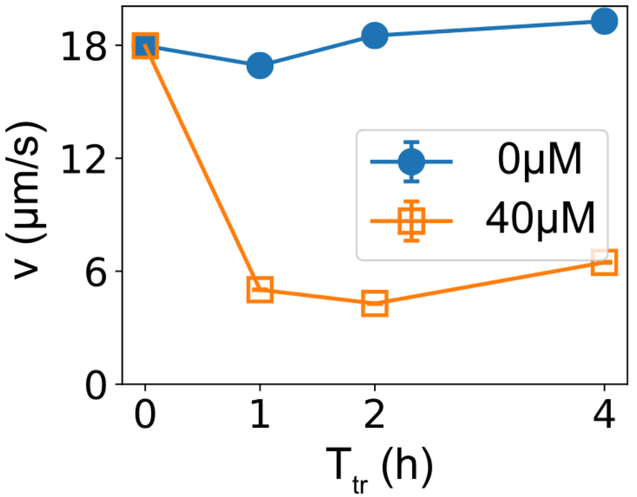

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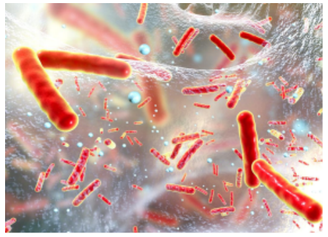

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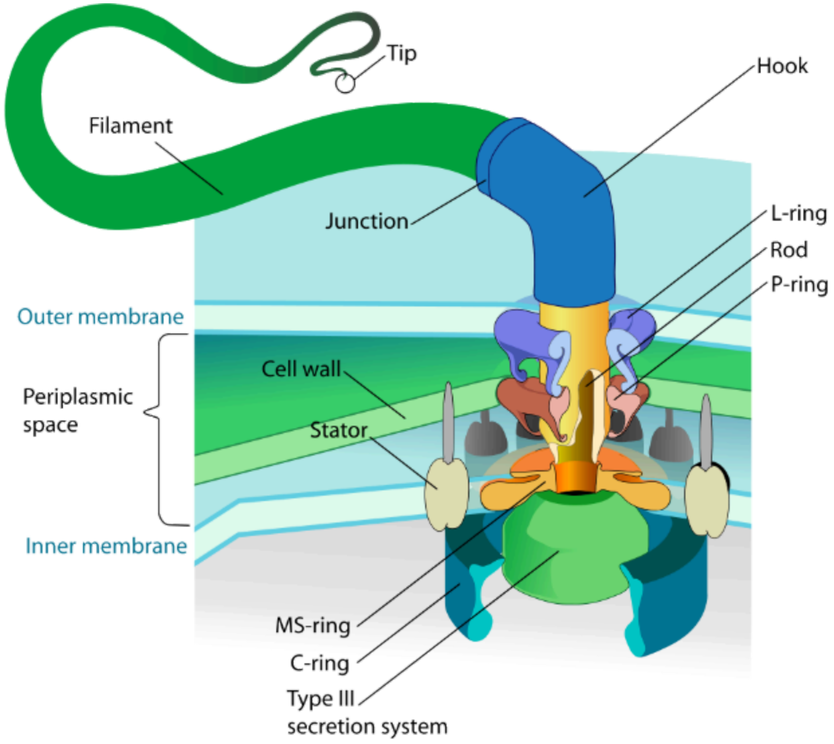
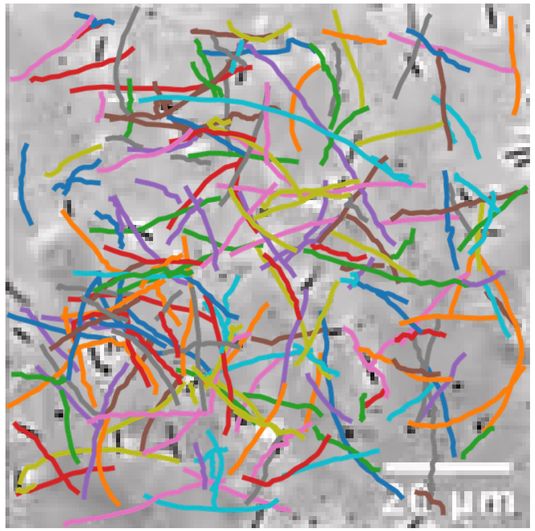
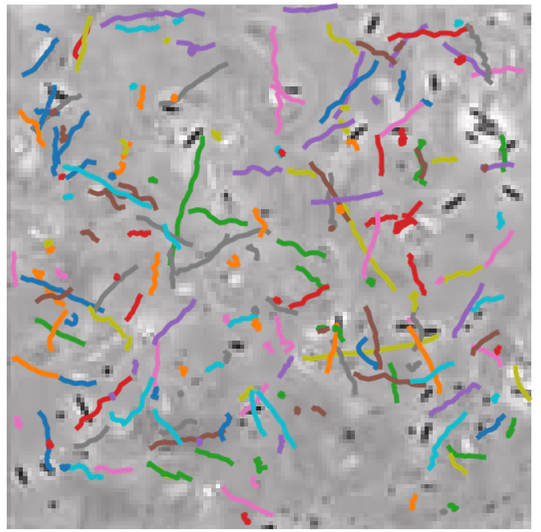


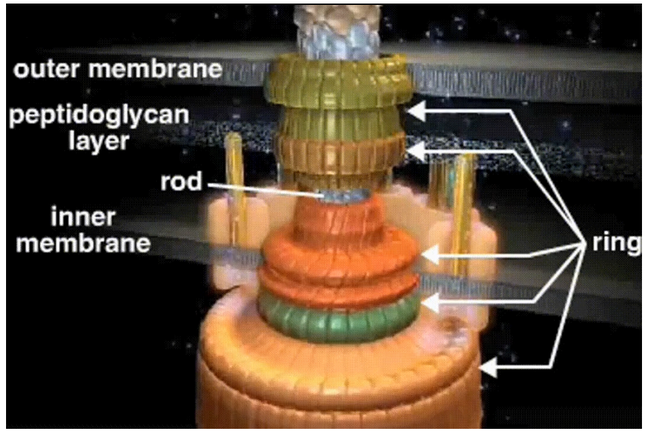
Image from Wikipedia

$-$ Ag^+ ions

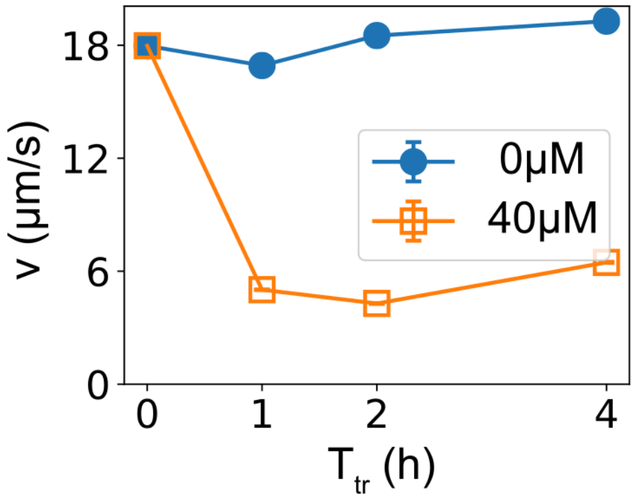


$+$ Ag^+ ions



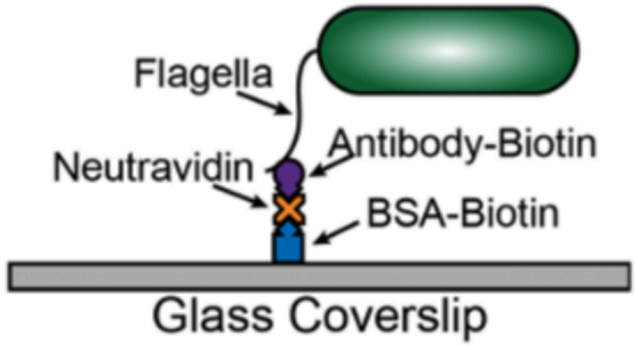


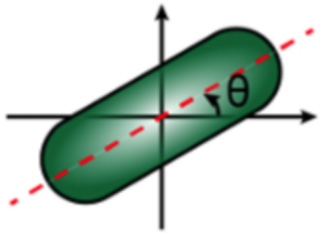
Video from Youtube



Russel et al., IJMS (2023)

Flagellar Motor Rotation for Bacterial Motility





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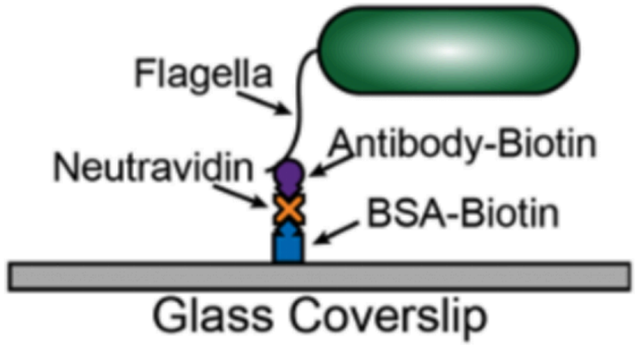

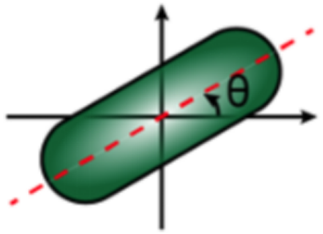

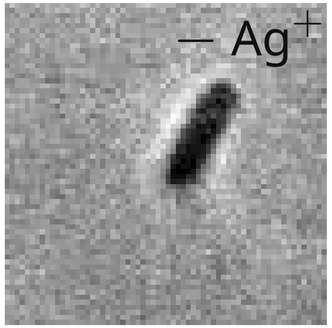

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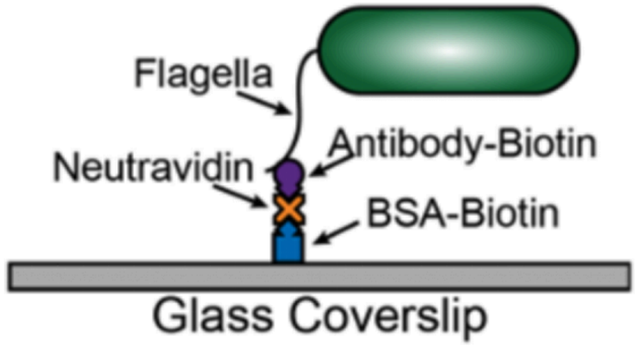

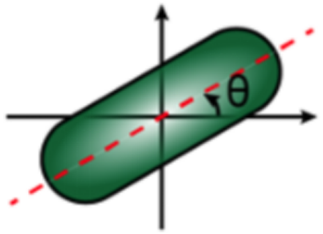

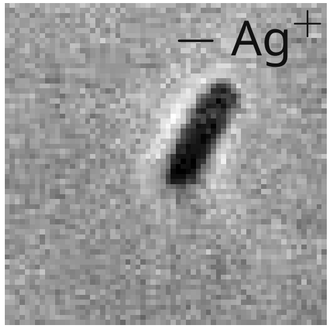

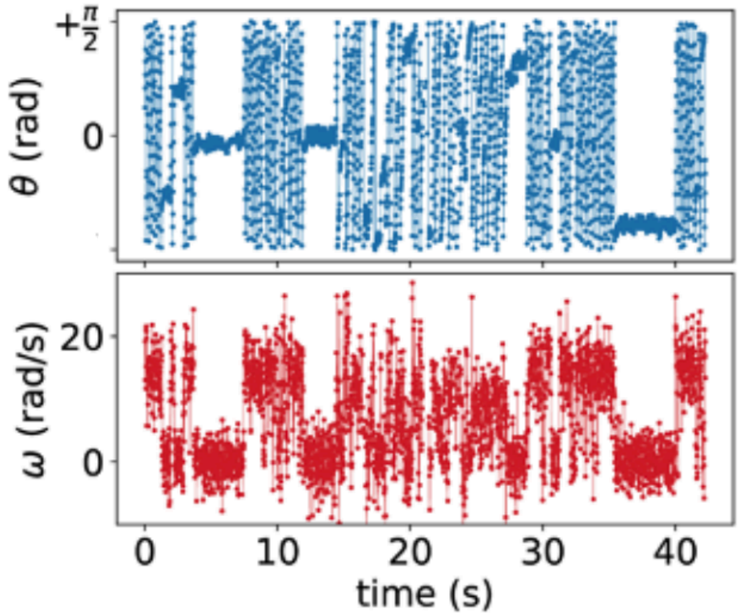

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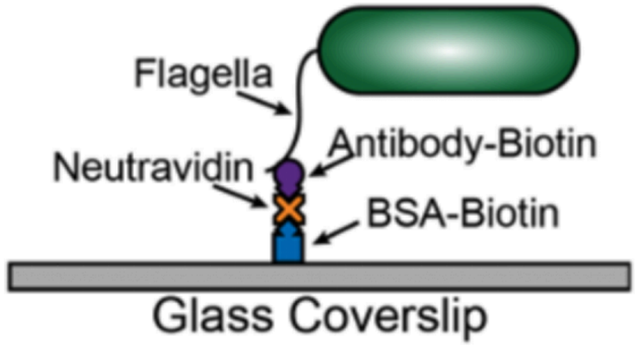

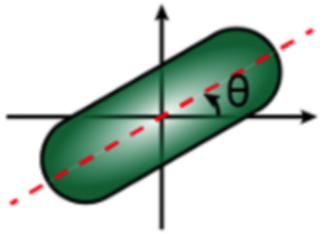

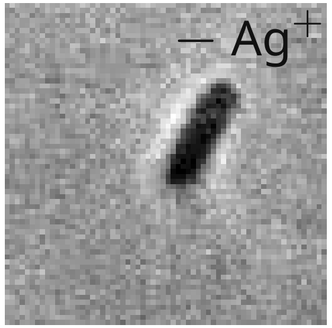

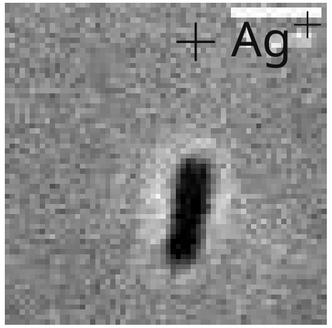

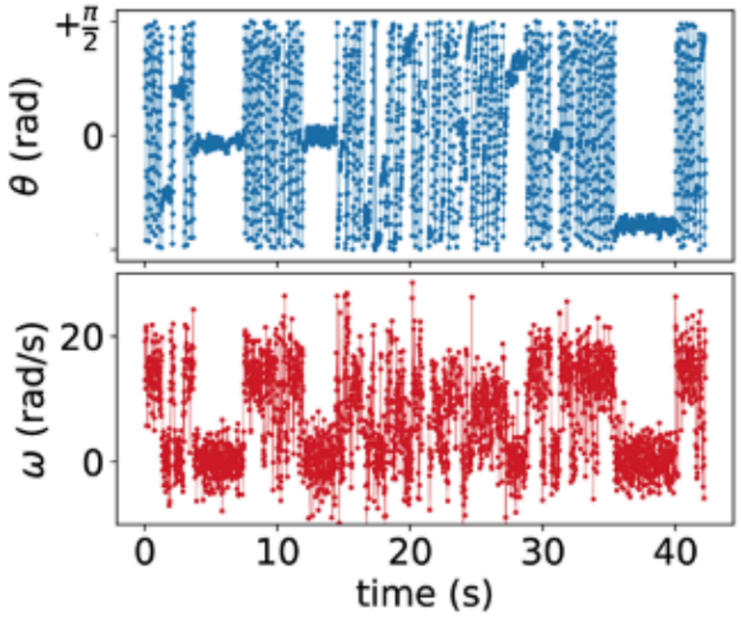

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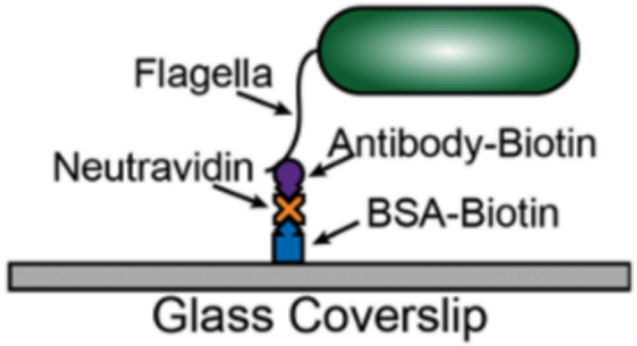

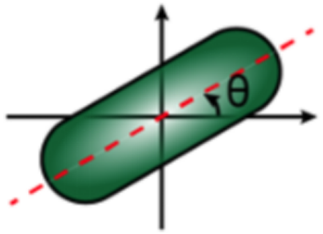

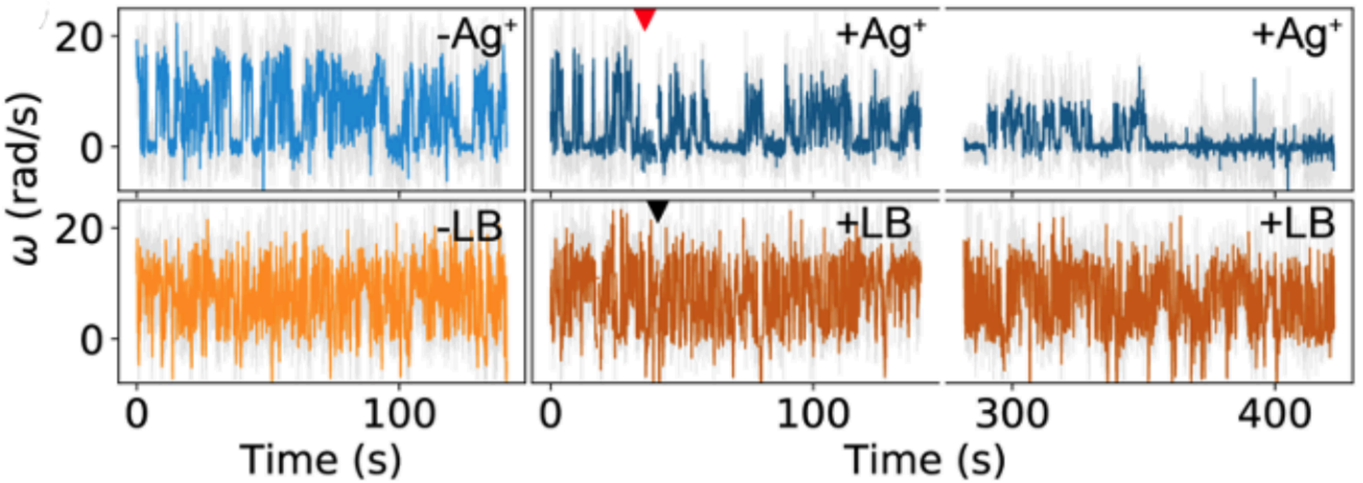

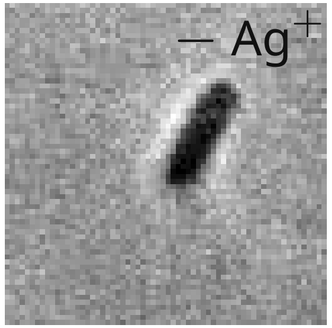

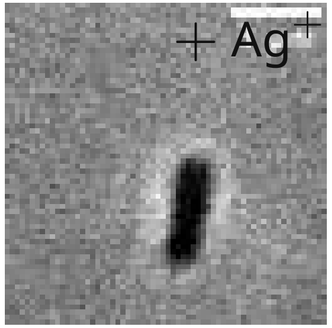

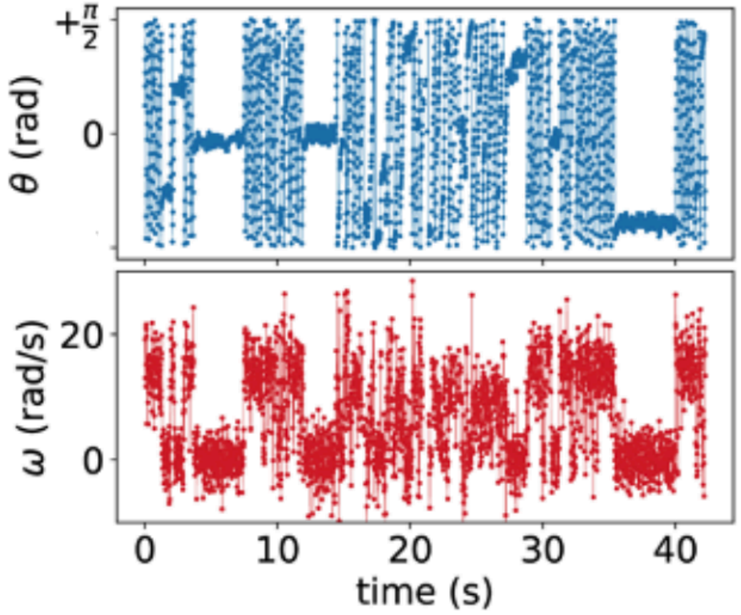

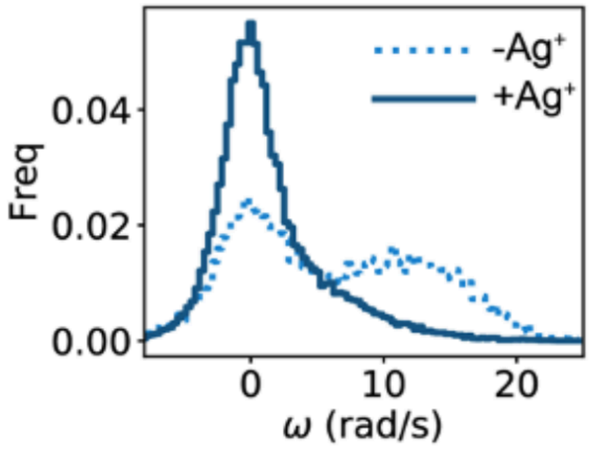

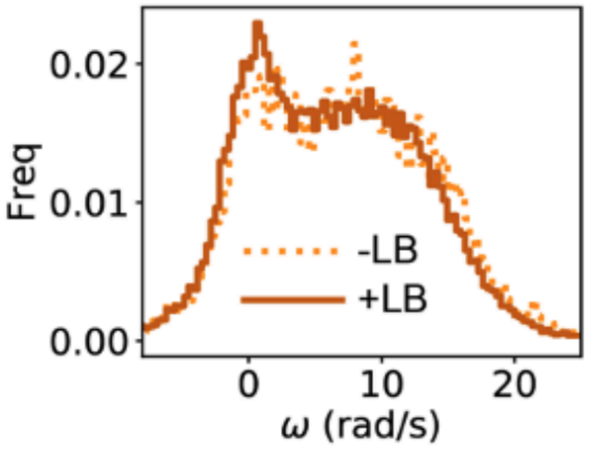

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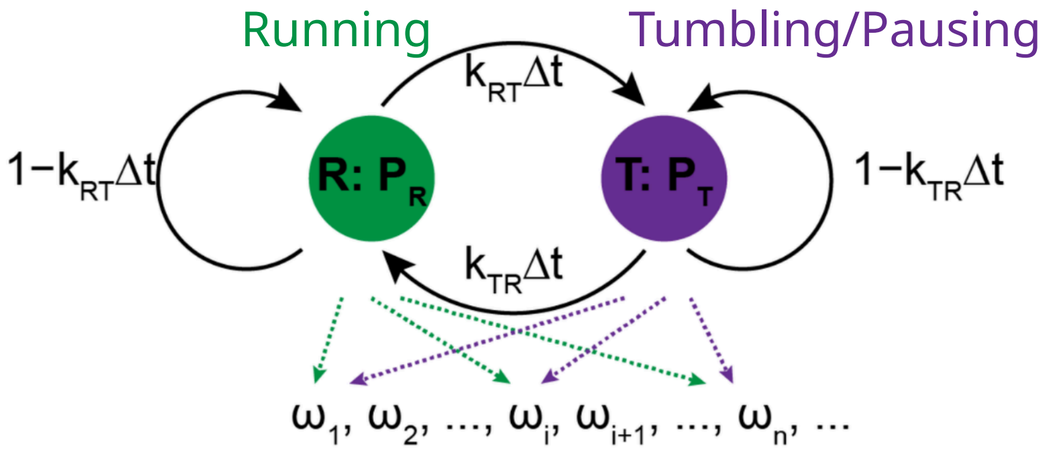
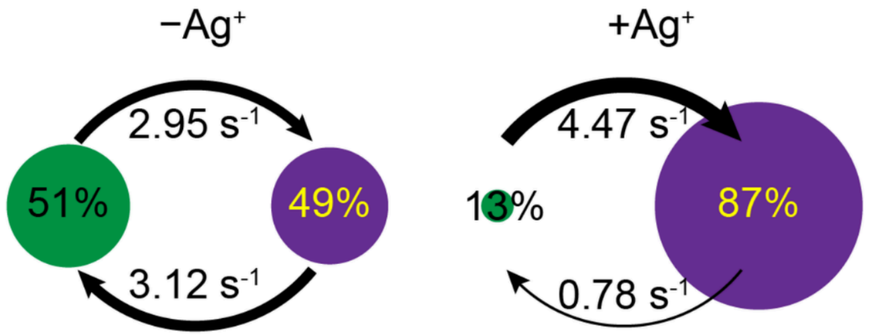
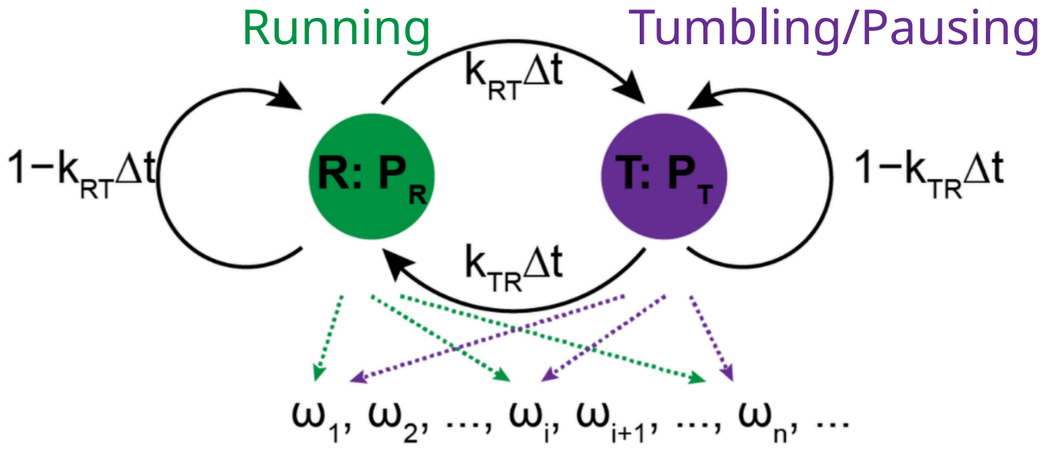
Hidden Markov Model (HMM) Analysis

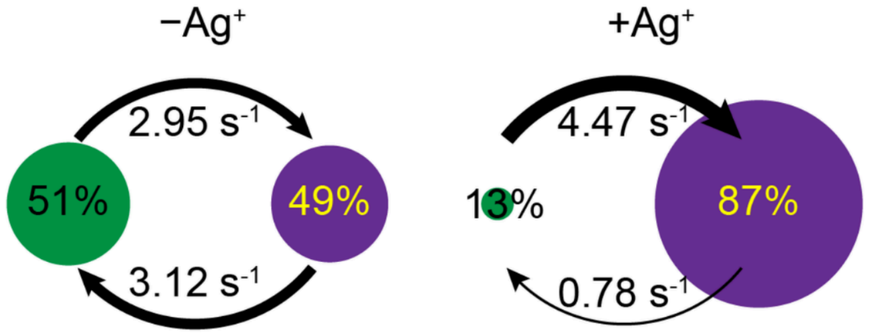


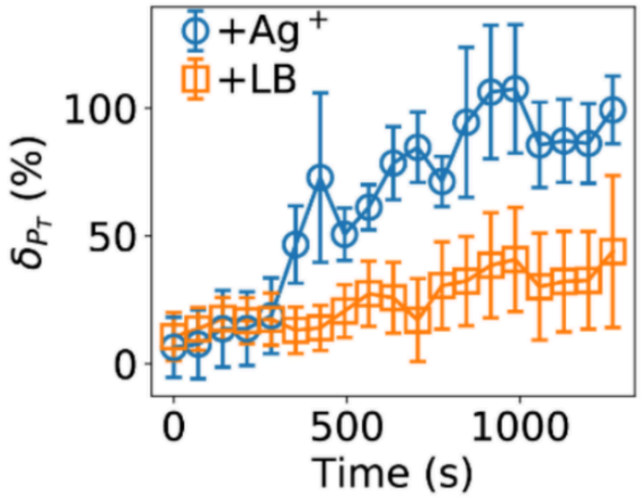


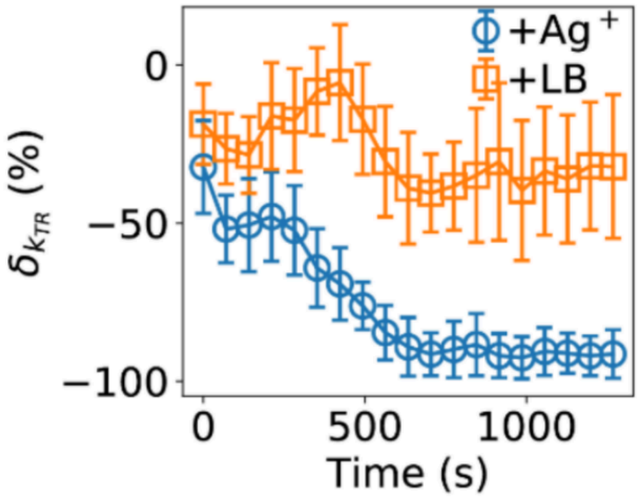
Russel et al., IJMS (2023)

Hidden Markov Model (HMM) Analysis



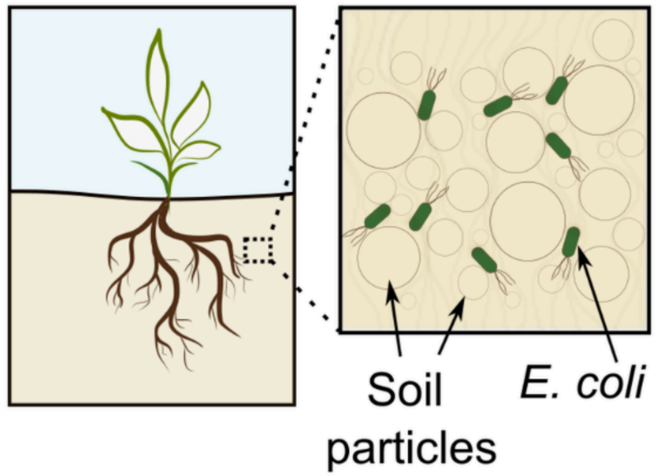


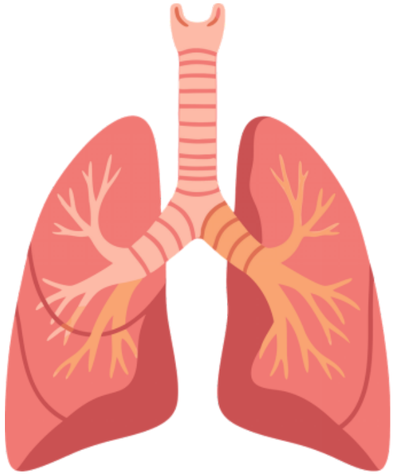


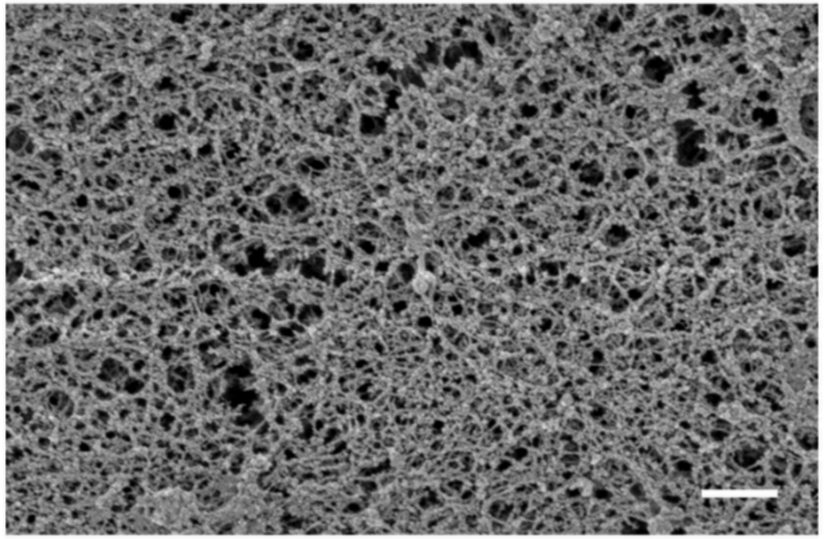


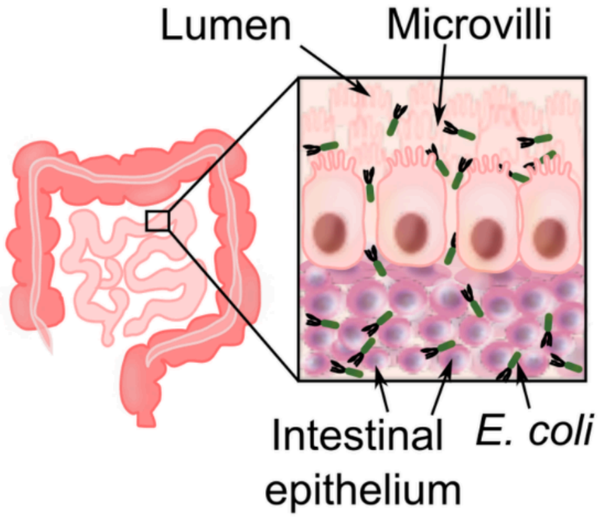
Russel et al., IJMS (2023)

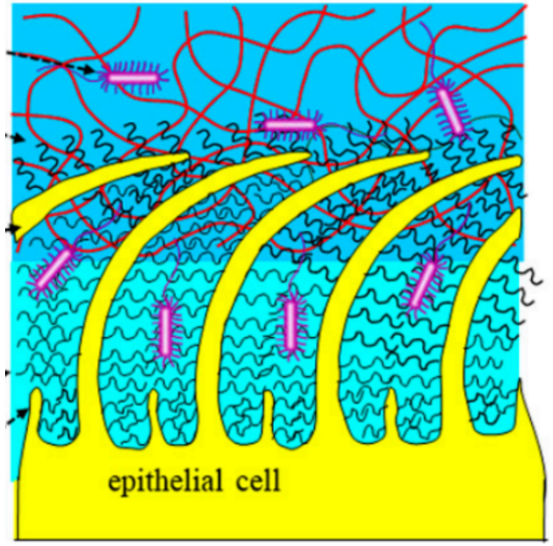
Bacteria Naturally Inhabit Complex Environments









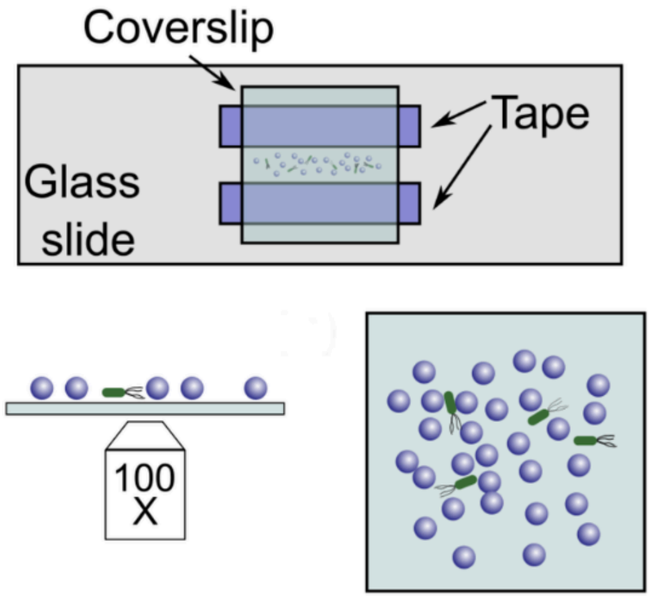


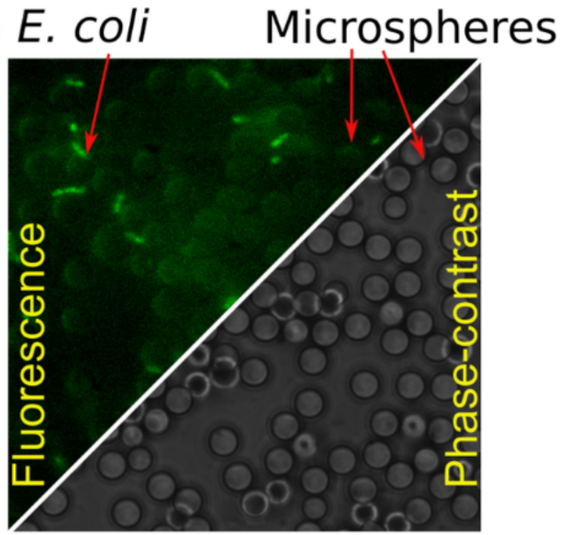
Shrestha et al. CSBB (2023)

Abrami et al., IJMS (2023)

Pangeni et al., Int. J. Pharm. (2023)

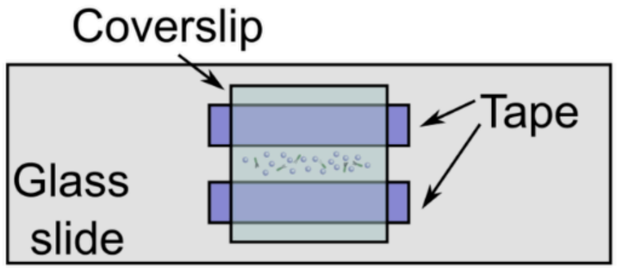
Bacterial Motility in Microsphere-Mimicked Soil

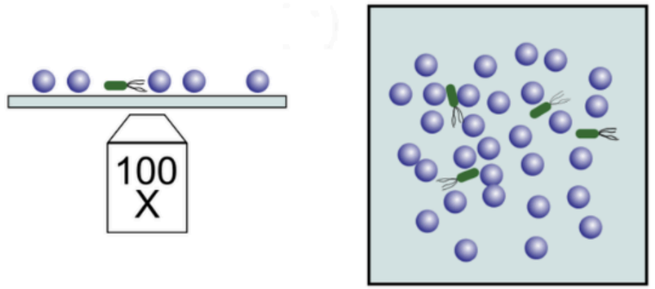


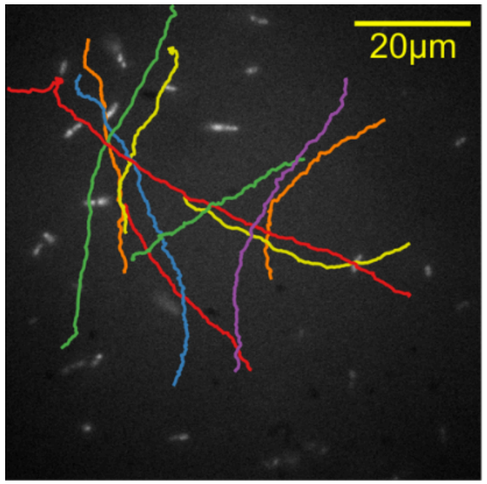


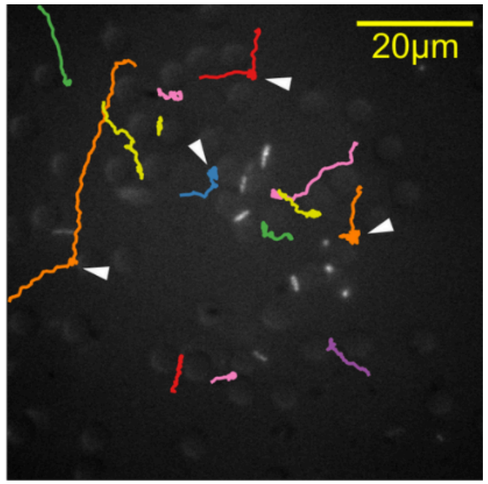
Shrestha et al. CSBB (2023)

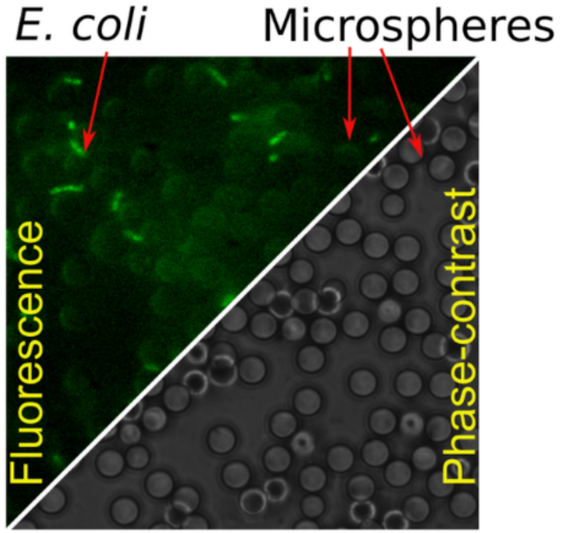
Bacterial Motility in Microsphere-Mimicked Soil





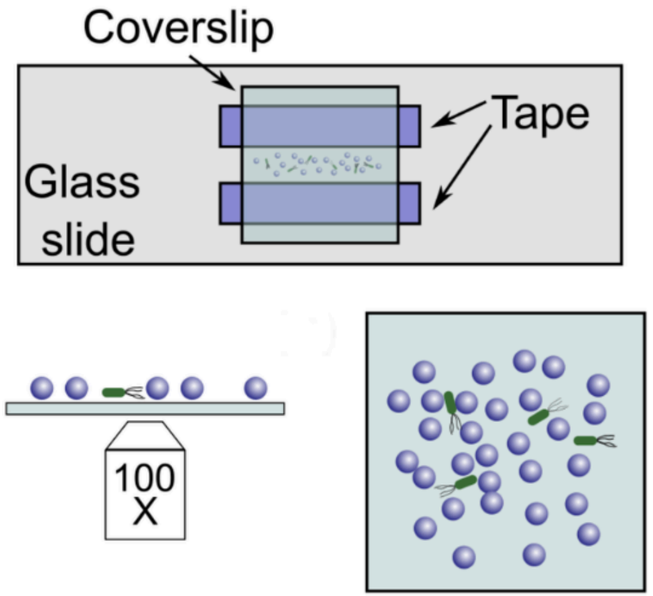


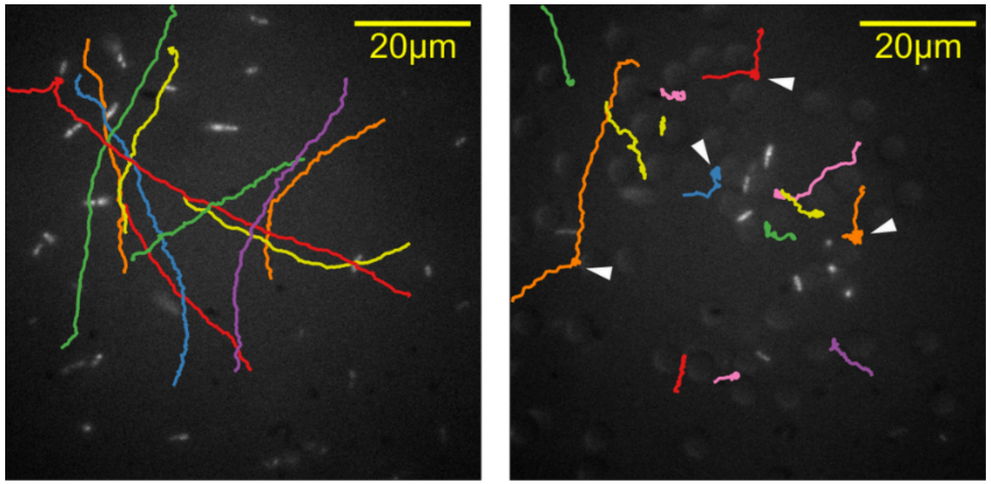




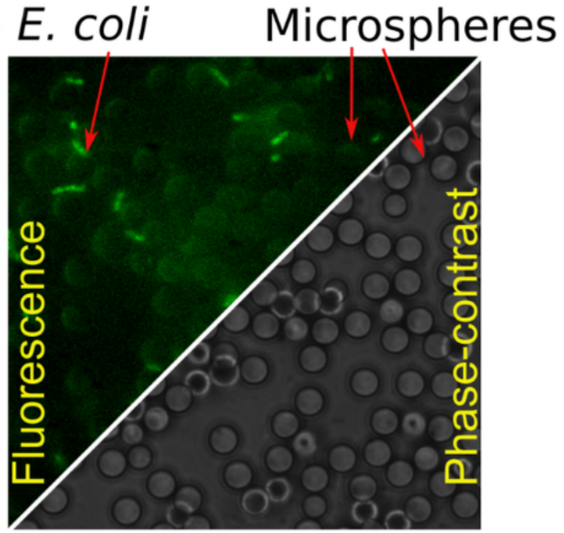
Shrestha et al. CSBB (2023)

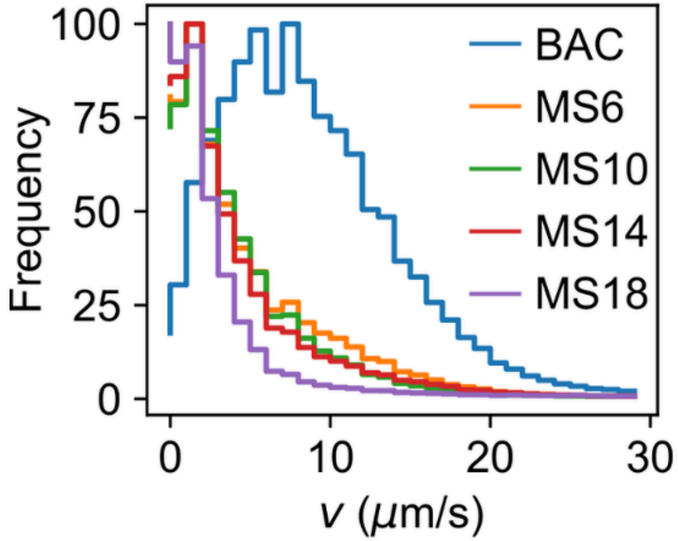
Bacterial Motility in Microsphere-Mimicked Soil

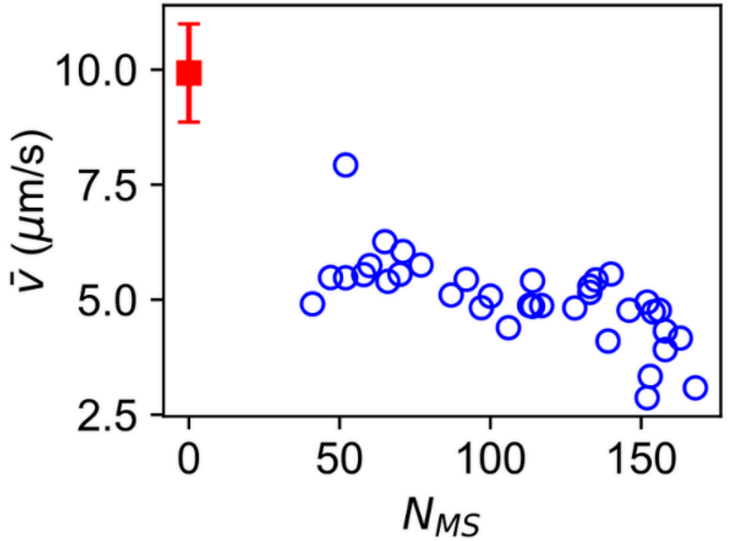




Bacterial Speed Decreases!

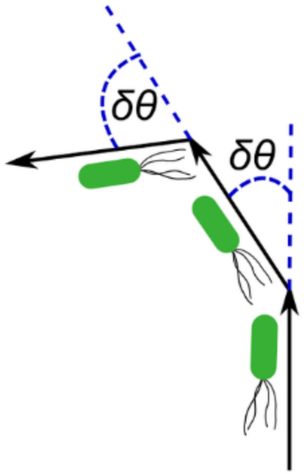






Shrestha et al. CSBB (2023)

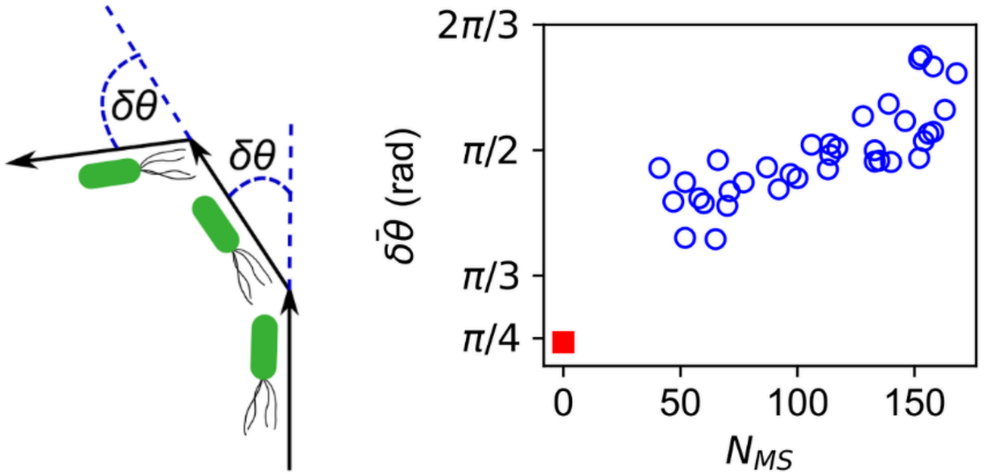
Bacterial Motility in Microsphere-Mimicked Soil

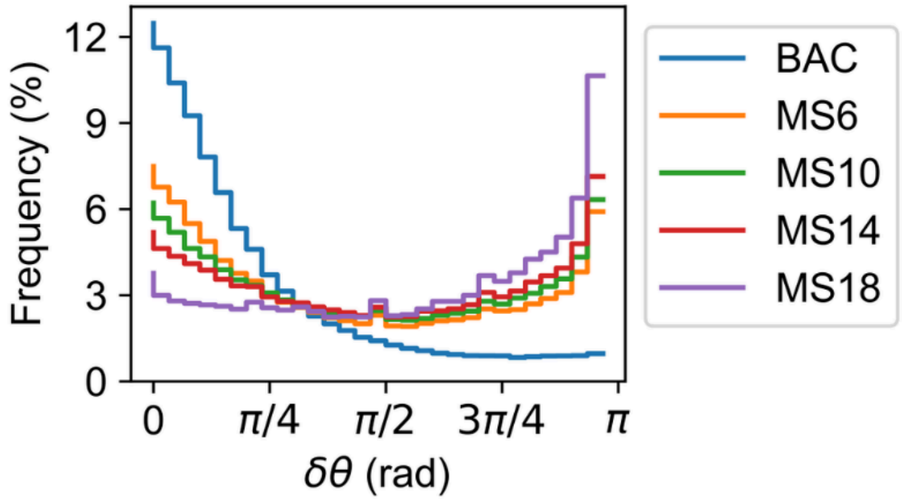


Shrestha et al. CSBB (2023)

Bacterial Motility in Microsphere-Mimicked Soil

Bacterial Directional Change Increases!

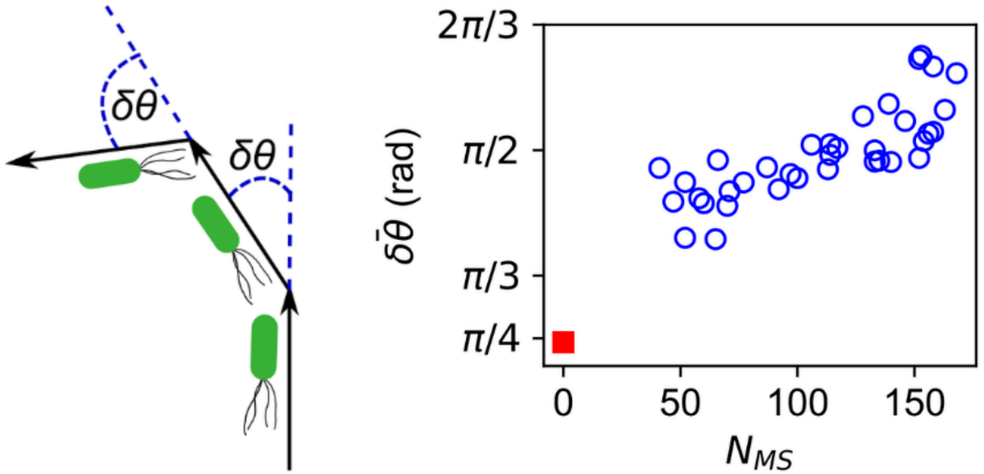


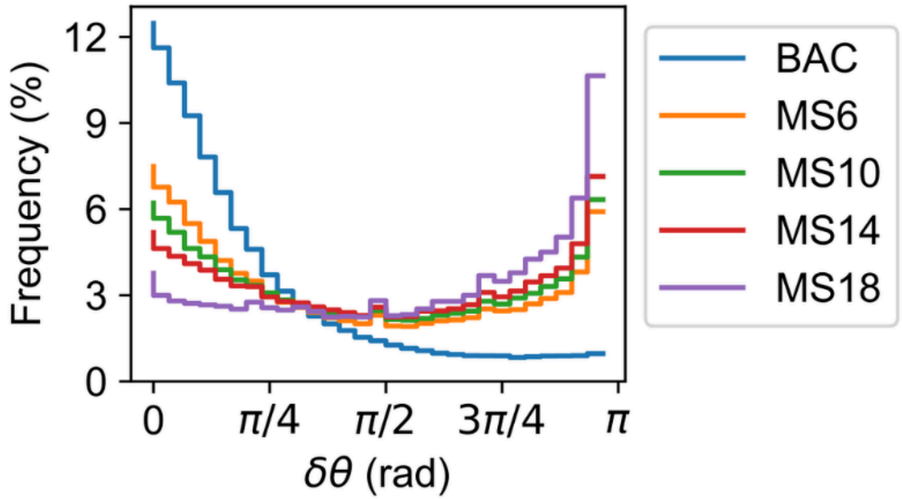


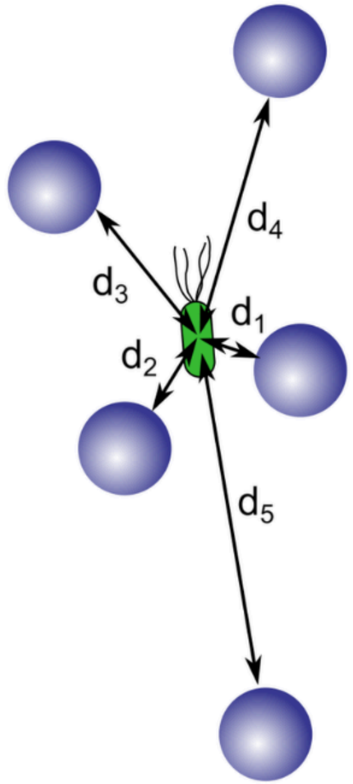
Shrestha et al. CSBB (2023)

Bacterial Motility in Microsphere-Mimicked Soil

Bacterial Directional Change Increases!



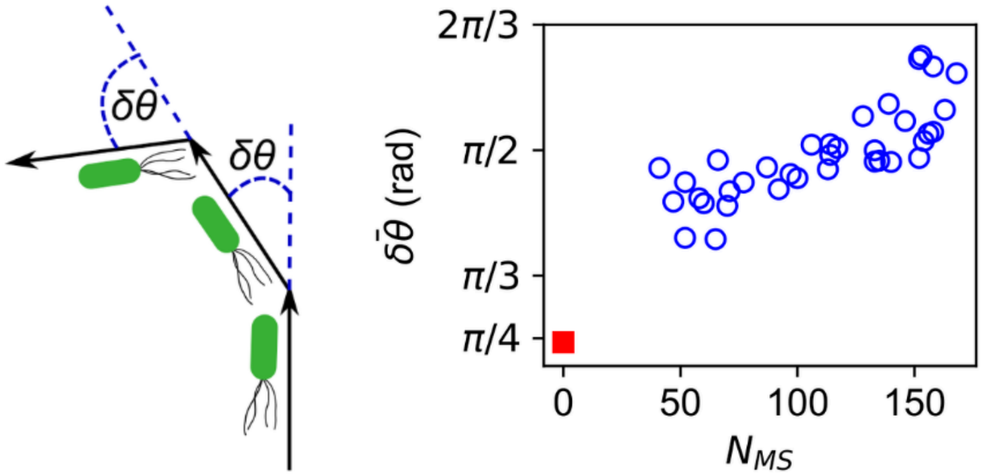


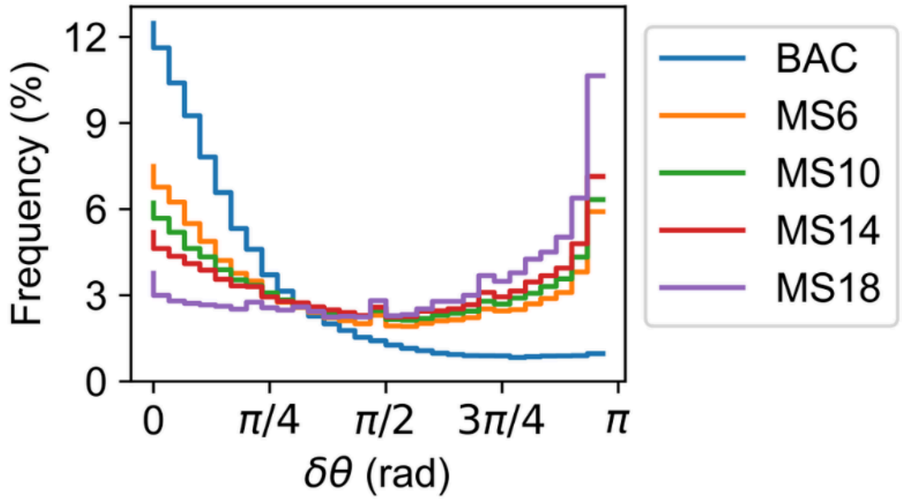


Shrestha et al. CSBB (2023)

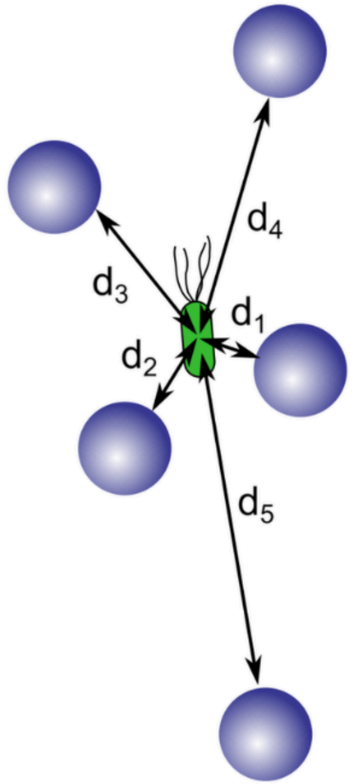
Bacterial Motility in Microsphere-Mimicked Soil

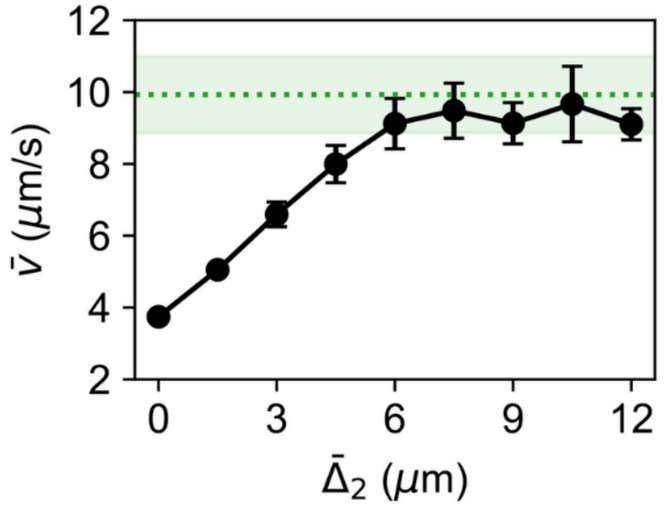
Bacterial Directional Change Increases!

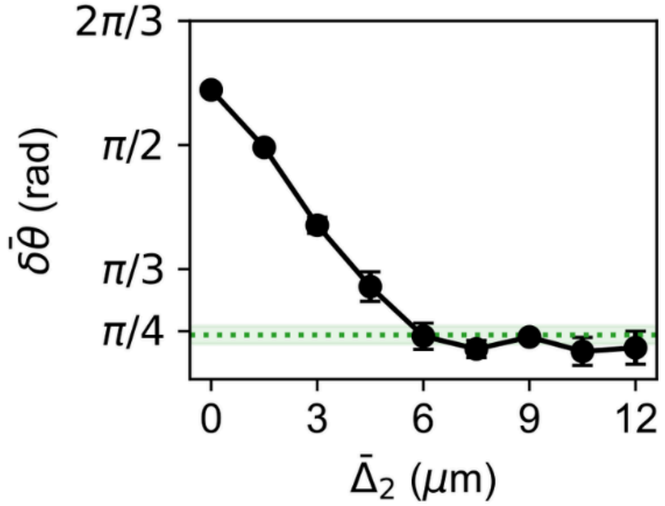




Bacterial Motility is Gap-Dependent!

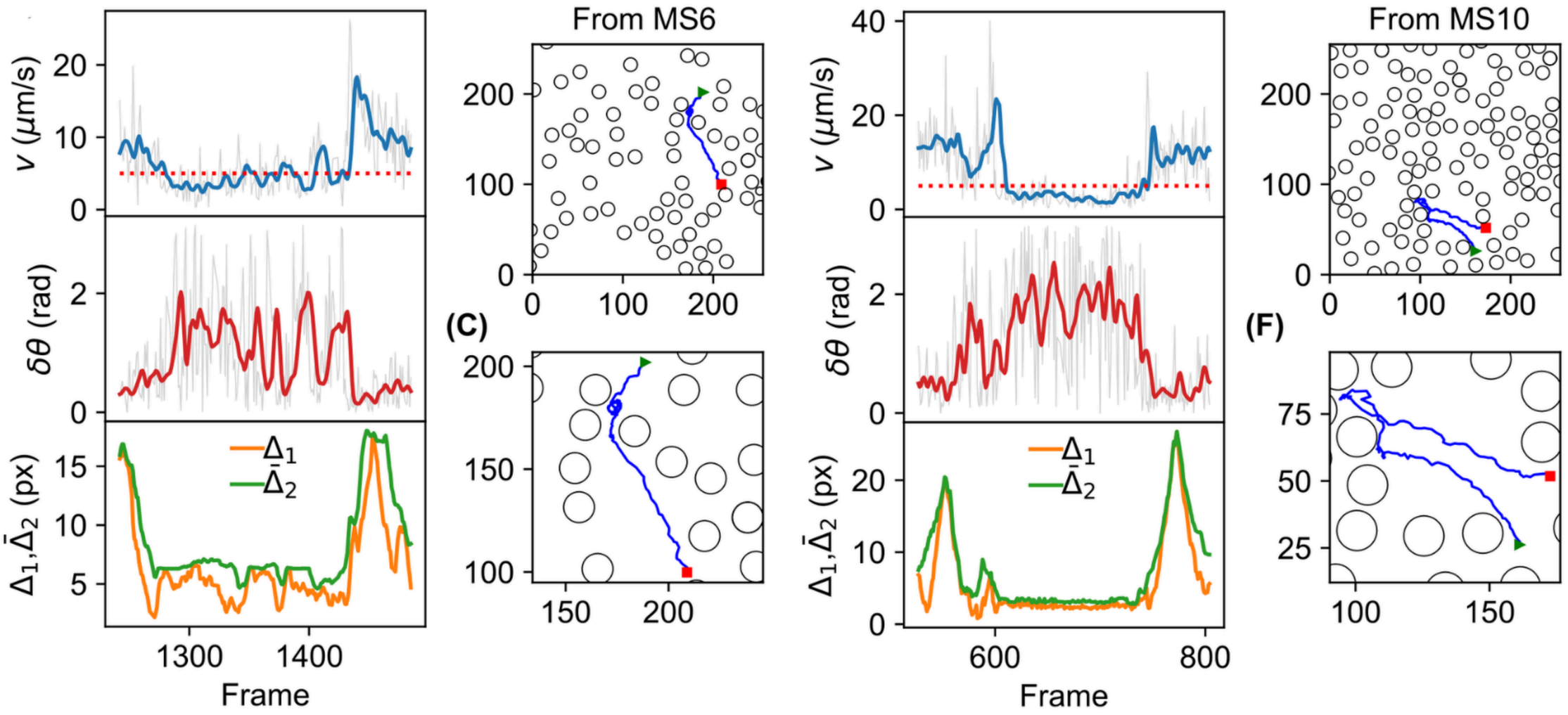






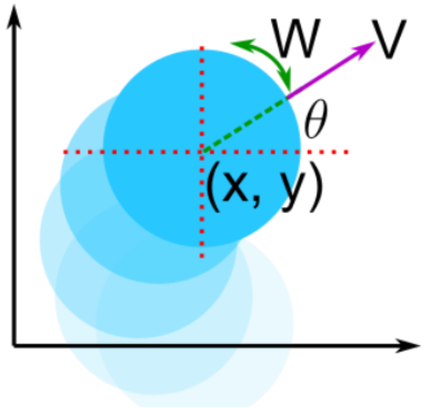
Shrestha et al. CSBB (2023)

Bacteria Trapping & Hopping



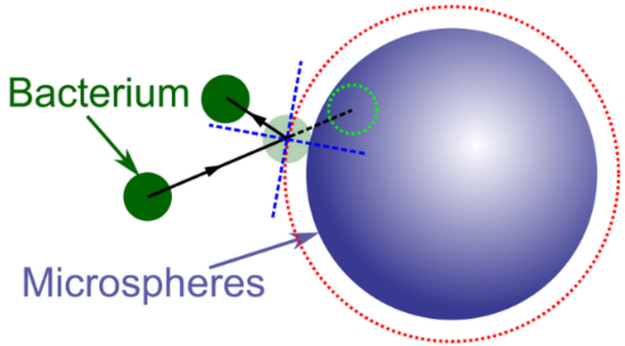
Shrestha et al. CSBB (2023)

Physics Model: Active Brownian Particle



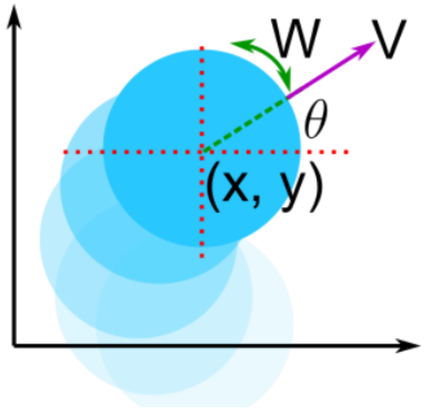
$$\Delta x = V\cos\theta\Delta t + \sqrt{2D_T\Delta t}\xi_x$$
$$\Delta y = V\sin\theta\Delta t + \sqrt{2D_T\Delta t}\xi_y$$

$$\Delta\theta = W\Delta t + \sqrt{2D_R\Delta t}\xi_\theta$$



Shrestha et al. CSBB (2023)

Physics Model: Active Brownian Particle



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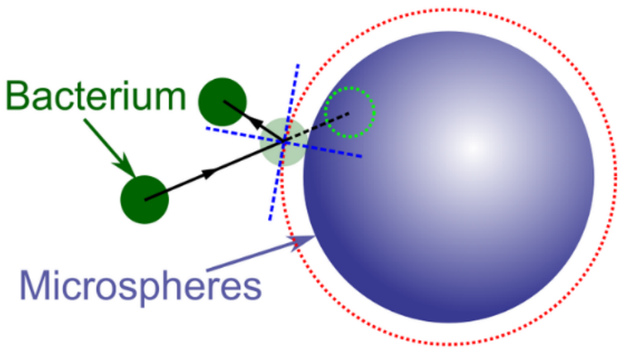
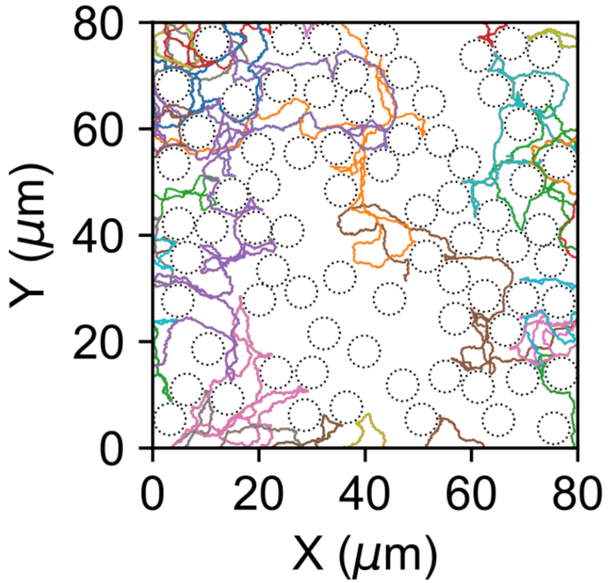
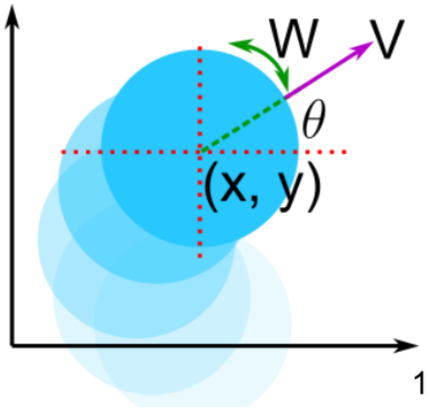




Shrestha et al. CSBB (2023)

Physics Model: Active Brownian Particle



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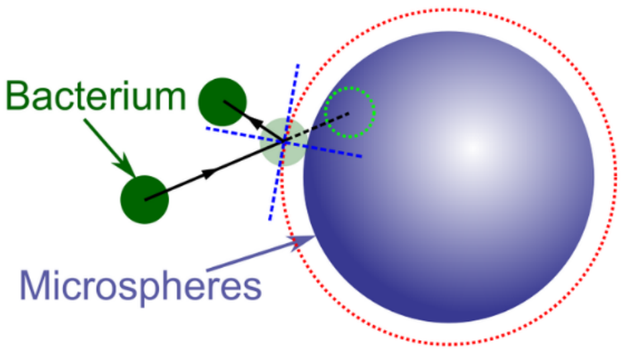
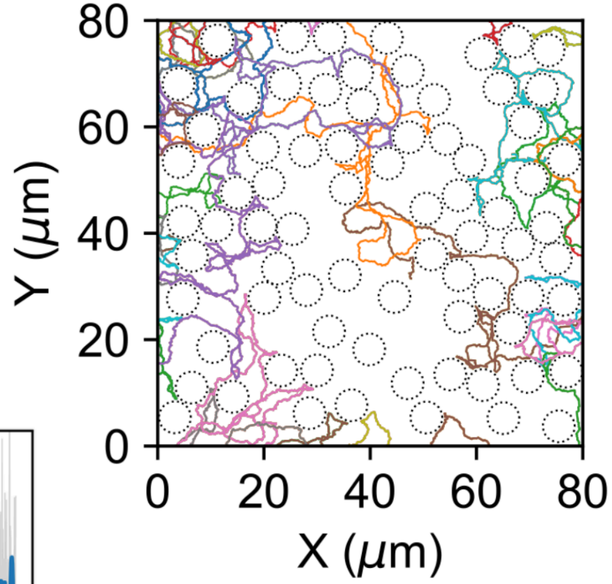
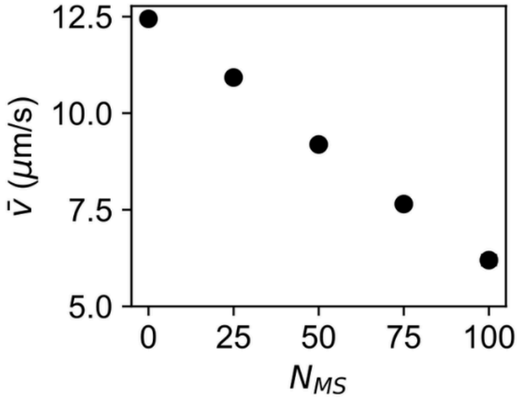
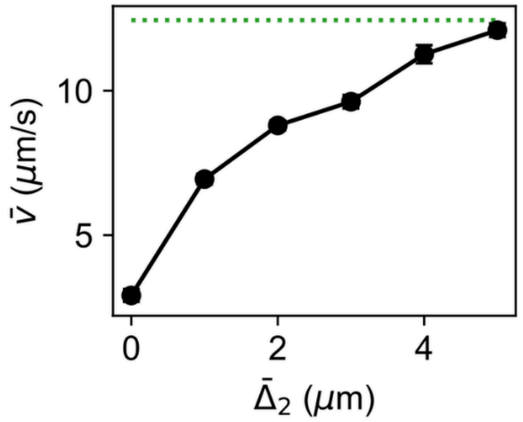
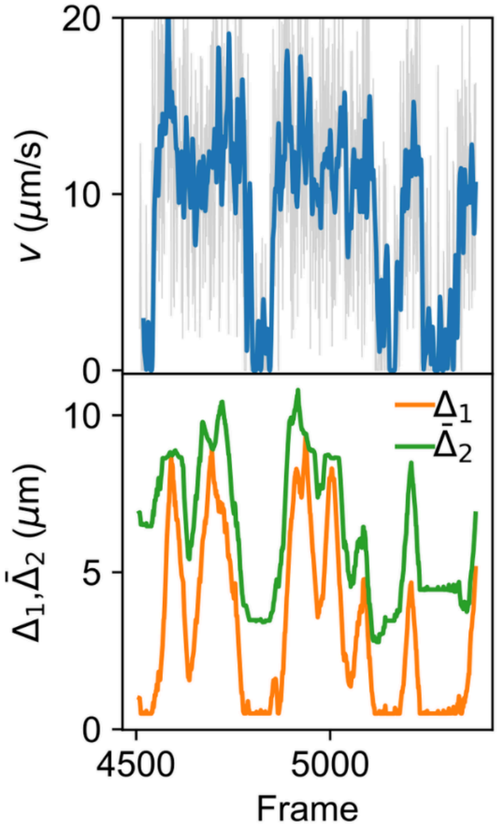
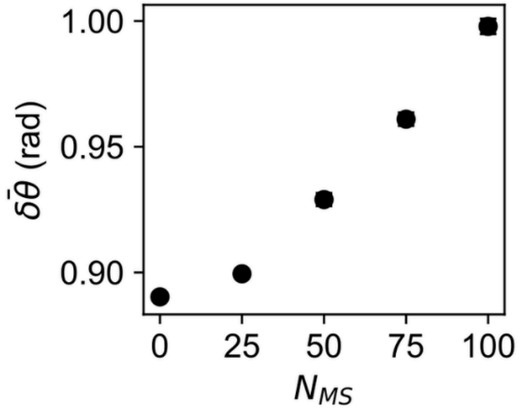


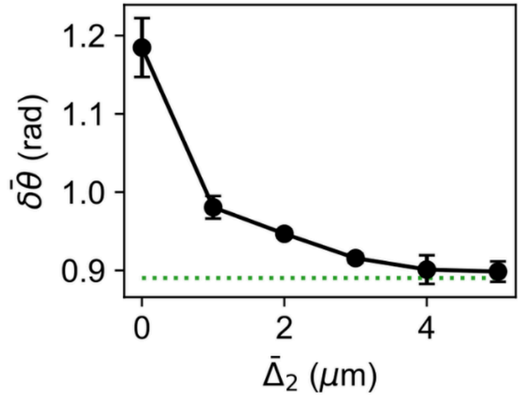






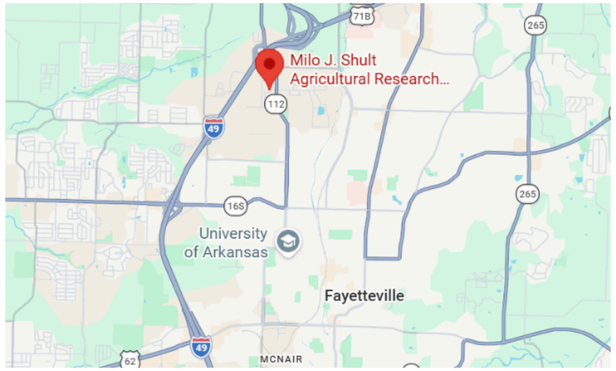




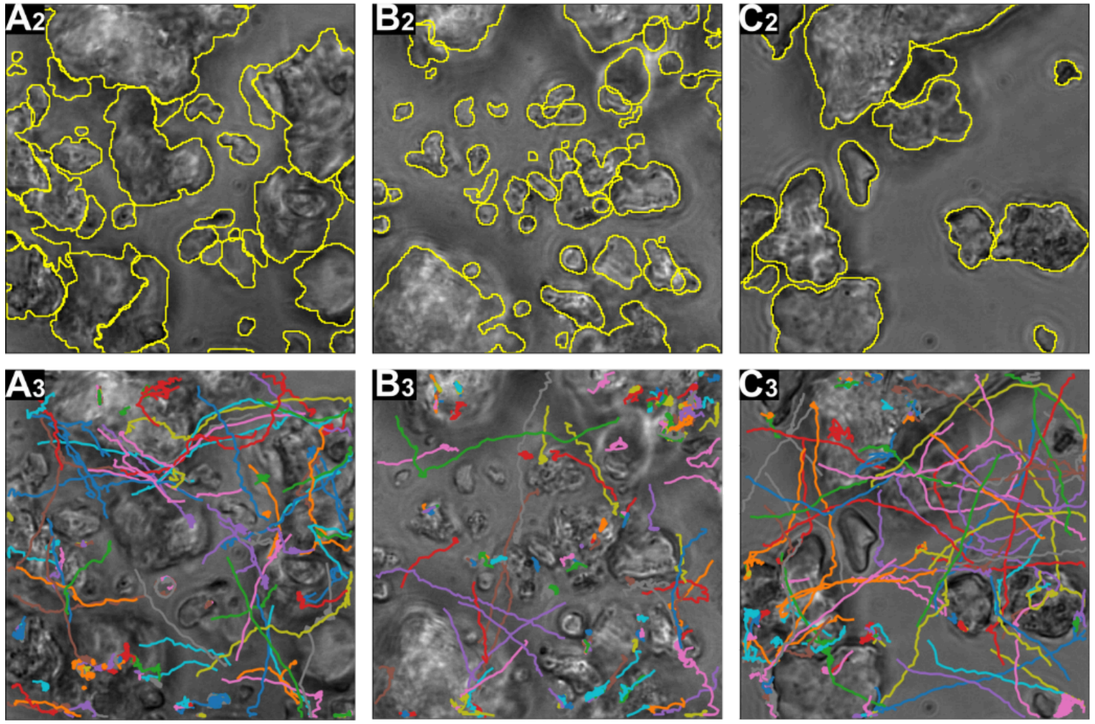


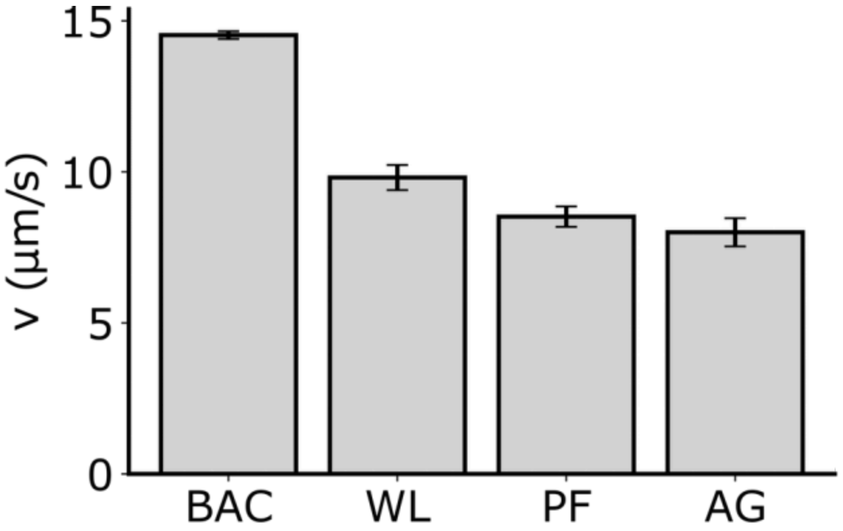
Shrestha et al. CSBB (2023)

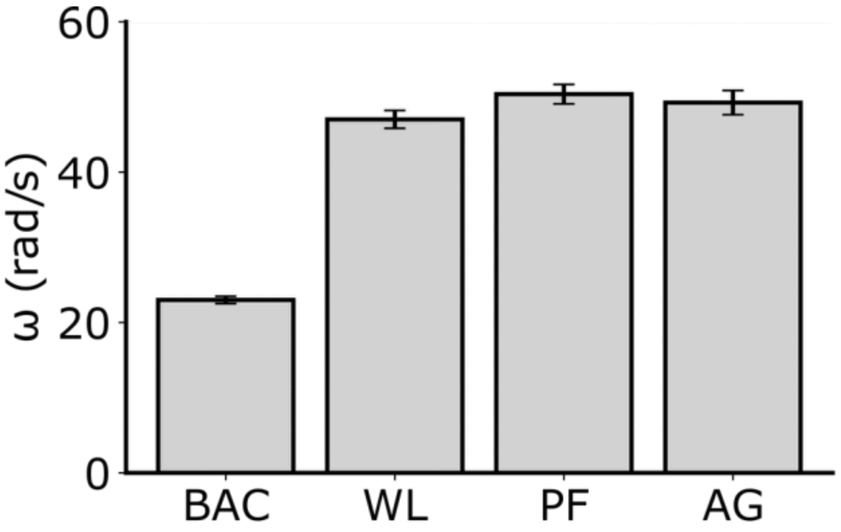
Bacterial Motility in Actual Soil Samples



Agricultural Land Pine Forest Wetland

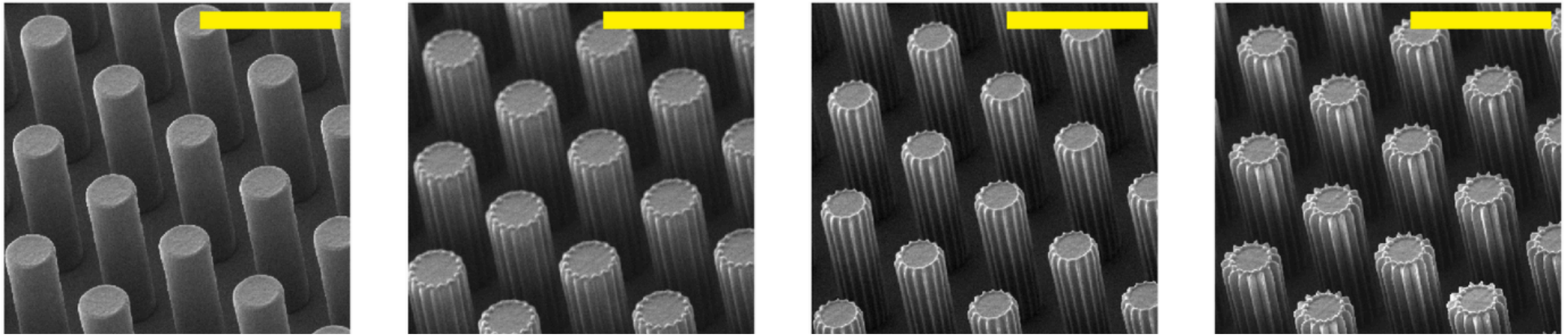


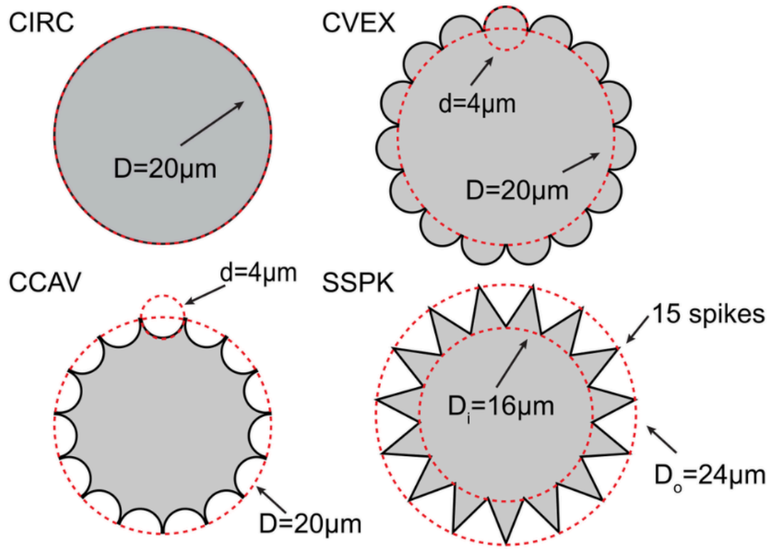


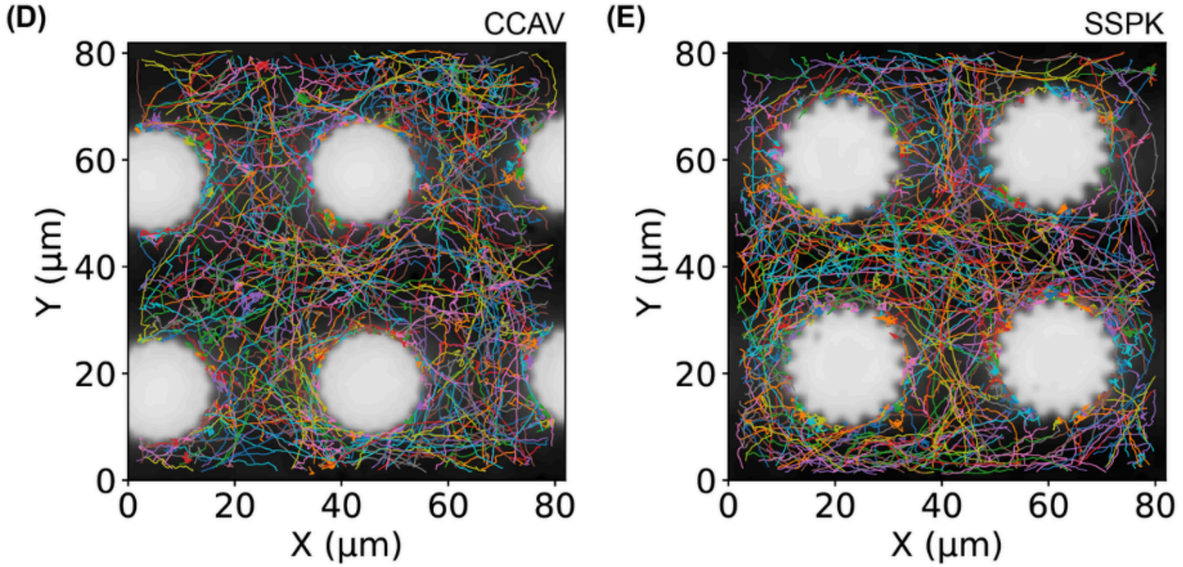


Shrestha et al. Bioterphases (2025)

Bacteria Motility around Micropillars





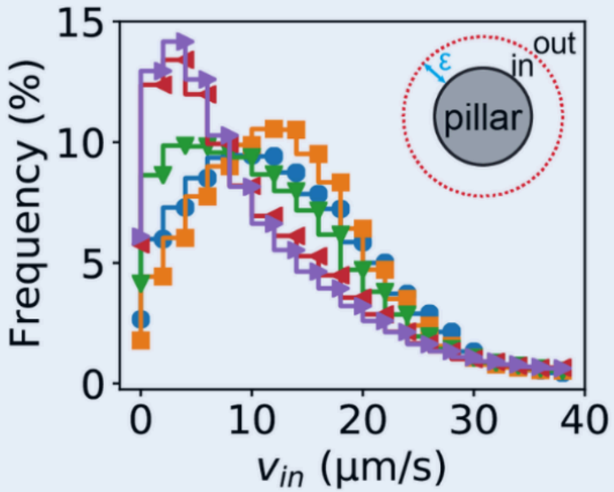


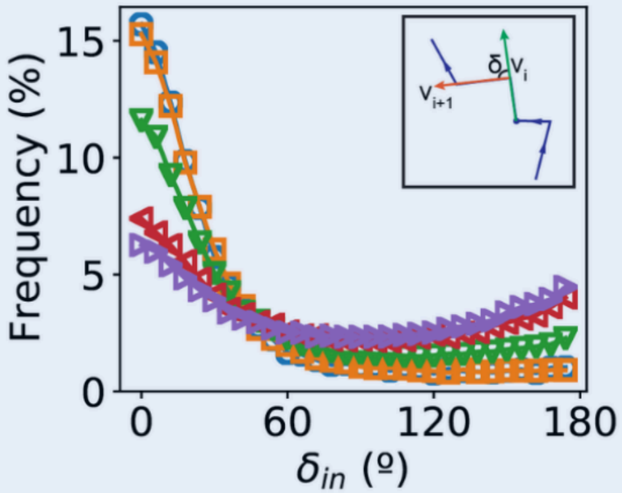
Krishnamurthi et al., CSB (2022)

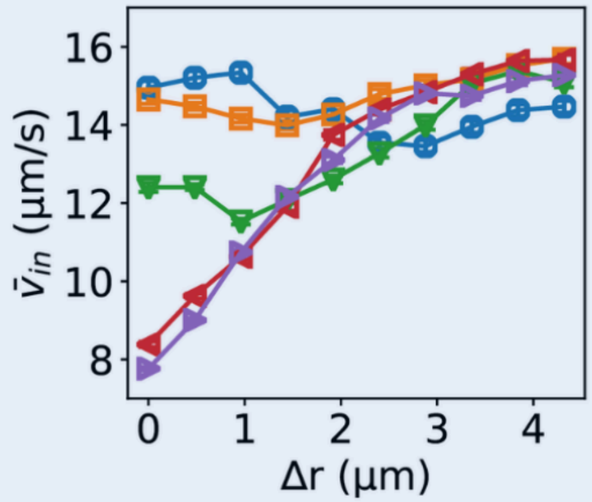
Bacteria Motility around Micropillars

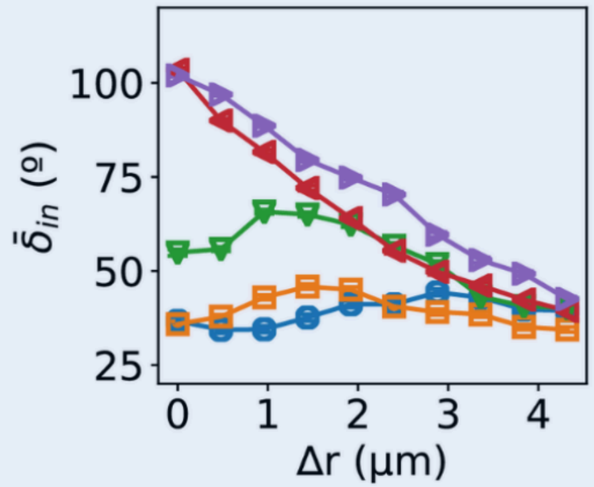
Krishnamurthi et al., CSB (2022)

Experiments

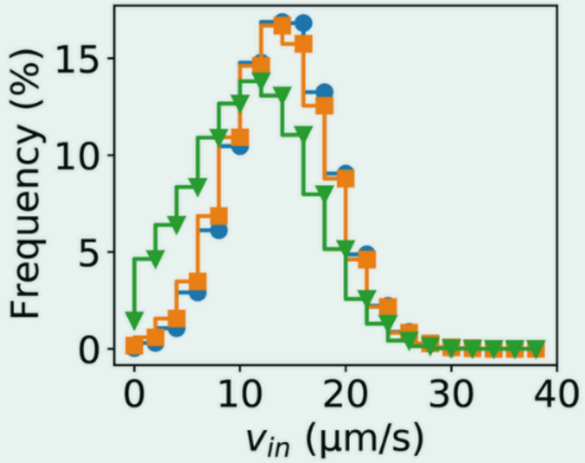


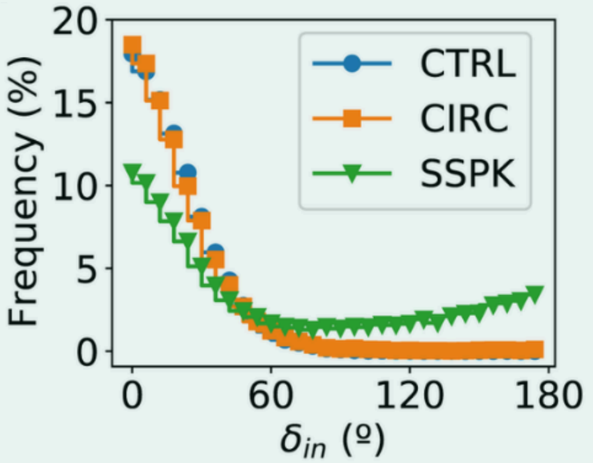


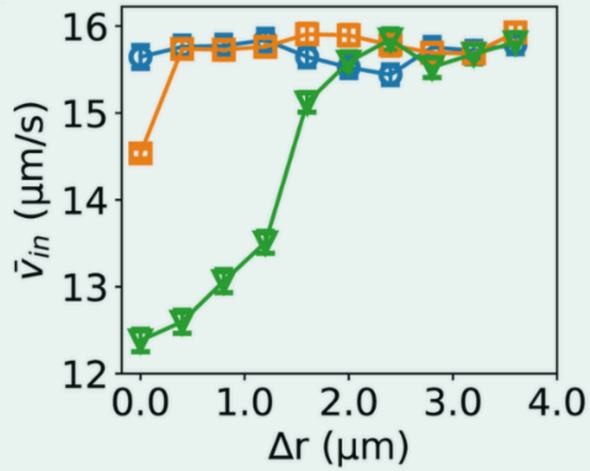


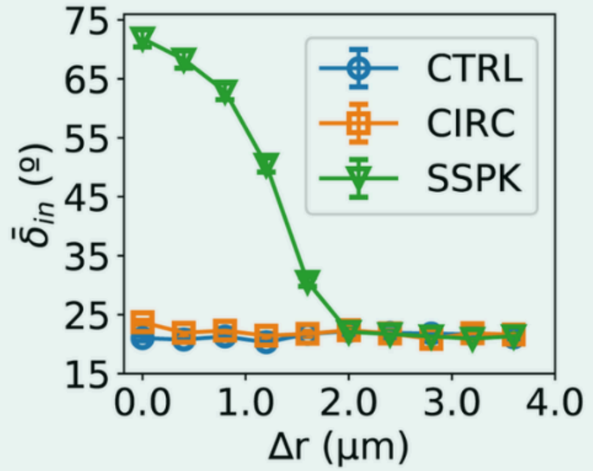


ABP-Based Simulations

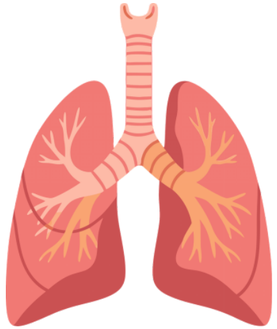


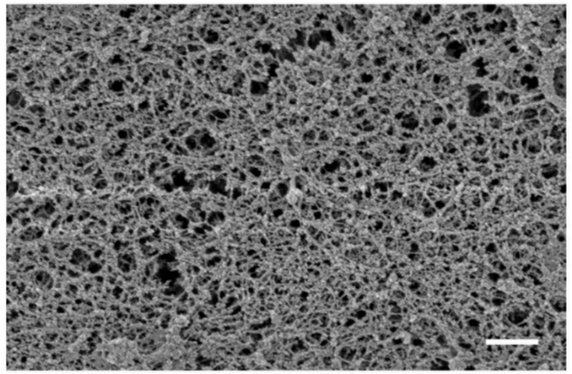


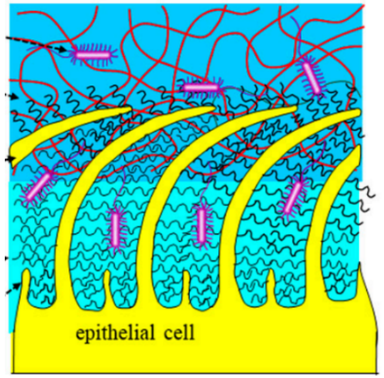




Bacterial Flagella Behavior in Soft Polymeric Network





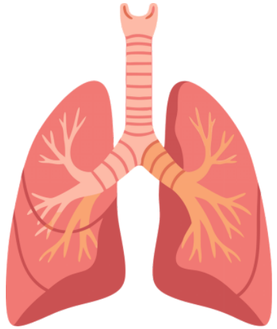


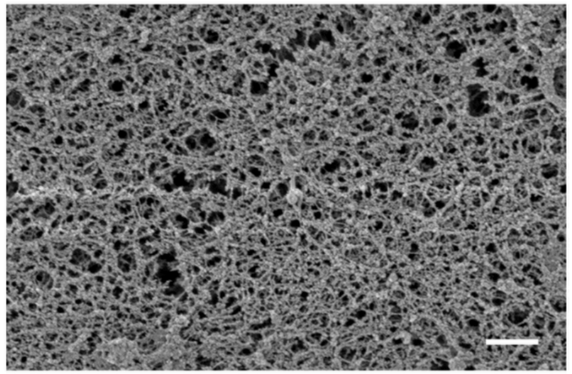
Abrami et al., IJMS (2023)

Pangeni et al., Int. J. Pharm. (2023)

Celli et al. PNAS (2009)
Bhattacharjee & Datta, Nat. Comm. (2019)
Liao & Aranson, PNAS Nexus (2023)
Grognot et al. AEM (2021)
Bassu et al. CSB (2024)
Martinez et al. PNAS (2014)
.....

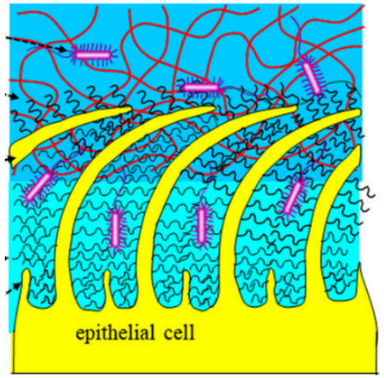
Bacterial Flagella Behavior in Soft Polymeric Network





Most studies focused on bacterial cell-body motility.

Behavior and dynamics of flagellar filaments in complex environments remain largely unexplored!



Abrami et al., IJMS (2023)

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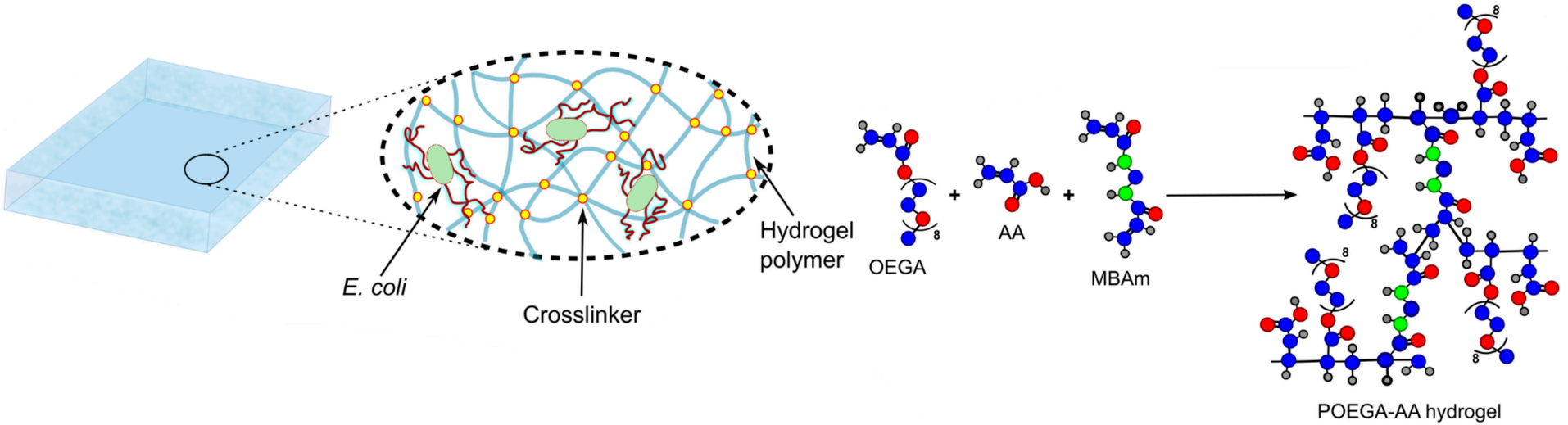
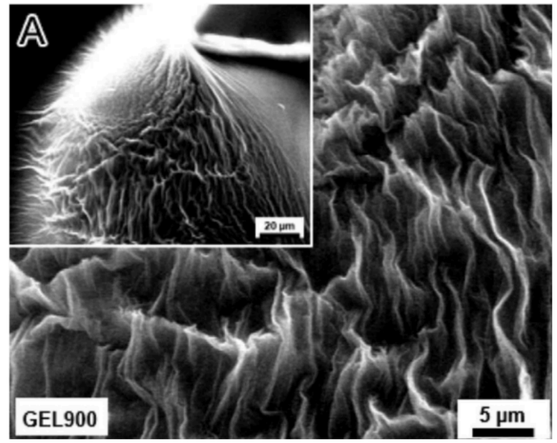
.....

Image from Shutterstock

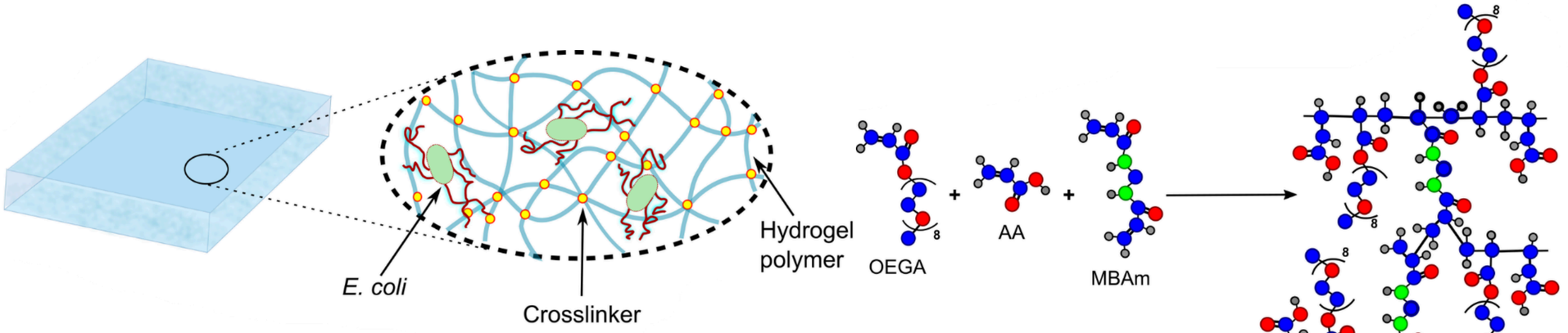
Model System: Polymeric Hydrogel + *E. coli* Bacteria

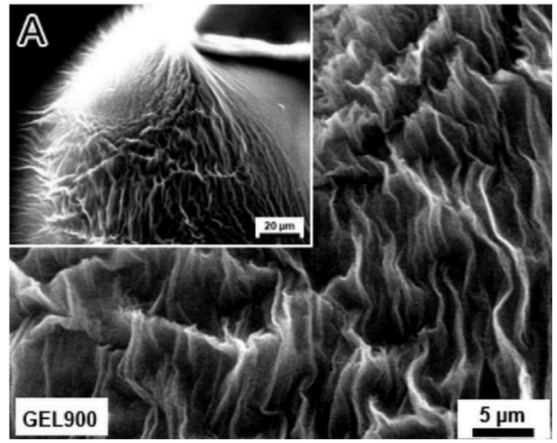


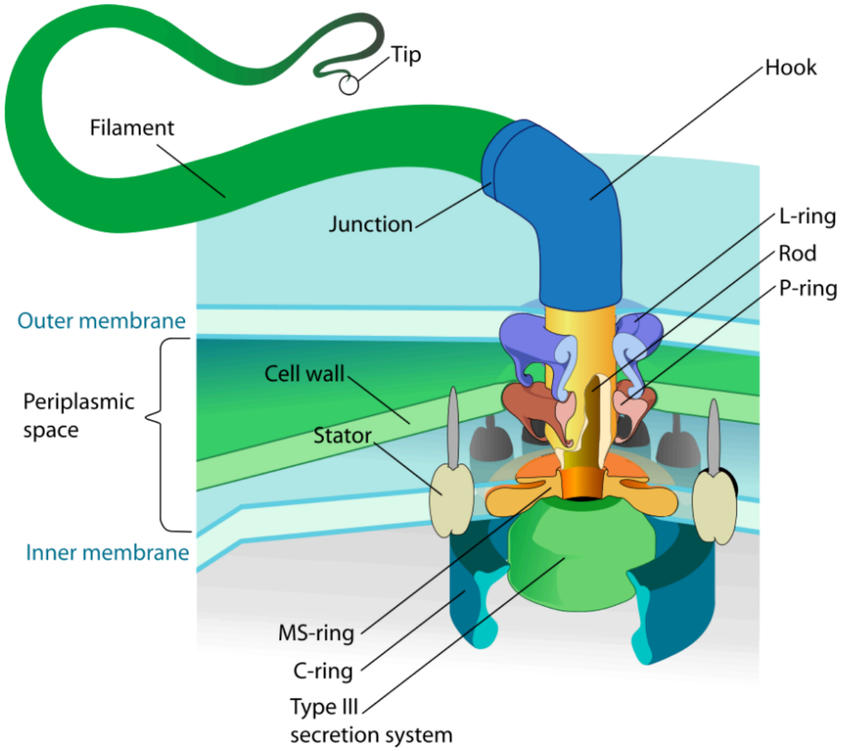


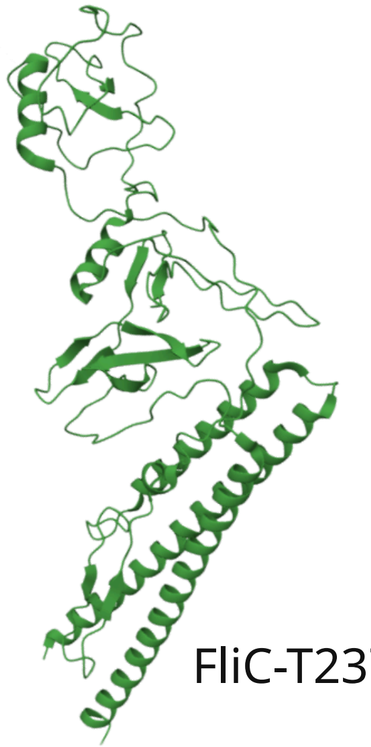
Shrestha et al. PRE (2025)

Model System: Polymeric Hydrogel + *E. coli* Bacteria

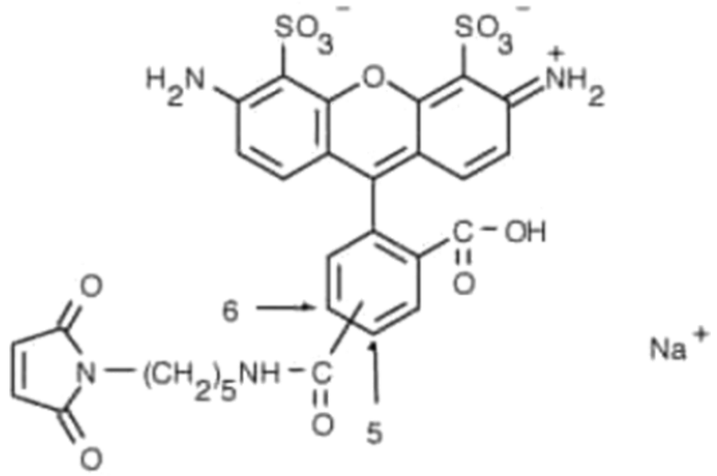








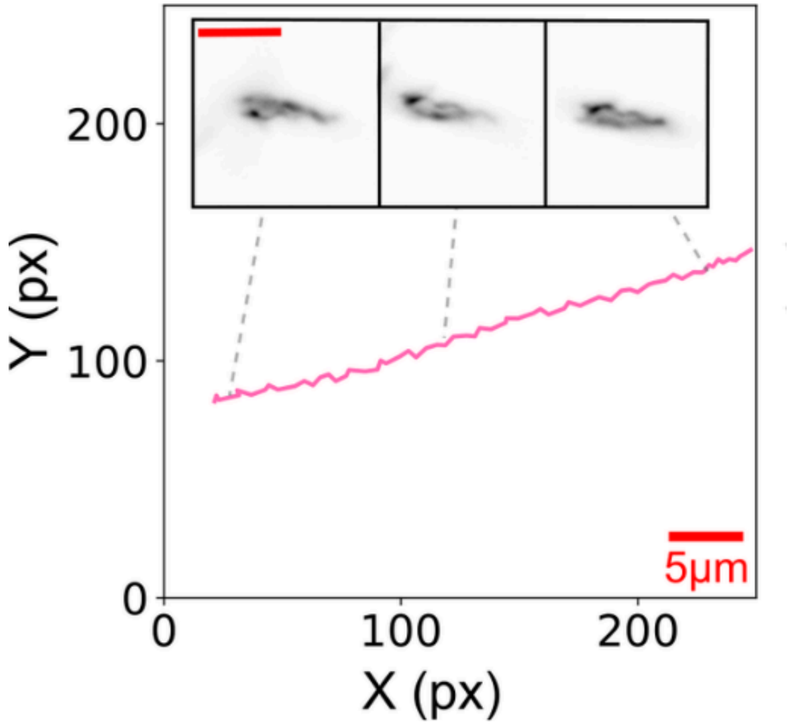
FliC-T237C



Maleimide AF488

Shrestha et al. PRE (2025)

Flagellar Dynamics



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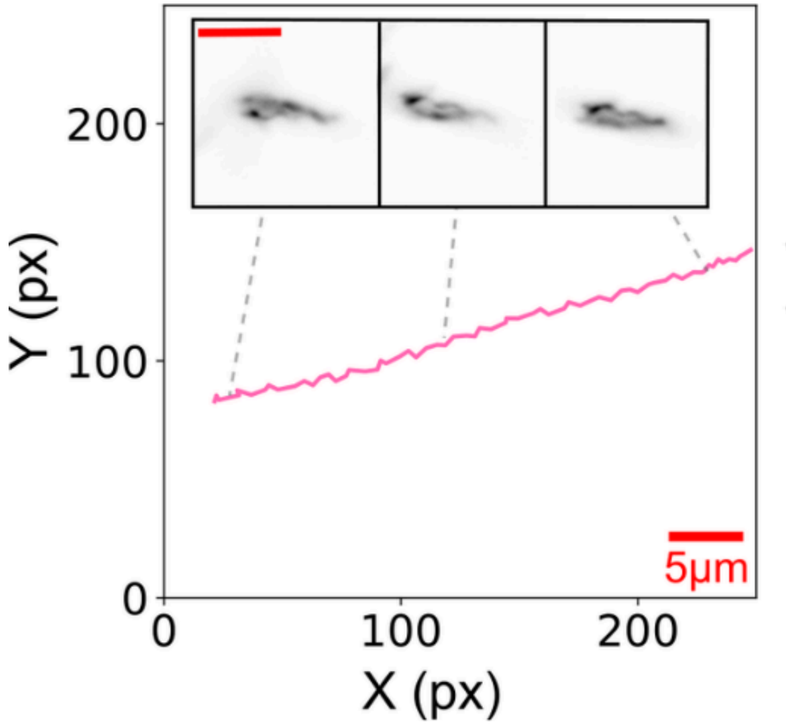
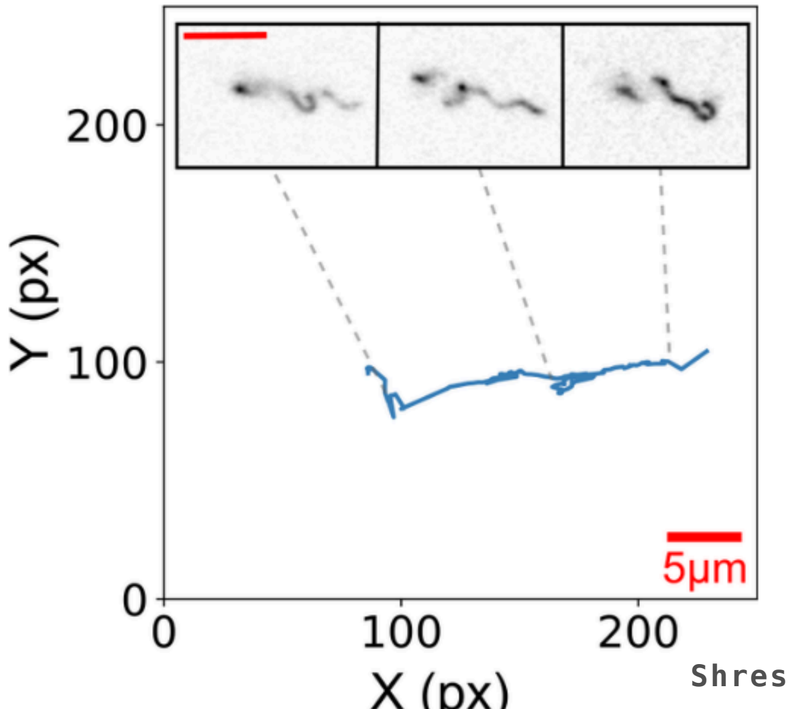
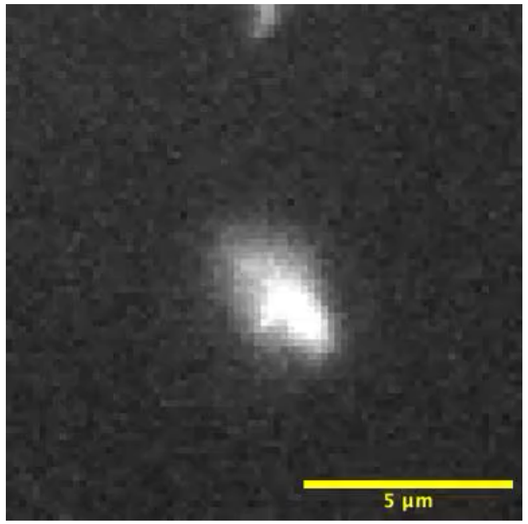
Flagellar Dynamics

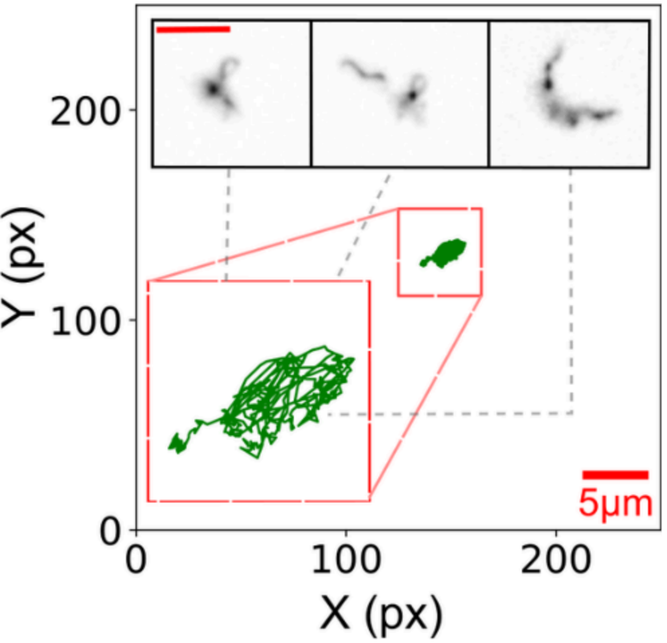




Shrestha et al. PRE (2025)

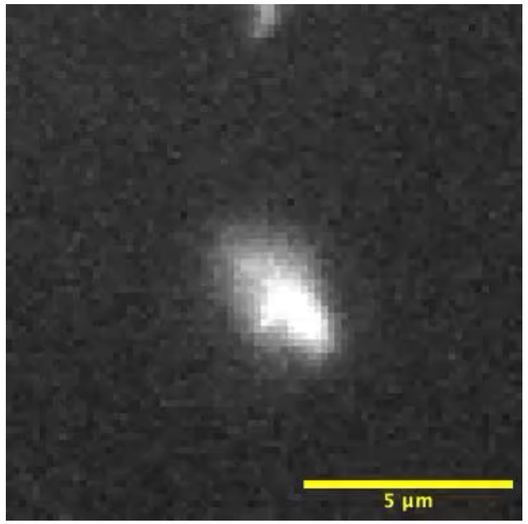
Flagellar Abnormalities - Kinks, Loops, Semi-coils

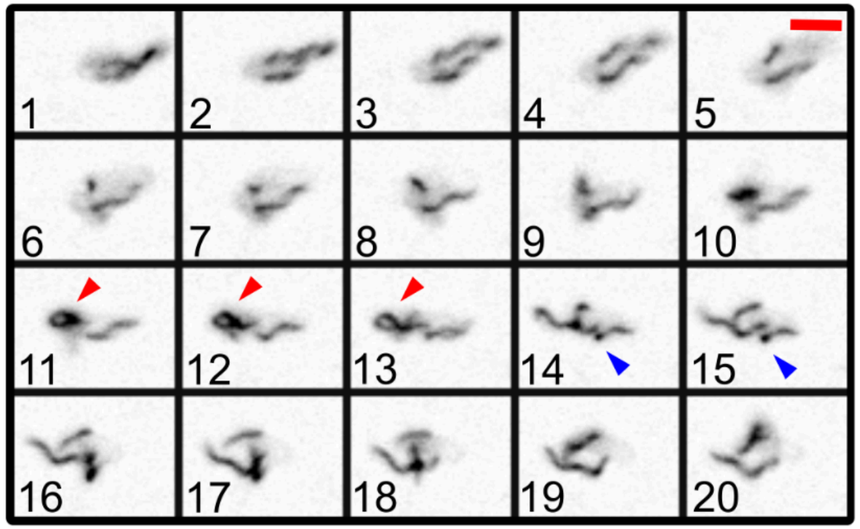


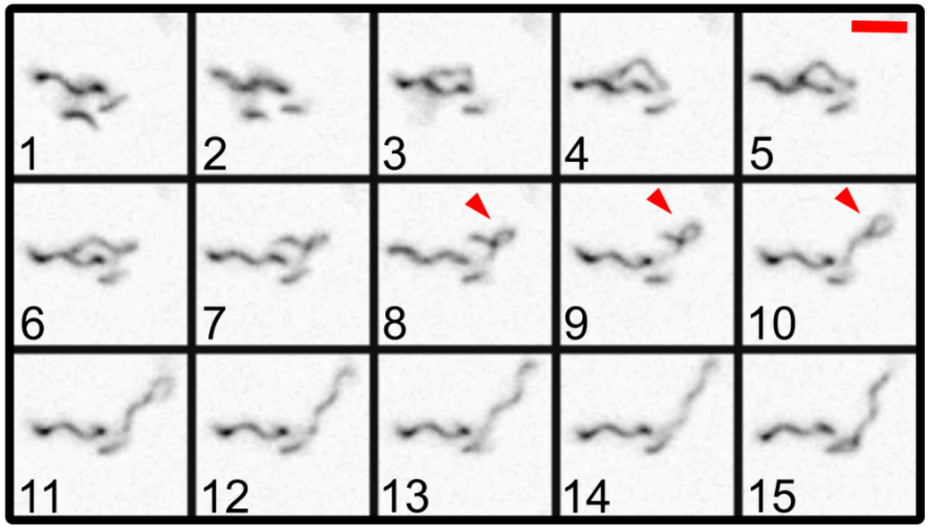


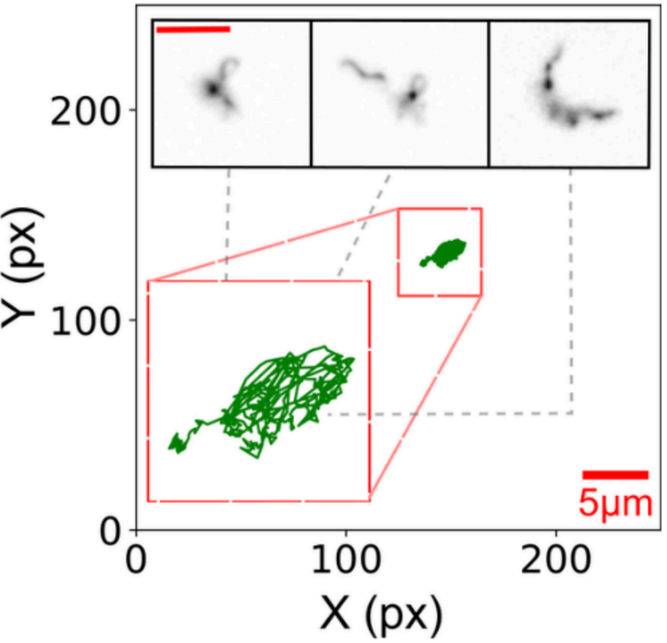
Shrestha et al. PRE (2025)

Flagellar Abnormalities - Kinks, Loops, Semi-coils









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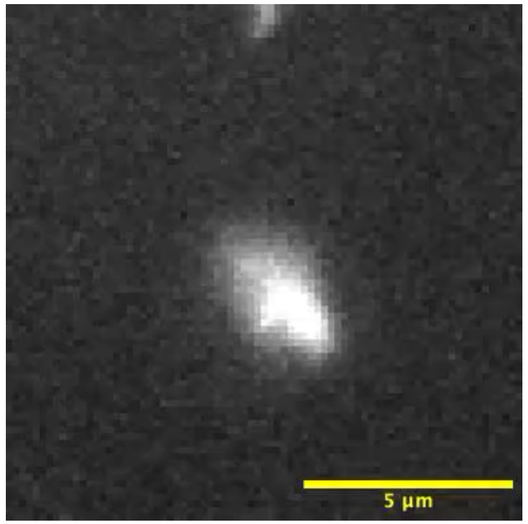
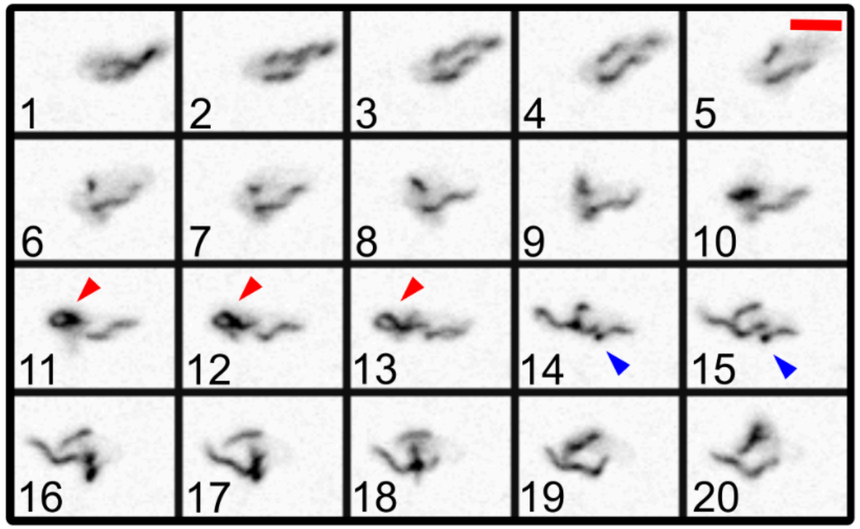
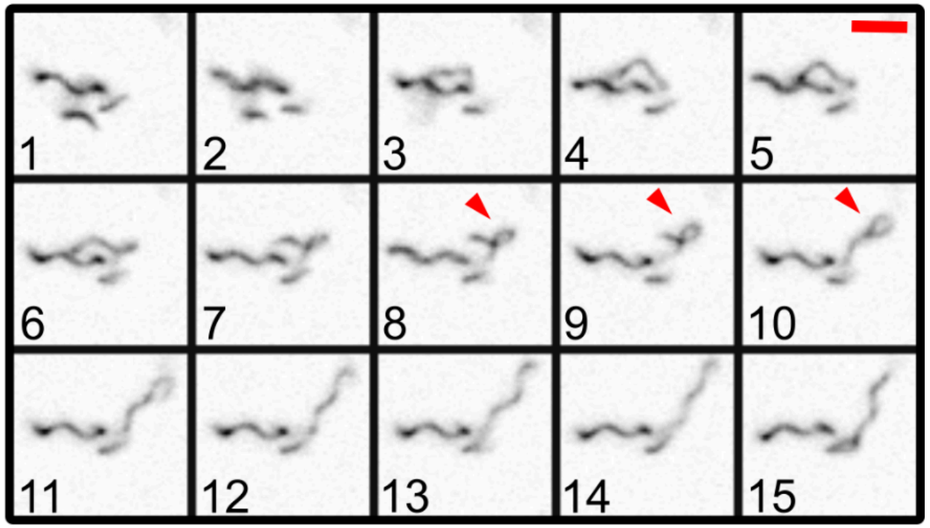
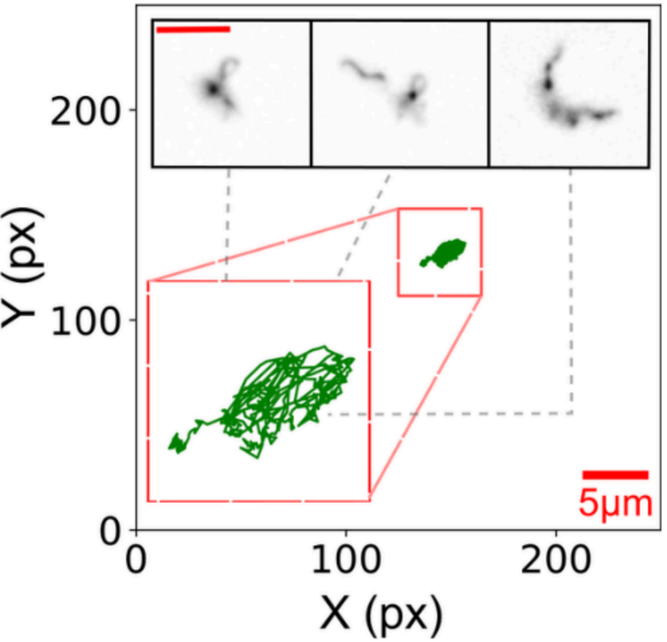
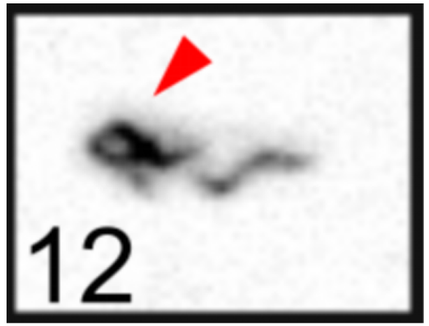
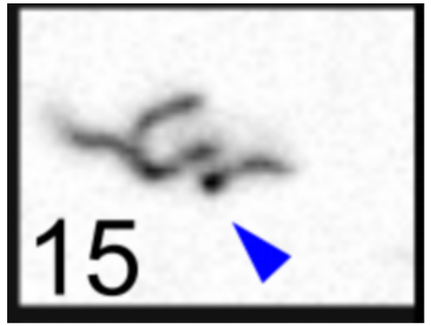


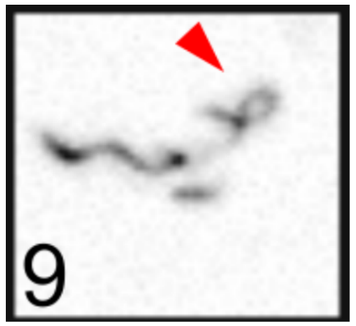






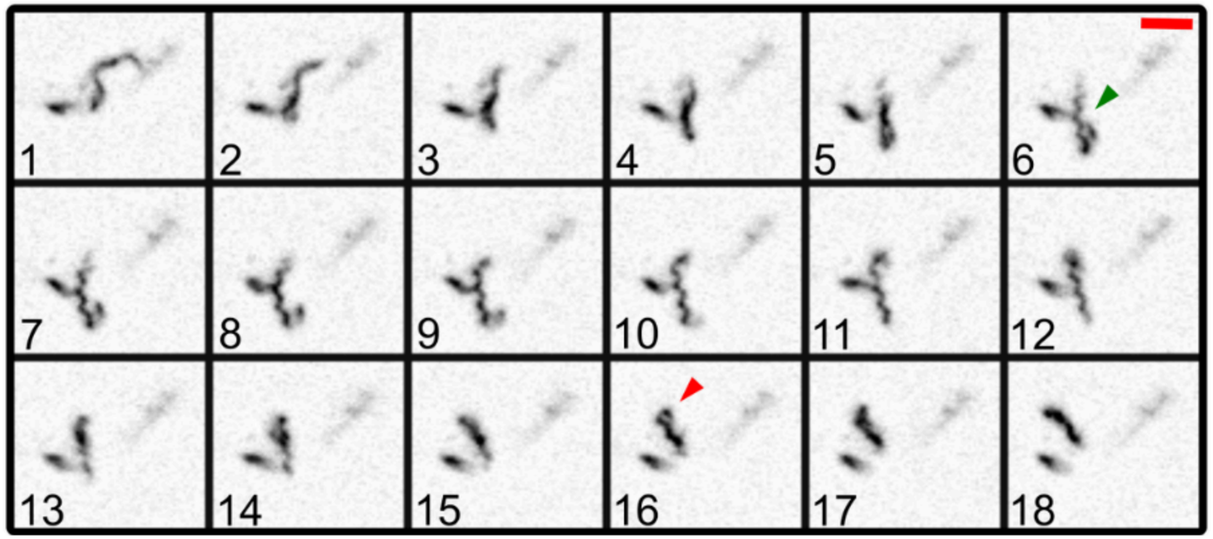


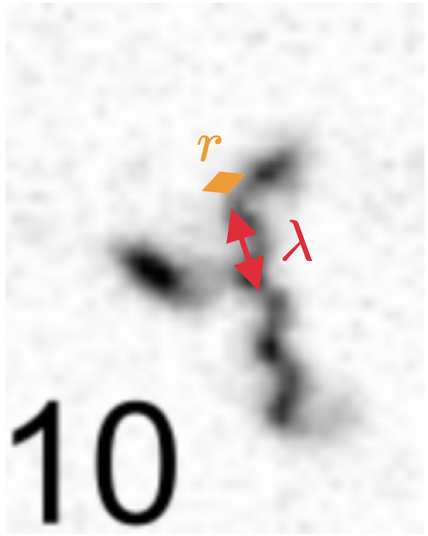




Shrestha et al. PRE (2025)

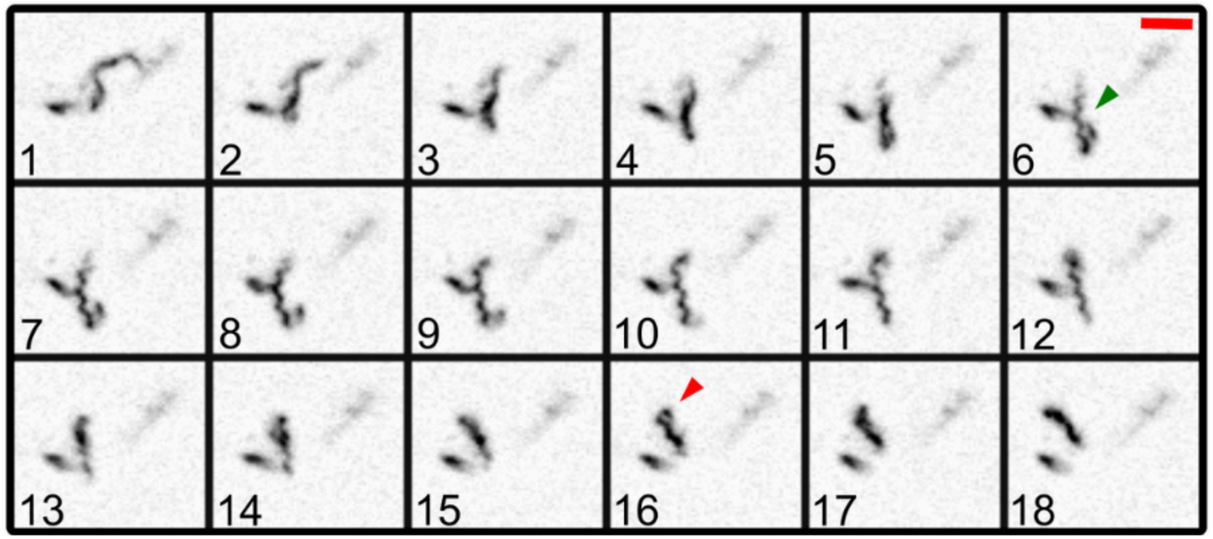
Flagellar Abnormalities - Kinks, Loops, Semi-coils

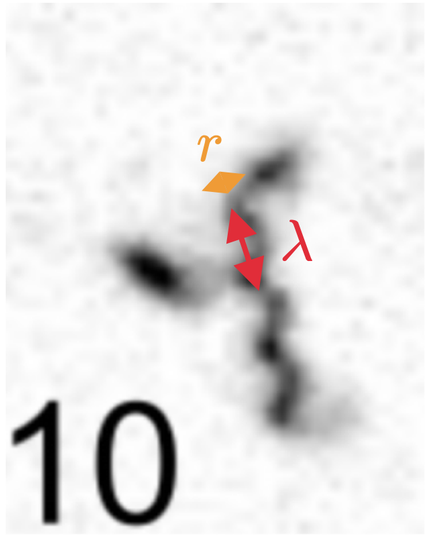




Shrestha et al. PRE (2025)
Calladine et al. JMB (2012)

Flagellar Abnormalities - Kinks, Loops, Semi-coils





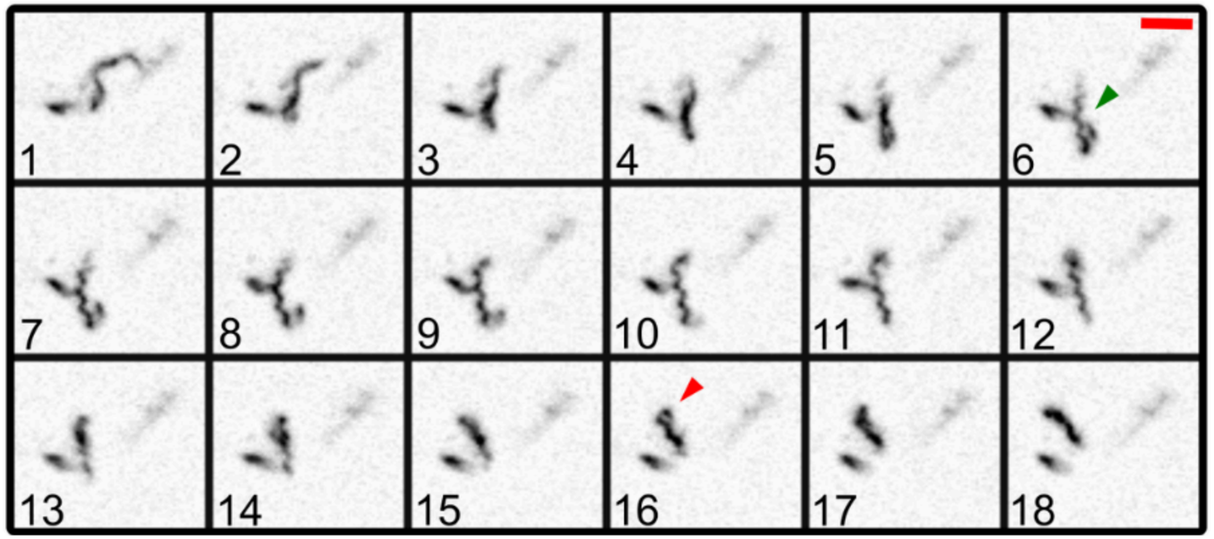
$$\ell = \frac{\lambda}{2\pi}$$

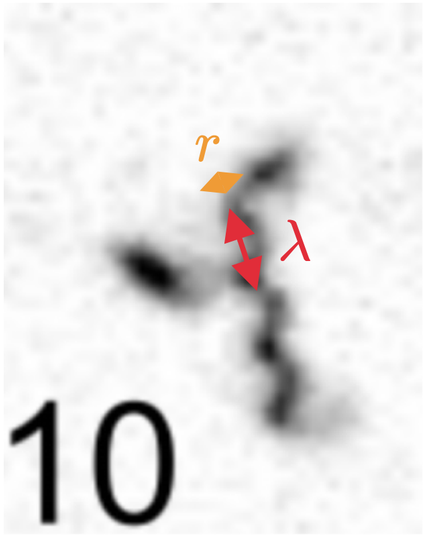
$$\kappa = \frac{r}{\ell^2 + r^2} \approx 3.2$$

$$\tau = \frac{\ell}{\ell^2 + r^2} \approx 2.1$$

Shrestha et al. PRE (2025)
Calladine et al. JMB (2012)

Flagellar Abnormalities - Kinks, Loops, Semi-coils





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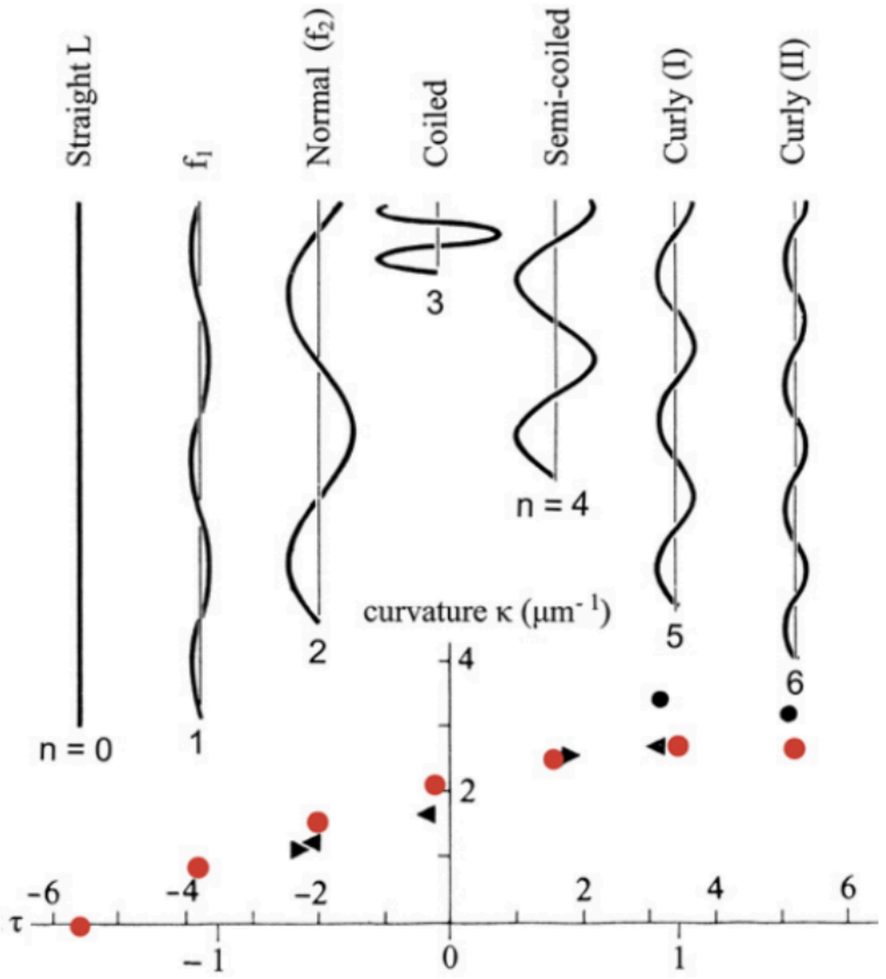

Shrestha et al. PRE (2025)
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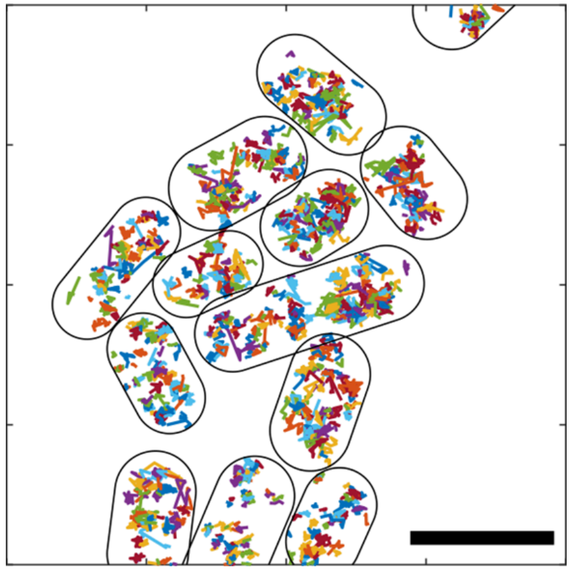
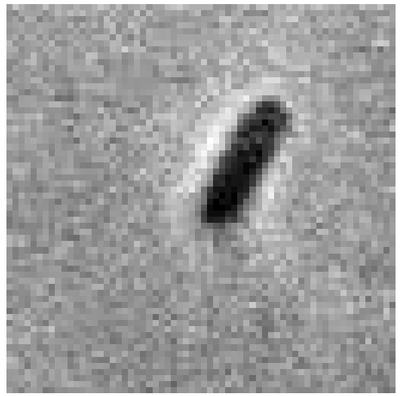
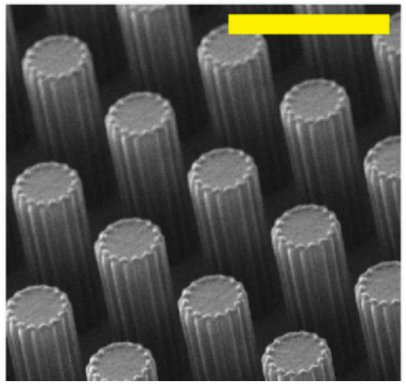
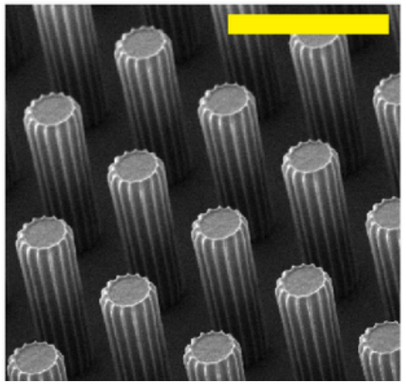
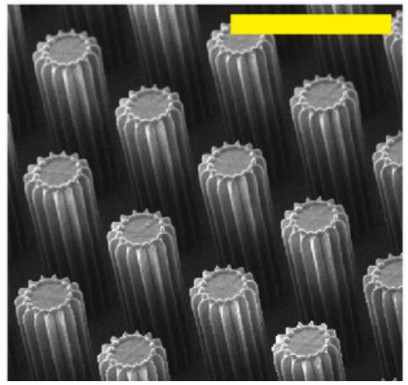
Summary

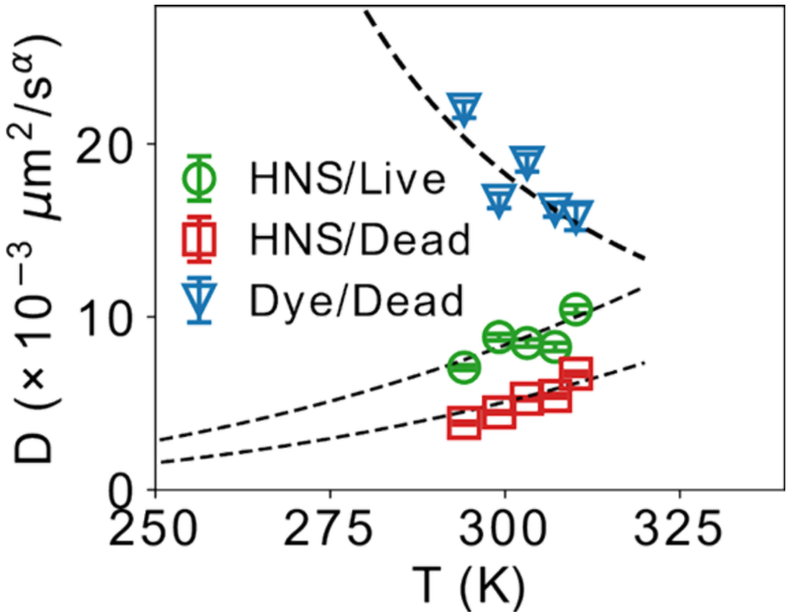


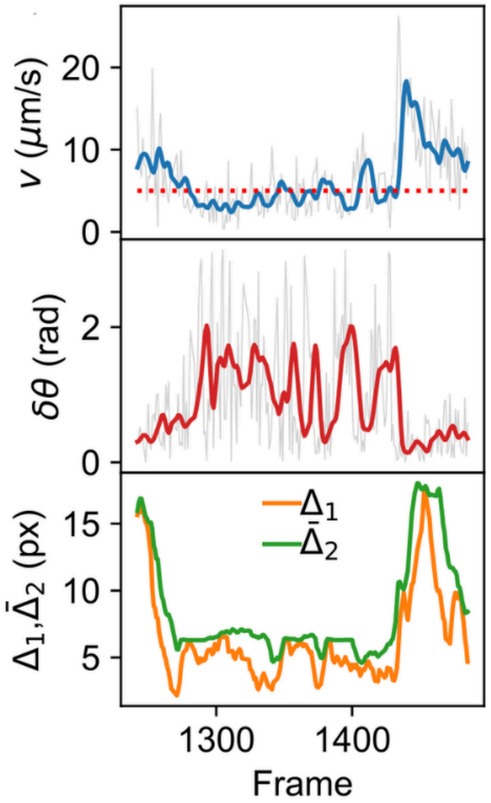


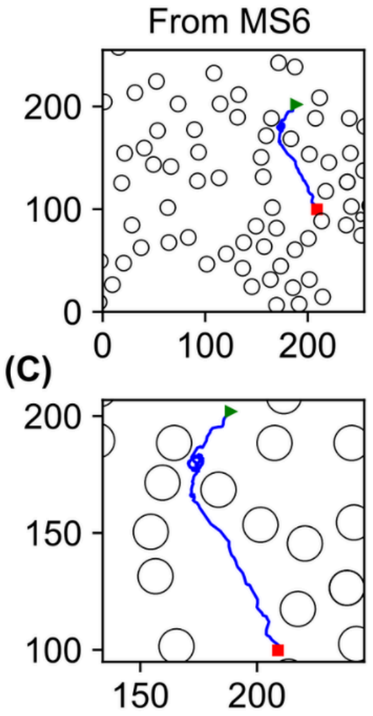


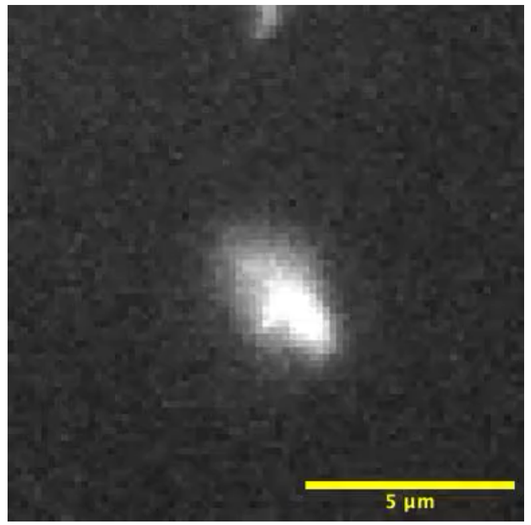





D (× 10⁻³ μm²/sᵅ)
20
10
0
250
275
300
325
T (K)
HNS/Live
HNS/Dead
Dye/Dead


v (μm/s)
20
10
0
δθ (rad)
2
0
Δ1,Δ̄2 (px)
15
10
5
Δ1
Δ̄2
1300
1400
Frame


From MS6
200
100
0
0
100
200
(C)
200
150
100
150
200


5 μm

Acknowledgment

Group Members

- Hemanta Pradhan (PhD)
- Michael Stewart (PhD)
- Rejeena Jha (PhD)
- **Sam Mortenson (PhD)**
- Brian Ewalt (BS)

Collaborators

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